



اونيفرسيتي مليسيا قهغ السلطان عبد الله
UNIVERSITI MALAYSIA PAHANG
AL-SULTAN ABDULLAH

BCS3263 SOFTWARE QUALITY ASSURANCE

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FINAL REPORT

PROJECT NAME: KOSSO FISHING STORE

LECTURER'S NAME:

PN. ROSLINA BINTI MOHD SIDEK

PROF. TS. DR. KAMAL ZUHAIRI BIN ZAMLI

Group Members:

No.	Name	ID
1.	Abdul Razzaq bin Abdul Aziz	CB22056
2.	Aina Jaleena binti Jusop	CB21089
3.	Mohamad Hatta Rahman bin Borhanuddin	CB21080
4.	Nazrin Naim Bin Ahmad Mahari	CB21072

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List of Abbreviations

Abbreviation	Definition
KFS	KOSSO Fishing Store
KFMS	KOSSO Fishing Management System

1. Project Proposal

1.1 Introduction

1.1.1. Current Issues

The KOSSO Fishing Store (KFS) is a store that operates business selling fishing equipment particularly for fishing enthusiasts. The store has been operating for 2 to 3 years, with the owner working alone and managing all aspects of the business without additional staff. The current operation of KOSSO Fishing Store (KFS) faces several challenges due to its physical-only business model. The daily sales exceed RM200, and monthly profit is roughly around RM5,000. This is due to the absence of an online platform restricting the potential sales for customers who prefer to purchase the equipment on online platforms.

Moreover, KFS faces seasonal challenges, particularly during the rainy season when stock movement slows down as fewer customers go fishing. This results in times of slower inventory and lowering profit potential. However, by not using an online platform, the company misses out on the possibility to sell the fishing equipment to customers in state unaffected by the rainy season which might help balance stock levels and sales during sluggish seasons. By implementing a management system with online accessibility will help to address these difficulties by expanding a customer base and improving inventory turnover.

1.1.2 Objectives of Study

1. To study the current challenges and limitations faced by the owner, KFS's staff and customers of the existing physical KFS.
 - i. Limited purchasing options: Customers only get to purchase products in physical store that is restricting access for online purchase.
 - ii. Location limitation: Customers cannot easily purchase products in physical store if they are located far from the store's location.
 - iii. Missed sales opportunities: KFS will miss out the sales opportunity that will reduce the sales of the business due to the inability to offer remote customers leads to a reduction in sales.
 - iv. Inventory management issue: Staff will face issues when handling inventory such as inaccurate stock counts as they are manually record-keeping.
 - v. Human error: Relying on manual record-keeping increases the possibility of errors in stock tracking, order processing, and handling customer data, leading to delays and a less efficient workflow.
2. To develop an online platform which is KOSSO Fishing Management System (KFMS) to enhance customer experiences and facilitate the owner and store's staff of KFS operations. It includes the functionalities such as managing users, products, tracking and managing stock levels, customer reviews and sales processes.
3. To evaluate the impact of the KFMS on operational areas such as inventory management, sales performance and customer satisfaction.

1.1.3. Project to be develop

The KOSSO Fishing Management System (KFMS) will be developed and used by any fishing enthusiast customer, owner and store's staff. It aims to develop the project that will facilitate the owner to run the business operations smoothly. To achieve the aim, Analyse Sales module will be designed for the owner to view the sales overview where it is breakdown into the amount of sales, revenue, profit and cost. Moreover, the owner can view the inventory summary which are the quantity of stock in hand and the stock that are on the way to be delivered to the store. Other than that, it can be viewed as sales by categories which will divided into 6 categories of sales equipment which are reels, rods, fishing lines, baits & lures, accessories and others. To view the sales & purchase and order summary, owner can view both analysis that are represent in the form of graph. To make a personalized sales target, top selling product summary will be developed according to the quantity sold followed by the respective customer's name, stock's remaining quantity and price.

The other module of KOSSO Fishing Management System (KFMS) will complement the functionalities that will facilitate the owner as mentioned above by developing module Manage User as user will register or login to access the system. Next module is Manage Stock where this module is a simple and effective tool for store administrators to manage stock. Admins can easily check current stock levels, view product details, add new products, update prices and information, and remove products that are no longer available. The system updates stock levels in real time, helping to prevent overstock or running out of items. After any changes are made, the updated product details automatically appear in the product catalogue for customers to see.

Other than that, Manage Cart module is designed to help customers easily manage their selected items after browsing the product catalogue. Once customers choose a product, they can add it to their cart, where they have full control over their selections. This includes viewing the items, updating quantities, or removing products entirely. The cart provides a clear summary of selected items, ensuring customers can review everything before proceeding to the next step. After finalizing their choices, users can proceed to checkout, where the payment process will take place. The payment page will display the subtotal price of items, shipping cost and total. User also have

an option to pay using bank or cash on delivery method and they can successfully place an order after filling in their billing details and make a payment.

Lastly, Manage Review module will be developed by displaying customer feedback on a specific product where user has bought and received it. Other users can read previous reviews or do a quick survey before buy the items based on user's feedback as this page will include a functionality such as sorting option to filter the reviews by rating or date.

The KFMS system will be created in accordance with recognised industry standards, ensuring that all processes satisfy the quality and performance requirements. Throughout the project, quality assurance techniques will be used to ensure the system's dependability, efficiency, and usability, resulting in a smooth experience for both the store's management and its customers.

1.2 Problem/Motivation

KOSSO Fishing Store lacks an online presence, which limits its reach to only local customers and hinders its capacity to interact with a larger group of fishing enthusiasts. While many stores these days use e-commerce platforms to sell their products, KOSSO Fishing Store mostly relies on in-store sales, which means it is missing out on prospects for growth and expansion. The store has been operating for 2 to 3 years, with no staff, relying solely on the owner. The store sees approximately 200+ sales per day, with monthly sales around 5,000, a business capital of 3,000, and a profit of about 2,000. However, it struggles during the rainy season when stock moves slowly because people tend not to go fishing. This situation highlights the potential for an online platform to reach customers in regions unaffected by rainy weather.

Currently, it is difficult for clients to simply browse and buy fishing rods, accessories, and fish food because the fishing business does not have a centralized, specialized website. Customers also lack access to extensive information, product comparisons, and reviews that could help them make decisions. In addition to negatively affecting consumer satisfaction, this online accessibility issue lessens the store's competitive edge in the expanding world of online retail.

In addition, manual tasks like tracking sales data and managing inventories without the use of an online system increase the risk of staff mistakes and inefficiency. The current configuration of the fishing store makes it difficult to manage orders effectively or to set up automatic updates, which makes it more difficult to give customers a seamless shopping experience.

As a result, the issue statement is to create a full online store for the fishing store that offers a consolidated platform for product display, purchase facilitation, and inventory management. Besides secure payment methods and user-friendly tools for product browsing, searching, and filtering, this website should also feature a streamlined customer support function, allowing customers to easily reach out for inquiries or assistance with their purchases.

The motivation for developing this online platform is to overcome the shortcomings of the fishing store's present sales strategy. The store has a great chance to reach fishing lovers across the country and increase its consumer base beyond the local community with the increasing trend

toward internet purchasing. Customers will find it easy to browse, compare, and buy fishing rods, accessories, and fish food on this platform, which will increase consumer satisfaction and encourage return business.

By implementing a website specifically tailored to the needs of fishing product buyers, the store may build a loyal customer base, maintain its competitiveness in the e-commerce market, and generate a strong online presence. In order to improve retail efficiency, the website will also automate several administrative tasks, including inventory control and order processing. The goal of this project is to give the fishing store the tools it needs to fulfill contemporary retail demands, increase operational effectiveness, and develop a convenient shopping environment that appeals to a wider range of fishing enthusiasts.

1.3 Background of Project

In today's digital age, online shopping has become popular for consumers looking for convenience and variety. However, KOSSO fishing store can only target customers who have access to the actual location due to the fishing store lacks online presence. Due to location limitations, this has caused a void between the store and a wider range of fishing lovers who could be interested in buying fishing rods, accessories, and fish food. In the absence of a digital platform, the store misses out on the chance to reach the ability of rising online market and potential sales opportunities.

While other retail shops might include fishing gear, they frequently don't have the same assortment or expertise as specialty fishing stores. Consumer evaluations, comparisons, and thorough product descriptions are often sought after by fishing enthusiasts, and they have a significant impact on their purchase choices. But without an online presence, the fishing shop is unable to showcase its distinctive goods and experience to customers who appreciate a reliable and informed source.

To overcome these challenges, the fishing store's website will offer a comprehensive online buying experience, allowing customers to explore and review products, do searches, apply filters, and safely finish transactions. Customers around the country will be able to shop on the website at any time thanks to its smooth and convenient design. The fishing shop can build a community around its products, boost customer engagement, and build a strong digital presence by setting up an online store. In the end, this will help the firm grow steadily and satisfy customers in a market that becomes more and more digital.

1.4 Development Methodology

1.4.1 Method of development

The Agile Methodology will be used as development process for the e-commerce platform of the KOSSO Fishing Store. Agile is well-known for its adaptability, customer-focused methodology, and incremental development, which makes it perfect for creating dynamic and changing systems. Through constant collaboration with stakeholders, such as store owners, staff members, and customers. This process will enable us to make sure the platform effectively satisfies their needs.

In accordance with agile principles, our team will start by establishing specific goals. Building an e-commerce platform to efficiently handle shop operations, process orders, and display fishing products as the main objective of this project. The platform's success depends on crucial elements including user management, stock management, product pages, review pages, and sales analysis. The most crucial things will be built and tested first in the development process.

To coordinate the efforts between developers, each sprint lasts between two to four weeks, during which time all the developers will work on their designated modules concurrently. For example, the Stock Management developer can configure inventory systems while the Sales Analysis developer creates dashboard architecture. Regular meetings will be held throughout the sprint to ensure alignment with project objectives.

At the end of each sprint, stakeholder reviews will be done to obtain feedback for future improvements. These reviews will assist to improve the platform based on real-time user feedback. After receiving feedback, the team will revise the schedule of next sprint, making sure that any new requirements are prioritised. This feedback loop is critical for producing a system that is always improving based on user needs.

In conclusion, this project will use a flexible methodology to plan activities and monitor progress, with frequent meetings and open communication. Through the implementation of agile principles, the team hopes to provide a scalable, fully functional online platform that satisfies

contemporary e-commerce requirements and offers a seamless experience for the company and its customers.

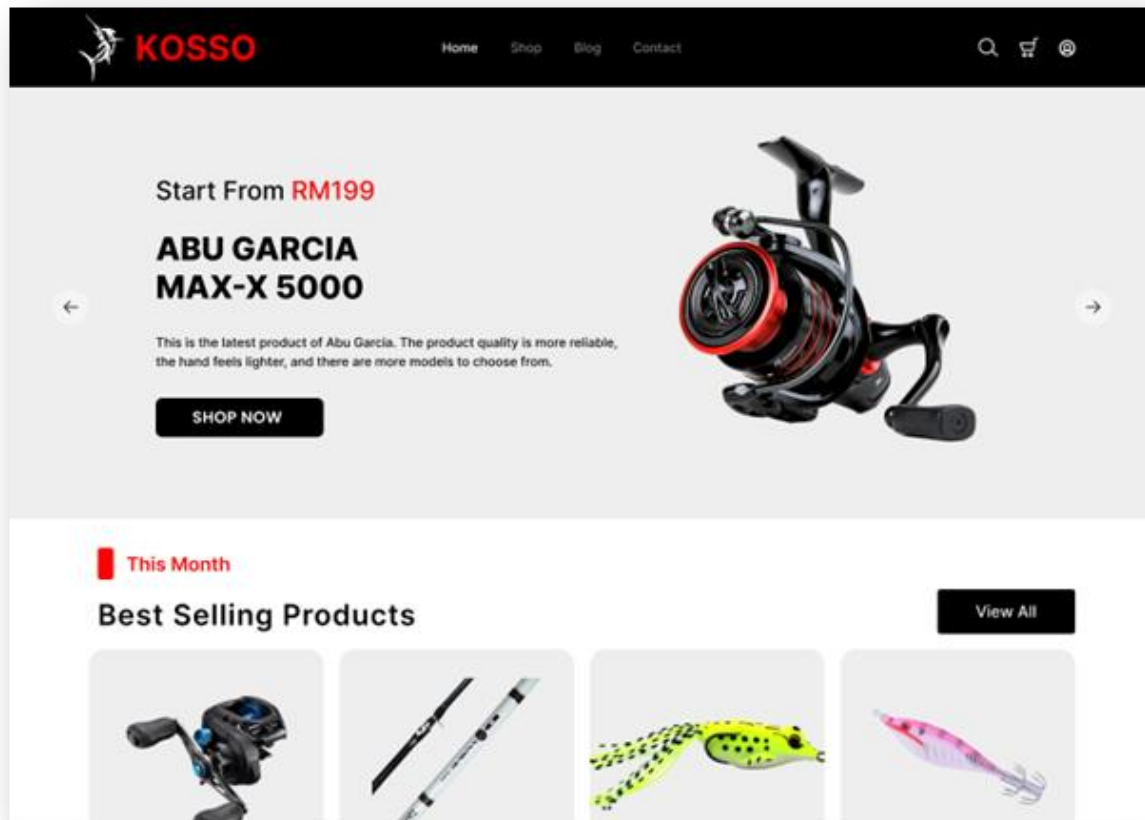
1.4.2 Module for each member

Module Name	Module Description	Person in Charge
Manage User	This module implements an effective user management system for the KOSSO Fishing Store. It enables new users to register easily by entering required information and creating a secure login. Once registered, individuals can access their accounts and manage their profiles, including changing personal information as needed. Furthermore, if a user decides they no longer want to be connected with the business, they may completely delete their account, maintaining complete control over their personal information. Admins may also have access to view or manage user accounts for administrative purposes	Abdul Razzaq bin Abdul Aziz (CB22056)
Analyse Sales	The Analyse Sales module is a user-friendly data analysis and visualisation tool created exclusively for store administrators. It gives real-time information about sales performance, stock levels, and other company data. Administrators can use dynamic charts and graphs to evaluate sales patterns, check inventory levels, and acquire important insights into client purchasing habits. This dashboard improves decision-making by displaying complex data in an intelligible fashion, ultimately assisting with strategic planning and business success.	Abdul Razzaq bin Abdul Aziz (CB22056)
Manage Stock	The Manage Stock module is a robust stock management solution designed specifically for store administrators. It allows for efficient control and oversight of all products featured on the KOSSO Fishing Store. Administrators may quickly view current stock, view stock details, add new products, update existing	Nazrin Naim Bin Ahmad Mahari (CB21072)

	product information, and delete products that are no longer available. This module uses real-time stock updates to guarantee that inventory levels are appropriately maintained, reducing the risk of overstocking or stockouts and increasing supply chain efficiency.	
Manage Cart	This module offers customers a seamless and user-friendly buying experience by managing the items they wish to purchase. After browsing the product listings and reading item descriptions, customers can easily add items to their cart. The module contains a fully functional cart system that allows users to manage their selections by adding products, adjusting quantities, or removing items before proceeding to checkout. This design ensures customers have complete control over their purchases and provides a smooth, uninterrupted shopping experience from product selection to final order confirmation	Aina Jaleena binti Jusop (CB21089)
Manage Review	The Manage Review module encourages customers to give input by allowing them to leave feedback on the things they buy. After an item has been delivered, users can write thorough reviews, share their experiences, and even attach product images. Customers can also update and delete their reviews as needed, ensuring that their feedback is accurate and up to date. This module not only improves customer happiness but also enables prospective consumers to make informed decisions based on real-world consumer experiences.	Mohamad Hatta Rahman bin Borhanuddin (CB21080)

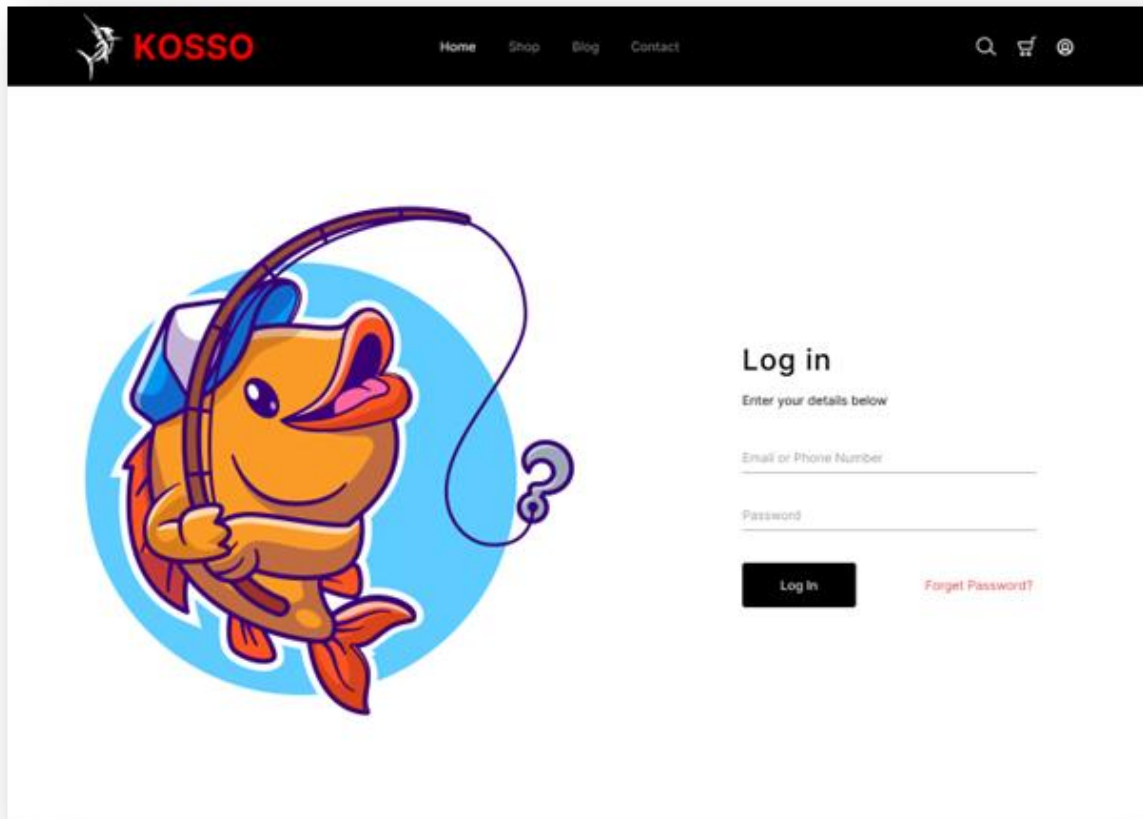
1.5 Expected outcomes

Landing Page



The landing page is main entry points of the website, designed to grab the attention of customers for the first enter the website. It features a navigation bar for easy access to different sections and a banner that showcases special promotions or highlighted items. Users will also see the top-selling products displayed with images and prices. Additionally, there are call-to-action buttons like "Shop Now" and a search bar for quick product searches, making it easy for customers to explore the store.

Login Page



The image shows a web browser window displaying the login page for 'Kosso'. The header is black with the 'Kosso' logo in red and white on the left. Navigation links 'Home', 'Shop', 'Blog', and 'Contact' are in the center, and search, shopping cart, and user profile icons are on the right. The main content area has a white background. On the left is a large illustration of a cartoon orange fish with a blue cap, holding a fishing rod with a blue lure. On the right is the login form, which includes the title 'Log in', a subtitle 'Enter your details below', two input fields labeled 'Email or Phone Number' and 'Password', a black 'Log In' button, and a red 'Forgot Password?' link.

KOSSO Home Shop Blog Contact

Q 🛒 👤



Log in

Enter your details below

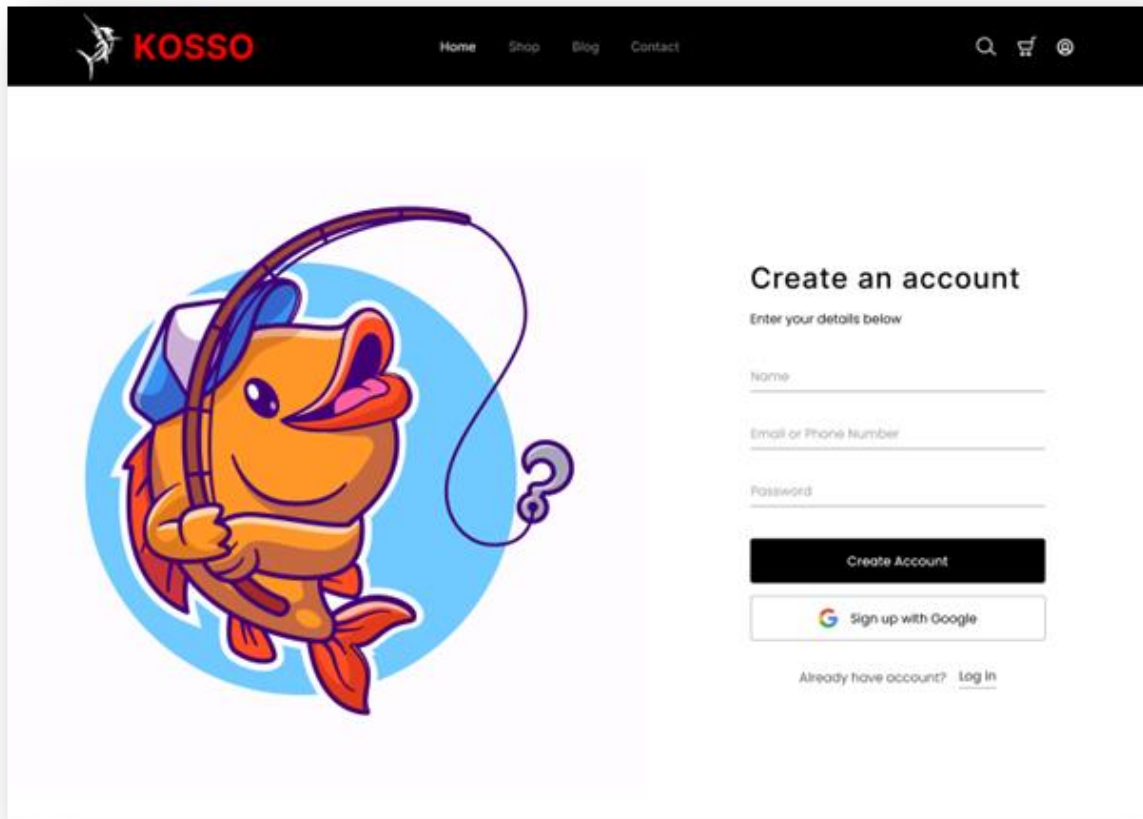
Email or Phone Number

Password

Log In [Forgot Password?](#)

On the login page, users can securely sign into their accounts. It features a straightforward sign-in form where they enter their email or phone number and password. For convenience, there is links for password recovery and registration for new users.

Registration Page



The image shows a web registration page for 'KOSSO'. The header is black with the 'KOSSO' logo in red and white on the left, and navigation links 'Home', 'Shop', 'Blog', and 'Contact' in white in the center. On the right of the header are icons for search, shopping cart, and user profile. The main content area has a white background. On the left is a large illustration of a cartoon orange fish with a blue cap, holding a fishing rod with a purple line and a purple hook. On the right is a 'Create an account' form. The form title is 'Create an account' in bold. Below it is the instruction 'Enter your details below'. The form contains three input fields: 'Name', 'Email or Phone Number', and 'Password'. Below these fields is a black 'Create Account' button. Underneath the button is a 'Sign up with Google' button featuring the Google logo. At the bottom of the form area, there is a link that says 'Already have account? [Log in](#)'.

The registration page lets new users create an account. The registration form captures essential information such as name, email or phone number, and password. A Google sign-up option makes things easier, and there's a button to submit the form. Users can also sign up using their Google account. If they already have an account, a link to the login page is provided for easy access.

Profile Page

The screenshot displays the KOSSO website's user profile page. The header features the KOSSO logo and navigation links: Home, Shop, Blog, and Contact. On the left, a sidebar menu includes 'Manage My Account' (with sub-links for My Profile, Address Book, and My Payment Options) and 'My WishList'. The main content area is titled 'Edit Your Profile' and contains the following form fields:

First Name	Last Name
Nazrin	Naim

Email	Address
nazrinnaime@gmail.com	Pekan, Pahang

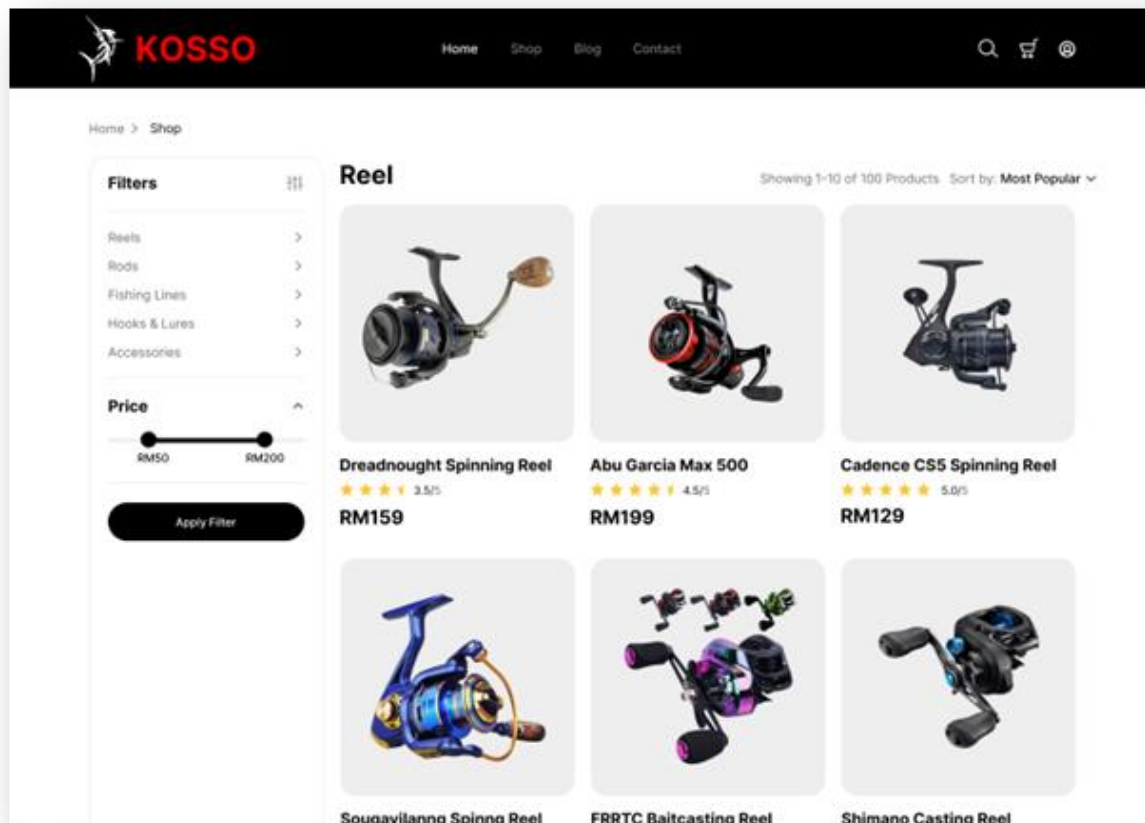
Password Changes

Current Passwod
New Passwod
Confirm New Passwod

At the bottom right of the form are two buttons: 'Cancel' and 'Save Changes'.

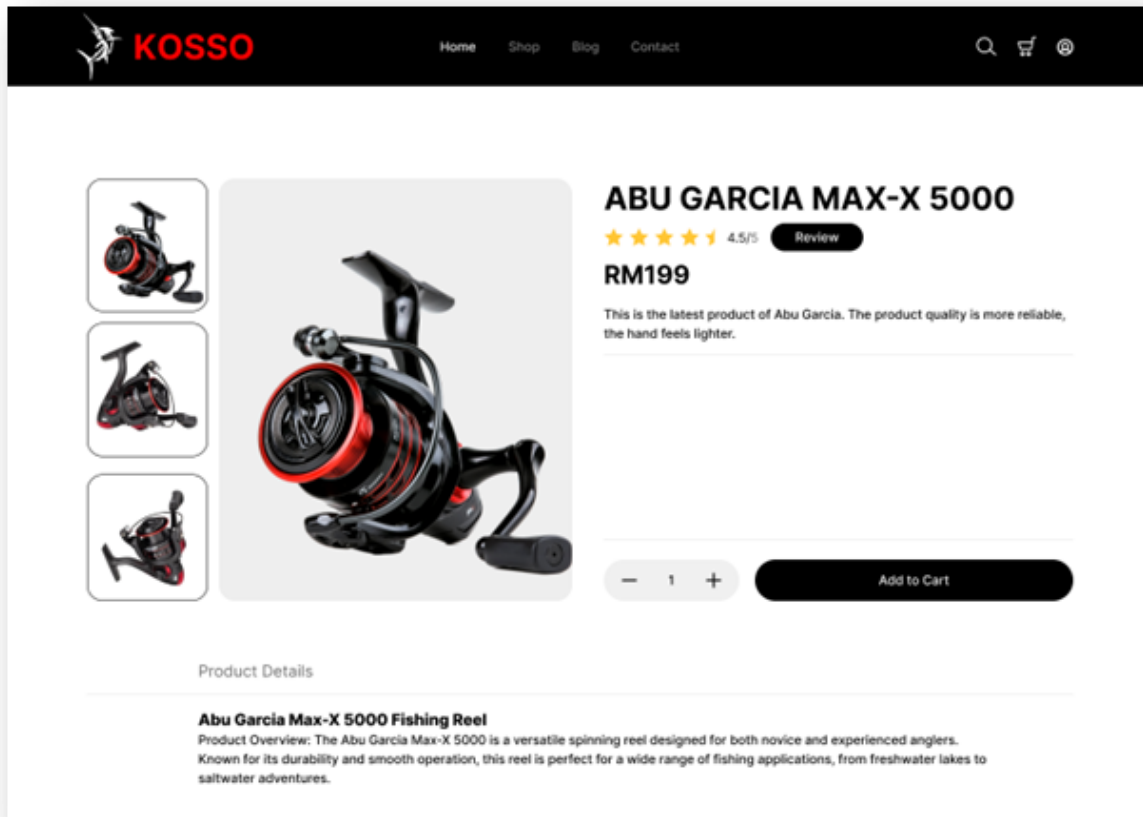
The profile page lets users view and edit their personal details. Users can update information like their name, email, phone number, and password. There's a "Save Changes" button to apply any updates they make, so they can keep their info up to date. If they decide not to change anything, they can hit the "Cancel" button to go back without saving any edits. Overall, this page makes it easy for users to manage their personal information.

Product List Page



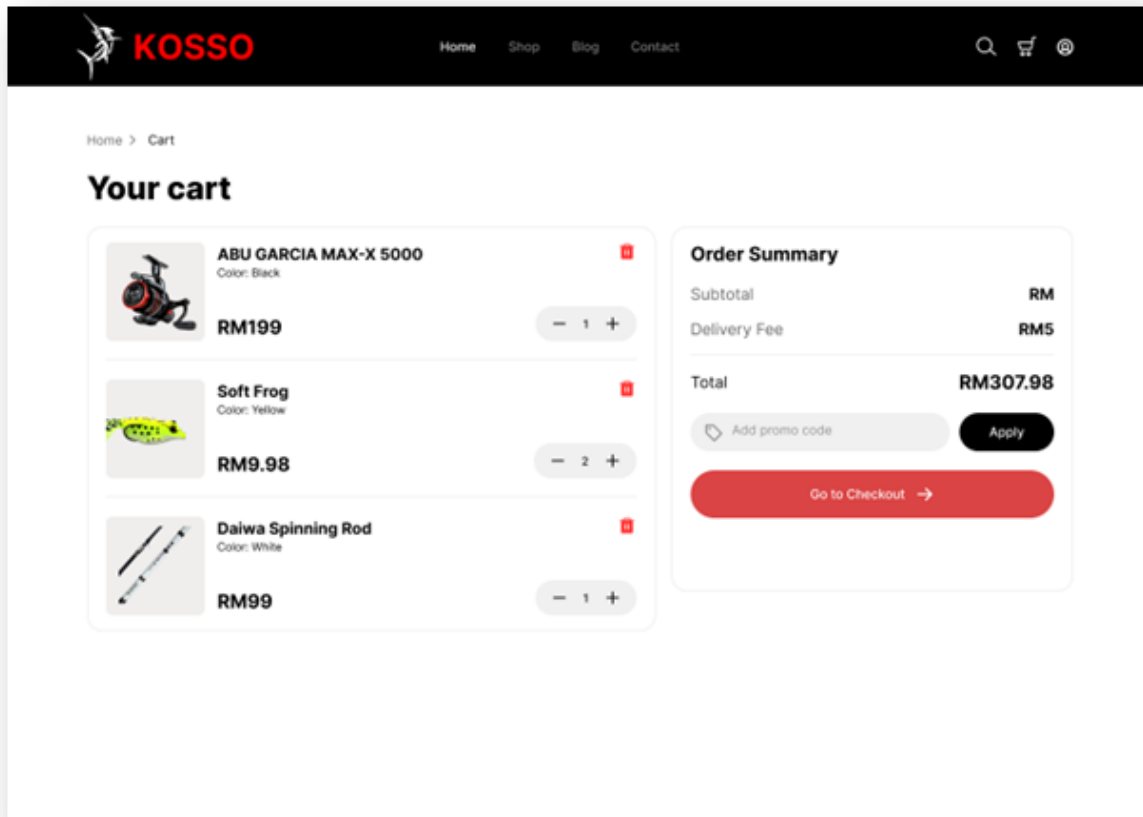
The product listing page showcases all the products available in the store. Customers can easily browse through items with images, names, and prices displayed. The page also allows users to filter and sort products by category, price, or rating, and there's a search function for finding specific items quickly.

Product Details Page



When users click on a product, they're taken to the product details page. Here, they can see high-quality images, detailed descriptions, pricing, and stock availability. There's also a quantity selector and an "Add to Cart" button to make purchasing straightforward.

Cart Page



The cart page shows users all the products they've added, complete with images, names, quantities, and prices. They can easily update quantities or remove items if needed. The page automatically calculates the subtotal, shipping costs, and the total amount due, and it includes a button to proceed to checkout.

Payment Page

The screenshot displays the 'Billing Details' section of the KOSSO website. The header features the KOSSO logo and navigation links: Home, Shop, Blog, and Contact. On the right, there are icons for search, shopping cart, and user profile. The main content area is divided into two columns. The left column contains form fields for 'First Name*', 'Company Name', 'Street Address*', 'Apartment, floor, etc. (optional)', 'Town/City*', 'Phone Number*', and 'Email Address*'. The right column shows a summary of the order: 'Subtotal: RM307.98', 'Shipping: RM5.00', and 'Total: RM312.98'. Below the summary, there are radio buttons for 'Bank' and 'Cash on delivery', with the latter being selected. To the right of these options are logos for payment methods: Alipay, VISA, Mastercard, and JCB. At the bottom of the right column, there are two buttons: 'Add Promo Code' and 'Apply Coupon', followed by a red 'Place Order' button.

KOSSO Home Shop Blog Contact

Billing Details

First Name*
[Input Field]

Company Name
[Input Field]

Street Address*
[Input Field]

Apartment, floor, etc. (optional)
[Input Field]

Town/City*
[Input Field]

Phone Number*
[Input Field]

Email Address*
[Input Field]

Subtotal: RM307.98

Shipping: RM5.00

Total: RM312.98

☐ Bank

☒ Cash on delivery

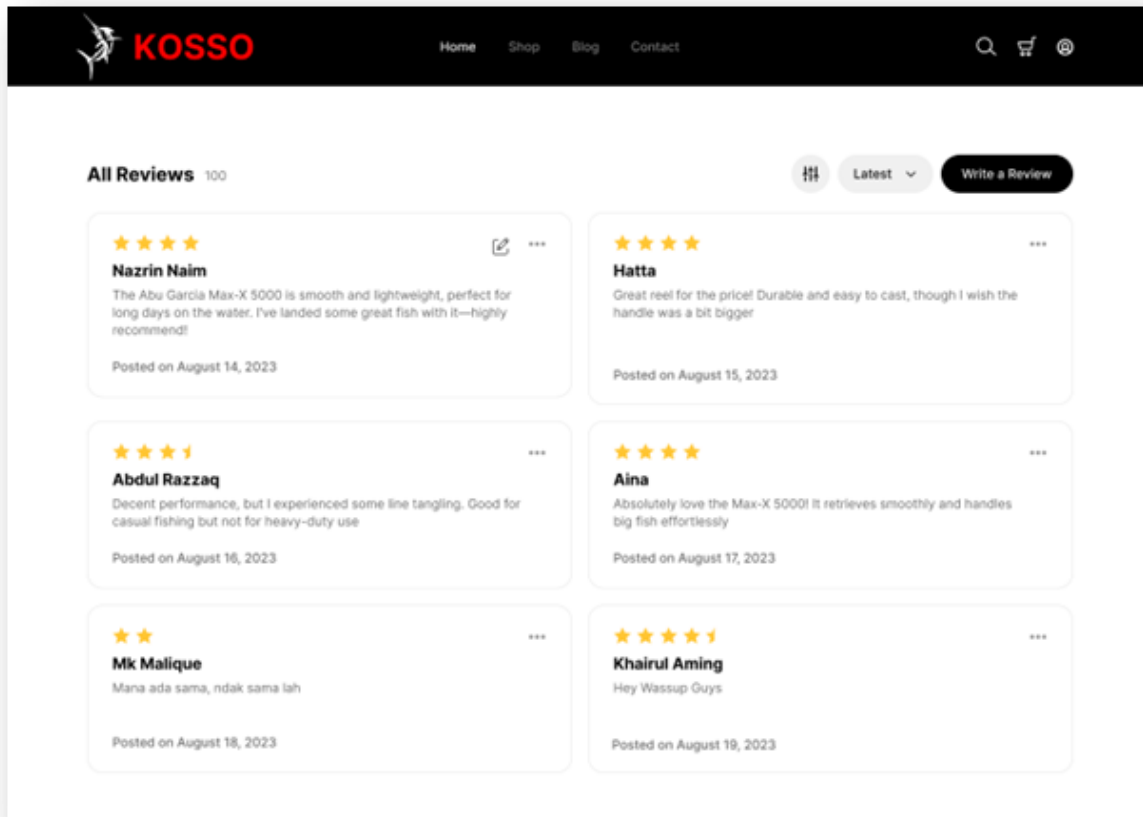
Alipay VISA Mastercard JCB

Add Promo Code Apply Coupon

Place Order

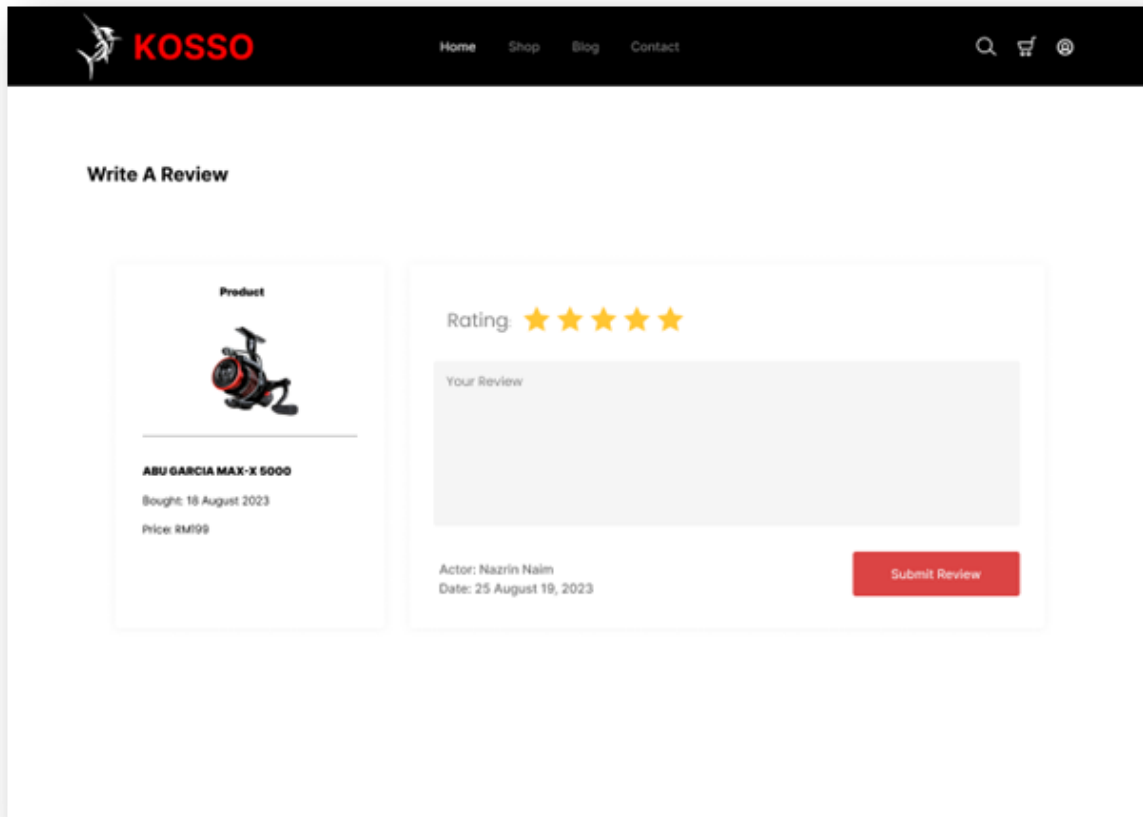
The payment page collects important billing information, payment methods, and shipping addresses. Users can see a summary of their order, including subtotals and shipping costs. If they have any promo codes, they can apply them here as well. A confirmation button lets them finalize their payment and place the order.

Review Page



The review page displays customer reviews and ratings for products. Users can read feedback from others, helping them make better purchasing decisions. This page summarizes the average rating for each product and allows users to sort reviews by date or rating for easier navigation.

Add Review Page



The screenshot shows the 'Write A Review' page for the product 'ABU GARCIA MAX-X 5000'. The page has a black header with the KOSSO logo and navigation links (Home, Shop, Blog, Contact). The main content area is white and contains a form for writing a review. The form is divided into two columns. The left column displays the product image, name, and purchase details. The right column contains a rating section with five stars, a text area for the review, and a submit button. The user's name and the date of the review are also displayed.

Write A Review

Product



ABU GARCIA MAX-X 5000
Bought: 18 August 2023
Price: RM199

Rating: ★★★★★

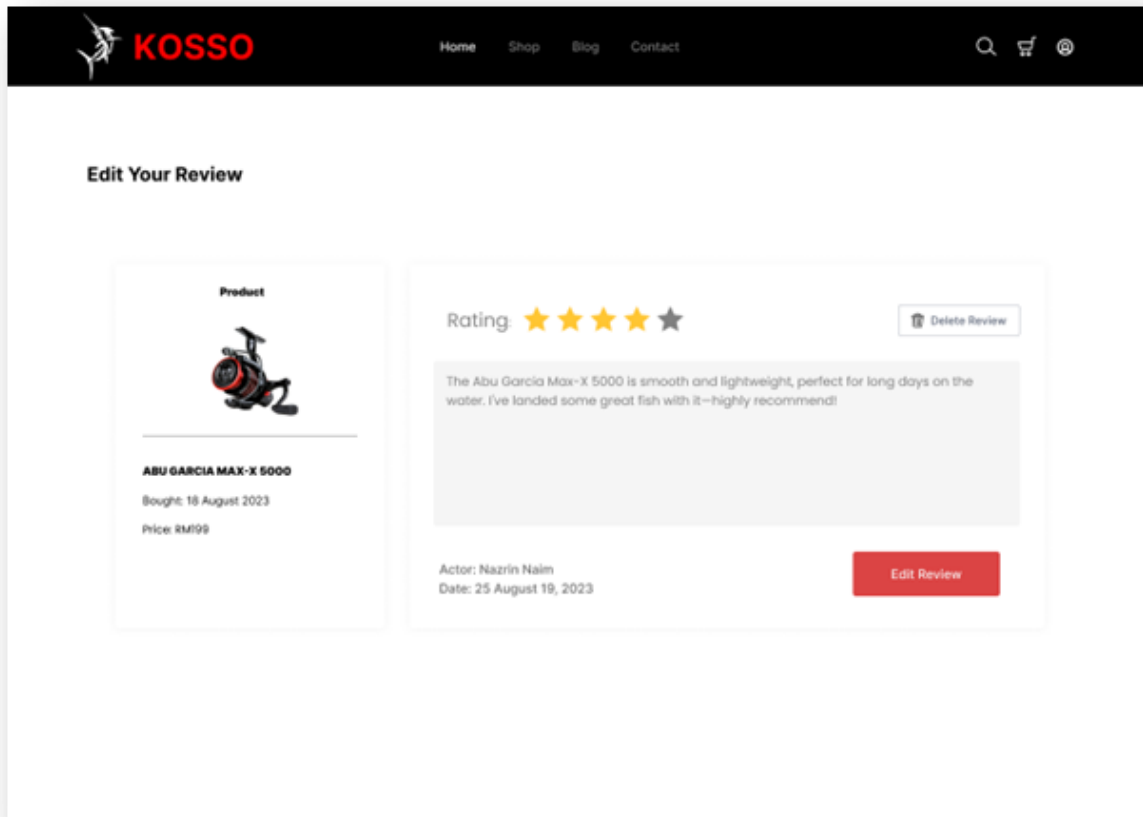
Your Review

Actor: Nazrin Naim
Date: 25 August 19, 2023

Submit Review

The add review page lets customers share their thoughts on products they've purchased. They can submit a rating and write comments about their experience. A submit button allows them to save their review, adding to the community insights about the product.

Edit Review Page



On the edit review page, users can change their previously submitted reviews. They can update their rating or adjust their comments based on further use of the product. A submit button confirms any changes they make to their review.

Stock Page

The screenshot displays the KOSSO Stock Page interface. On the left is a dark sidebar with navigation links: Dashboard, Inventory, Reports, Orders, Settings, and Log Out. The main content area features a search bar at the top. Below it is the 'Overall Inventory' section with four summary cards: Categories (6), Total Products (156, RM5200 Last 7 days Revenue), Top Selling (5, RM700 Last 7 days Cost), and Low Stocks (12 Ordered, 2 Not in stock). The 'Products' table below lists items with columns for Product, Buying Price, Selling Price, Stock Quantity, Restock Date, and Availability. The table includes 10 rows of product data. At the bottom of the table are 'Previous' and 'Next' pagination buttons, and 'Page 1 of 10' is displayed.

Products	Buying Price	Selling Price	Stock Quantity	Restock Date	Availability
Bushido Rod 5.5ft	RM65.00	RM89.99	12	11/12/22	In stock
Dawal Rod 6ft	RM45.00	RM69.99	0	21/12/22	Out of stock
Deukie AC5000	RM30.00	RM49.99	13	5/12/22	In stock
Deukie AE4000	RM35.00	RM54.99	0	8/12/22	Out of stock
Deukie AE3000	RM30.00	RM49.99	1	9/1/23	In stock
Shimano TK3000	RM80.00	RM99.99	8	9/1/23	In stock
Abu Garcia M5000	RM150.00	RM199.99	0	15/12/23	Out of stock
FRRTC EK3000	RM25.00	RM34.99	10	6/6/23	In stock
Soft frog 4.5cm	RM1.50	RM2.99	3	11/11/22	Low stock

The stock page includes several key features for managing inventory. It displays the current stock levels for all products, making it easy to see what's available at a glance. There's an "Add New Stock" button for quickly updating inventory, along with filter options to sort products by category. The page provides an overview of total products and alerts for low stock levels. Additionally, a search functionality allows for easy lookup of specific products.

Add Stock Page

The screenshot displays the 'Add Stock' page in the KOSO system. A 'New Product' modal form is open, allowing users to add new inventory items. The form is divided into two main sections: 'New Product' and 'Product Details'.

New Product Section:

- Image Upload:** A dashed box for dragging an image, with a 'Browse image' link.
- Product Name:** Text input field.
- Product ID:** Text input field.
- Category:** Dropdown menu labeled 'Select product category'.
- Buying Price:** Text input field.
- Selling Price:** Text input field.
- Quantity:** Text input field.
- Date:** Text input field with the value '24 Aug 2024'.

Product Details Section:

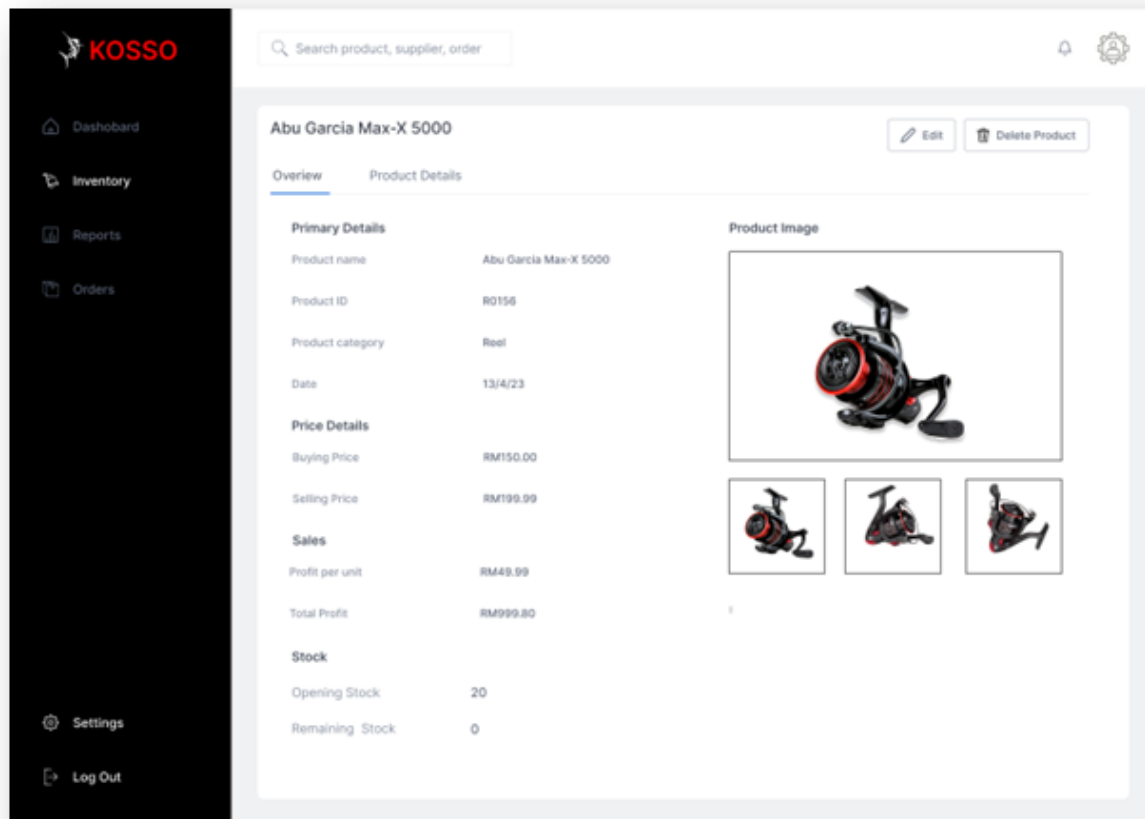
- Product Color:** Text input field.
- Product Description:** Large text area for entering the product description.

At the bottom of the modal, there are two buttons: 'Discard' and 'Add Product'.

The background shows a blurred view of the inventory management interface, including a search bar, navigation menu, and a table of existing products with columns for product name, quantity, and availability status (e.g., 'In stock', 'Out of stock', 'Low stock').

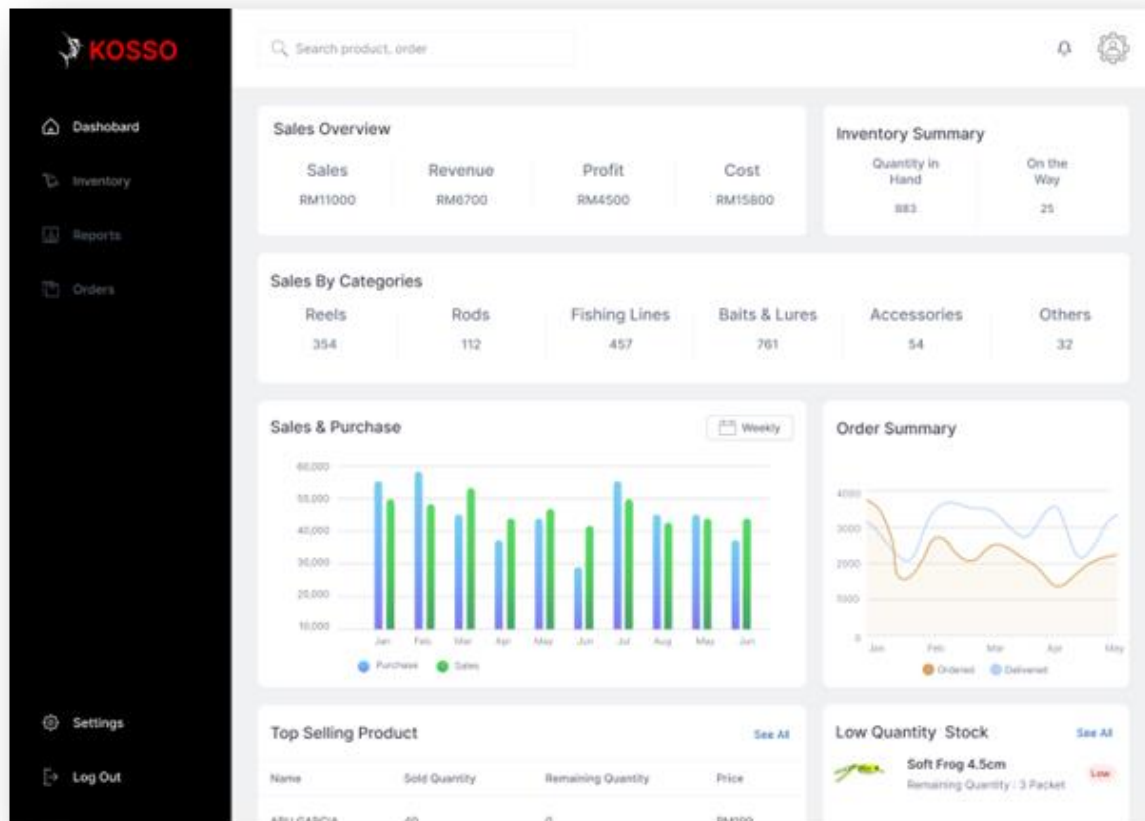
The add stock page includes fields for entering essential product information, such as the product name, quantity, and description. There's an "Add Product" button to submit the new stock information, along with a "Discard" option to cancel the add product process. Additionally, the page allows for uploading an image of the product, providing a visual representation of items in the inventory.

Stock Details Page



The stock details page provides an overview of product stock details, presenting key information at a glance. Admin can edit stock details to update quantities or other relevant information. There's a delete option available for removing items that are no longer in inventory. Additionally, the page displays an image of the product, making it easier to identify items visually.

Sales Page



The sales page is designed to provide a clear view of total sales revenue. It includes a summary of the inventory, showing current stock levels and highlighting any items that are running low. The page breaks down sales by categories, making it easy to spot which areas are doing well. Visual charts and graphs display sales trends over time, helping recognize patterns and make informed decisions. Additionally, it highlights the top-selling products, offering insights into what customers are buying the most.

1.6 Conclusions

In conclusion, an online store like KOSSO Fishing Store provides a great chance to reach more fishing fans and improve the purchasing experience by being accessible and convenient. The store's ability to expand and attract clients who value specialized goods and knowledge is now limited by the constraints of operating only in a physical location. Customers may explore, search, and buy fishing rods, accessories, and fish food at their convenience thanks to KOSSO Fishing Store's online presence. A seamless purchasing experience is also made possible by the platform's simplified features, which include customer review and safe payment methods. In the end, the e-commerce website of KOSSO Fishing Store will increase customer satisfaction, broaden the company's consumer base, and offer a specific online area for the fishing community.

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2. Software Quality Assurance Plan (SQAP)

2.1 Purpose and Scope

2.1.1 Purpose

The purpose of the Software Quality Assurance Plan (SQAP) for the KOSSO Fishing Management System (KFMS) is to provide a clear blueprint for ensuring the system is developed to meet high standards of quality. This plan aims to guide the team in delivering a reliable, efficient, and user-friendly system that aligns with business goals and customer expectations. It outlines the key quality objectives and details the processes for checking and improving the software throughout its development. The SQAP also specifies the roles and responsibilities of the team, ensuring that quality assurance activities are well-organized and tracked at each phase of the project. By following this plan, the system's performance, security and usability will be closely monitored and evaluated, resulting in a final product that fulfils the required standards and provides a positive user experience. The KFMS will include functionalities such as user registration, stock management, sales analysis, cart management, and customer feedback. SQAP will ensure these features are thoroughly tested and optimized to provide a seamless experience for both store administrator and customer.

2.1.2 Scope

This SQAP covers the entire development process for the KFMS, from initial planning through deployment and beyond. Key areas of focus include:

- **Requirements Validation:** Ensuring all business and customer requirements are fully understood and accurately documented.
- **Design Evaluation:** Reviewing the system's design to confirm it will meet performance, scalability, and security expectations.
- **Development Practices:** Implementing coding standards, conducting regular code reviews, and ensuring thorough unit testing for quality assurance.
- **Testing Process:** Carrying out extensive testing, including functionality tests, system integration tests, and user acceptance testing (UAT), to validate the system's performance.
- **Risk Management:** Identifying potential risks such as data security threats or performance issues and developing strategies to mitigate them.
- **User Engagement:** Involving end-users like store owners and customers to test the system and ensure it meets their needs and expectations.

2.2 Definition and acronym

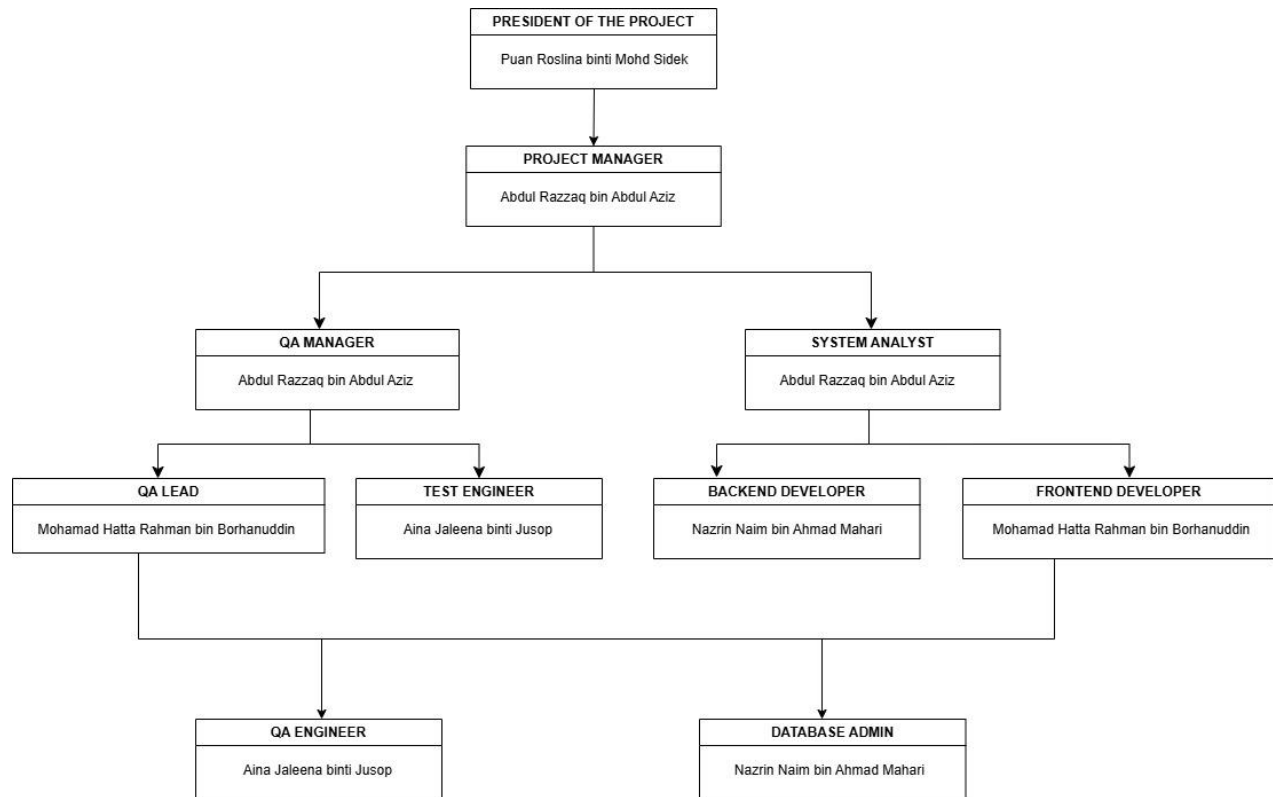
Acronym	Definition
KFS	KOSSO FISHING STORE
KFMS	KOSSO FISHING MANAGEMENT SYSTEM
SQA	Software Quality Assurance
SQAP	Software Quality Assurance Planning
SDLC	Software Development Lifecycle
SRS	Software Requirements Specification
SDD	Software Design Document
UAT	User Acceptance Testing
IEEE 730	IEEE Standard for Software Quality Assurance Plans
ISO/IEC 25010	International Standard for Software Product Quality Models

2.3 Reference document

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2.4 SQA Plan Overview

2.4.1 Organization and Independence



Role	Assigned to	Responsibilities
Project Manager	Abdul Razzaq bin Abdul Aziz	- Manage resources efficiently, allocate tasks to team members, and optimize the use of resources. - Monitor the KFMS budget, expenses and make budget adjustment for financial sustainability. - Ensure that the QA team properly tests each module, ensuring meet the requirement of KFS

		- Manage the deployment of sales analytics, ensure the data is displayed clearly to be used to make inventory and sales decisions. - Handling Module Manage User and Module Analyze Sales
QA Lead	Mohamad Hatta Rahman bin Borhanuddin	- Lead the QA team, delegate tasks, and ensure efficient test execution. - Design specific test plans for the KFMS sales and stock modules, focusing on accurate real-time updates and order processing. - Handling Module Manage Review. - Provide assistance to the QA Engineer and Test Engineer, particularly in managing possible KFMS challenges such as order updates, product availability, and customer review integration.
Backend Developer	Nazrin Naim Bin Ahmad Mahari	- Develop backend logic for inventory updates, order processing, and customer data management. - Integrate third-party payment APIs and provide safe data processing procedures for customer's sensitive information. - Handling Module Manage Stock
Frontend Developer	Mohamad Hatta Rahman bin Borhanuddin	- Develop user-friendly interfaces and implement responsive designs.

		- Ensure effective interaction between the front-end and back-end components
QA Engineer	Aina Jaleena binti Jusop	- Perform functional, regression, and integration testing to ensure that all parts of the system operate effectively. - Report and track defects, working closely with the development team to verify fixes. - Conduct testing for KFMS modules, focusing on user experience elements like smooth navigation, accurate product displays, and reliable checkout. Handling Module Manage Cart.
System Analyst	Abdul Razzaq bin Abdul Aziz	- Analyze the business requirements, document system specifications, and develop functional requirements. - Collaborate with stakeholders to gather feedback and make design improvements. - Document the specifications for each KFMS module, ensuring features meet KFS's requirements.
Database Administrator	Nazrin Naim Bin Ahmad Mahari	- Design, implement and manage database of KFS interactions. - Set up a reliable database structure to store stock data, sales records, customer information, and reviews
QA Manager	Abdul Razzaq bin Abdul Aziz	- Define quality standards and methods to ensure compliance with organizational requirements.

		- Manage the QA team's resources and timeline for testing phases. - Establish quality criteria for each KFMS module for accurate stock tracking.
Test Engineer	Aina Jaleena binti Jusop	- Conduct various tests such as functional, integration and performance testing to validate the KFMS stability and functionality. - Document test results, identify defects and track KFMS fixes through the whole process. - Test user interactions such as customer orders, payment processing, and stock updates.

2.4.2 Software product risk

2.4.2.1. Risk identification and risk management

Evaluating potential risk to the development process helps the team member to create effective tactics for minimizing or handling these risks.

Module	Potential Risk
Manage User	Data Security Risk
Analyse Sales	Data Accuracy Risk
Manage Stock	Inventory Discrepancy Risk
Manage Cart	Transactional Integrity Risk
Manage Review	Content Moderation Risk

2.4.2.2. Risk Assessment

Every risk that was detected has been evaluated based on its significance and the potential of occurrence:

1. Data Security Risk

This risk includes the possibility of sensitive user data disclosure, data breaches, or illegal access. A data security breach can result in major privacy breaches, legal ramifications, and a decline in user trust in modules that handle personal information including names, addresses, or payment details. Financial losses, brand harm, and regulatory fines can all arise from a data security breach. It is a critical danger for every system that handles private information. Sensitive user data, such as account information and perhaps personal details, are contained in the Manage User module and may be subject to data breaches or illegal access.

Risk Level: High

Reason: Data breaches or unauthorized access to user data can lead to serious privacy violations and legal implications. Protecting user data is crucial to maintaining trust and compliance with data protection regulations.

Mitigation Measures: Protect critical information with encryption, implement stringent access rules, update security patches often, and do vulnerability assessments to identify and address possible flaws.

2. Data Accuracy Risk

The potential for mistakes in data collection, processing, or reporting that might result in erroneous analysis and insights is known as data accuracy risk. Modules that compile or analyze huge datasets, such as sales or performance data, require this. Poor decision-making, resource misallocation, and a lack of confidence in the system's reporting capabilities can all result from inaccurate data. Although it might not have an immediate effect on consumers, it does have an influence on strategic planning and operational effectiveness. Accurate sales data is necessary for this module to produce insights, and mistakes in data aggregation or analysis may result in inaccurate reporting, which might influence decision-making.

Risk Level: Low

Reason: Although making informed decisions requires accurate data, errors in sales analysis data usually have no direct influence on the customer experience. Here, mistakes are less serious and may be found and fixed more easily over time.

Mitigation Measures: To guarantee data consistency, use procedures for data validation, put error-checking systems in place, and carry out routine audits. Automated data quality checks can also assist in identifying disparities early on.

3. Inventory Discrepancy Risk

This risk entails the potential for inaccurate inventory counts to result in either excess inventory or stock shortages. Errors in human data entry, a lack of real-time updates, or inadequate system synchronization can all lead to this. Inventory inconsistencies can raise holding costs for surplus goods, have a detrimental effect on customer satisfaction, and delay order fulfillment. Accurately forecasting demand and managing demand are similarly hampered by inaccurate stock levels. Order fulfillment may be impacted by inventory shortages or overstocking caused by stock management mistakes, such as inaccurate stock counts.

Risk Level: Medium

Reason: Customer satisfaction and order fulfillment may be impacted by inventory mistakes. However, this risk may be successfully managed to avoid major effects with frequent audits and stock monitoring systems.

Mitigation Measures: In order to resolve discrepancies, conduct routine physical stock audits, correlate stock levels with sales data, and implement real-time inventory management. Human error in stock updating can also be decreased by automation.

4. Transactional Integrity Risk

When a system handles transactional data incorrectly, it might lead to duplicate entries, missing products, or inaccurate orders, which is known as transactional integrity risk. Modules that manage the processing of orders and payments are particularly vulnerable to this vulnerability. Financial disparities, unhappy clients, and higher administrative expenses because of refunds or order adjustments might result from any problems with transaction integrity. Transactional accuracy is an essential component of the purchase process. To ensure that users' selections appear correctly in their cart items, this module must preserve transactional integrity. In this section mistakes might lead to orders with the wrong goods or amounts.

Risk Level: High

Reason: Inaccurate orders, unhappy clients, and possible revenue loss might result from cart system errors. Integrity is essential for a seamless customer experience because this module has a direct impact on the purchase process.

Mitigation Measures: Verify data at every stage, ensure atomicity in transactions by completing each one completely or partially, and record transaction activities for traceability. In situations with high demand, regular testing helps guarantee dependability.

5. Content Moderation Risk

The possibility that user-generated material such as reviews might be published without enough control and be offensive, damaging, or improper is known as content moderation risk. This might contain offensive language, false information, or even harmful connections that go against community standards. Depending on the type of information, unmoderated negative content might result in legal or compliance problems, ruin a brand's reputation, and erode consumer confidence. Community norms and user happiness are also affected. User-generated content is permitted by this module and may contain inappropriate language, inaccurate information, or malicious content such as dangerous links.

Risk Level: Medium

Reason: Reviews that include offensive or negative information may damage a brand's reputation. In contrast to security and transaction risks, this risk is less significant because it can usually be controlled with the use of efficient moderation techniques.

Mitigation Measures: Use content moderation technologies that identify risky content and censor offensive language. Provide reporting tools and community standards so that

users may contribute to maintaining a pleasant atmosphere. Combining automatic and manual moderation works well for more control.

2.4.2.3. Monitoring and Reporting

Every member will continue to keep an eye on the identified possible risks as the project develops. To stay up to date on the status, trends, and mitigation activities of risks, the Project Quality Assurance Team will create an annual risk report. To demonstrate the success of system mitigation actions, the report will additionally reflect any changes to the degree of risk.

2.4.2.4. Contingency Planning

In the event of the worst risk scenario, we must carefully consider risk mitigation strategies to reduce the influence of the risk on the system's development process, including its schedule, cost, and deliverables. By handling this circumstance with caution and success, it demonstrates that risk preparedness planning is successful, as it will assist to maintain order and raise the likelihood that the system will be developed effectively.

2.4.3 Tools

Tools	Description
WordPress	Adaptable content management platform, powers the KFMS's frontend, making it simple to build, organize, and update content. It keeps navigation smooth and delivers a user-friendly experience for everyone interacting with the system.
Microsoft word	The KFMS team relies on Microsoft Word to keep project documentation in check. It's handy for drafting and organizing everything from requirements to reports and meeting notes, helping to stay clear on details and track progress easily.
Github	A platform for version control with Git is used in developing the KFMS to store source code, track changes, and collaborate with the development team.
Google Drive	For the KFMS, we keep everything on Google Drive. It's the central hub for storing, sharing, and updating files, so everyone has easy access and stays aligned.
Draw.io	The diagramming and flowcharting tool helps create different types of diagrams. For the KFMS development, we used these tools to design a database structure that organizes our system's data and to create a use case diagram that maps out user relationships within each module.
Figma	A design tool built for collaboration plays a key role in shaping the KFMS's UI. It lets the team bring interface ideas to life, fine-tuned design details, and work together in real-time, helping create a polished and intuitive user experience.
MySQL	A programming language used for managing and manipulating relational databases. We use Structured Query Language (SQL) to handle database queries, requests and validation of KFMS.

2.4.4 Standards, Practices, and Conventions

Throughout the software development lifecycle, the following guidelines will be followed to guarantee quality and consistency throughout the project:

- Coding Standards:
 - The project will adhere to WordPress Coding Standards to ensure consistency, readability, and maintainability across the codebase, including PHP, HTML, CSS, and JavaScript standards. All developers must follow these standards during development and peer reviews to align with WordPress best practices.
- Documentation Standards:
 - Project documentation will follow established standards for clarity, and consistency across all phases, from proposal to testing. Key documents will be created, reviewed regularly, and updated as needed to reflect any changes, ensuring effective project management and quality assurance.
- Testing Standards:
 - Testing processes will follow industry standards for comprehensive documentation, including test cases, test plans, and test reports. Both automated and manual testing methods will be employed to ensure the software meets functional and non-functional requirements, guaranteeing quality and reliability throughout the project.
- Version Control Practices:
 - GitHub will manage custom theme and plugin development within the WordPress project, facilitating code tracking and updates. Semantic Versioning (MAJOR.MINOR.PATCH) will be applied to custom elements, while standard WordPress versioning will apply to core WordPress and external plugins.
- Project Management Conventions:
 - The project team will follow agile methodologies, with weekly stand-up meetings and regular sprint reviews to maintain transparency and prioritize tasks effectively. Task progress will be tracked throughout development to ensure alignment with project goals.

2.4.5 Effort, resources, and schedule

The development of this project is structured into sprints, with each sprint lasting approximately two weeks. The follow are the effort distribution, key resources, and schedule for completing the project:

➤ Effort Estimation:

- Effort is estimated based on each module's complexity and team members' availability, with adjustments for learning and potential challenges. Since this is our first project, we're using similar projects as benchmarks to gauge required resources and pacing, refining estimates as the project progresses.

➤ Resources:

- Human Resources: The project team of four developers will manage all roles, including development, quality assurance, and project management, working in parallel on assigned modules to ensure efficient coverage of all tasks.
- Hardware/Software Resources: Development will be conducted using standard computers with WordPress as the primary platform. Version control for custom themes or plugins will be managed on GitHub, with collaborative assets stored in cloud storage.
- Budget Allocation: The project budget is allocated at zero, prioritizing the use of free software, cloud storage, and resources necessary to achieve project goals without incurring additional costs.

➤ Schedule:

- Sprint 1: Initial setup and all module development (5 days)
- Sprint 2: Integrate secure payment gateways, add filtering tools (5 days)
- Sprint 3: Customer review and feedback modules, order tracking integration (3 days)
- Sprint 4: User interface optimization and performance testing (2 weeks)

- Final Review and Testing: Comprehensive testing and stakeholder feedback (1 week)

This timeline will be adjusted based on progress reviews at the end of each sprint, and any necessary schedule adjustments will be communicated to all stakeholders.

2.5 Activities, Outcomes and Tasks

2.5.1 Product assurance

Manage User & Analyse Sales – Abdul Razzaq Bin Abdul Aziz (CB22056)

Product assurance activities ensure that each module meets quality standards and project objectives. The Manage User module is concerned with the security of user data, specifically confidentiality, data integrity, and access control. Next, product assurance in the Analyse Sales module focuses on the quality and consistency of sales indicators, ensuring accurate data for decision-making. These procedures ensure the integrity and dependability of important system functions.

Module	Quality attribute	Product Assurance Activity
Manage User	Confidentiality	Determine whether security needs are met.
	Integrity	Conduct document review for user data handling procedures
	Access Control	Perform software validation testing for user access levels
Analyse Sales	Consistency	Confirm module alignment with expected sales metrics
	Reliability	Perform validation testing to confirm the accuracy of sales reports

Table 2.5.1: Product Assurance for KFMS Modules

2.5.1.1 Evaluate Plans for Conformance

Evaluating plans for conformance ensures that the strategies and processes described for each module meet quality standards and project specifications. The Manage User module evaluates the efficiency of security mechanisms and user data management processes to ensure confidentiality, integrity, and access control. The Analyse Sales module evaluation looks at how data analysis processes and reporting structures correspond with defined sales metrics to provide consistency and dependability in data representation. These evaluations serve to detect any gaps in the planning process, allowing for prompt remedial actions to improve overall project quality.

Module	Evaluation Activity	Objective
Manage User	Review security measures in the user data management plan	To ensure plans protect sensitive user data.
	Assess processes for handling user information	To confirm the plans guarantee proper handling of user data.
	Verify access control in user access plans	To ensure plans provide clear guidelines for authorized access.
Analyse Sales	Assess methodologies for data analysis and reporting	To ensure plans enable seamless and accurate sales reporting.
	Review data visualization standards	To confirm that plans ensure accurate representation of sales data.

Table 2.5.1.1: Evaluate Conformance Plans for KFMS Modules

2.5.1.2 Evaluate Product for Conformance

Evaluating the product for conformance involves determining if the final outputs of each module fulfill the established quality criteria and project specifications. For example, the Manage User module must be evaluated whether it has implemented the security elements correctly so that the confidentiality, integrity, and access control of the user's data are ensured. Analyse Sales module evaluates if the sales data and reports produced by the system are correct and dependable. This way, the results from both modules are of high quality and just exactly as designed and expected by the users. Thus, improve the overall integrity of the system.

Module	Evaluation Activity	Objective
Manage User	Conduct security testing on user data management	To ensure that sensitive user data is protected from unauthorized access.
	Validate user information accuracy	To verify that all user data is accurately stored and processed.
	Test access control features for all roles	To ensure only authorized users can access restricted areas of the system.
Analyse Sales	Perform data accuracy checks on sales reports	To confirm that sales reports are accurate and consistent with expected metrics.
	Review data visualization outputs	To ensure that sales data visualizations accurately reflect the underlying data.

Table 2.5.1.2: Evaluation Conformance Product for KFMS modules

2.5.1.3 Evaluate Plans for Acceptability

Evaluating plans for acceptability ensures that proposed modules fulfill set quality benchmarks and are consistent with user needs and expectations. The Manage User module's actions are concerned with ensuring that security, data integrity, and access control measures are in place and functioning properly. the analytic tools output is correct, consistent, and reliable data congruent with the requirement. This evaluation is critical to ensure that all product features are not only built but also acceptable to stakeholders before final deployment.

Module	Evaluation Activity	Expected Outcome
Manage User	Perform SQL injection tests on the database to expose security vulnerabilities.	User data remains intact, preventing unauthorized manipulation.
	Use equivalence partitioning to validate data entry processes for valid and invalid inputs.	The system does not accept invalid data, ensuring stability and no crashes occur.
	Implement state transition testing to verify role-based access control.	Each user is only able to view the pages associated with their roles.
Analyse Sales	Evaluate the consistency of data presentation methods through black box testing.	Sales data is shown consistently across all reporting interfaces.
	Review the accuracy of sales reporting by comparing actual results against expected metrics.	Sales reports accurately reflect performance metrics, ensuring reliability in decision-making.

Table 2.5.1.3: Evaluate Acceptability Plans for KFMS modules

2.5.1.4 Evaluate Product Life Cycle Support for Conformance

Evaluating product life cycle support for conformance involves looking through the Manage User and Analyse Sales modules to ensure they are consistent with recognised life cycle management methods. This evaluation focuses on the efficacy of processes across the whole product life cycle, from development and deployment to maintenance and final decommissioning. By doing this evaluation, we can verify that each module is long-term, maintainable, and meets user requirements and quality standards.

Module	Evaluation Activity	Goal
Manage User	Assess documentation practices for user data management throughout its life cycle.	Comprehensive documentation exists, supporting maintenance and user training.
	Review change management processes to track modifications to user data handling.	Changes are documented and evaluated for impact, ensuring conformance with user requirements.
	Perform regular maintenance checks to identify and rectify potential security vulnerabilities.	The system remains secure and responsive to evolving user needs.
Analyse Sales	Evaluate data validation processes to ensure accuracy in sales reporting over time.	Sales data is accurately processed and presented, maintaining its integrity throughout the life cycle.
	Assess the usability of sales analysis tools for adapting to changing business needs.	Sales reporting tools remain effective and user-friendly as organizational requirements evolve.

Table 2.5.1.4: Evaluate Product Life Cycle Support for KFMS System

2.5.1.5 Measure Products

Product measurement guarantees that the Manage User and Analyse Sales modules fulfill predefined performance and quality requirements. This method includes quantitative assessments that focus on critical criteria including reaction speed, data accuracy, and resource efficiency. By measuring these factors, we can verify that both module provides dependable and efficient functioning that meets the entire system's usability and performance criteria.

Module	Measurement Activity	Expected Outcome
Manage User	Measure the response time of login, logout, and edit profile.	Users may access it without unnecessary delays.
	Monitor data integrity during update and deletion operations	Data remains accurate and uncorrupted through all changes.
	Monitor unauthorized access attempts	Confirm access control restricts unauthorized users reliably.
Analyse Sales	Verify the accuracy of sales report calculations	Data processing provides correct and reliable outputs.
	Assess the consistency of data presentation in reports	Guarantee uniformity and clarity in sales report displays.

Table 2.5.1.5: Measure Products for KMNS system

Manage Cart – Aina Jaleena binti Jusop (CB21089)

2.5.1.1 Evaluate plans for conformance

Description: This procedure ensures that all scheduled activities for the cart module follows to project and quality guidelines. This involves examining design documentation, development timelines, testing plans to ensure they meet the required requirements and follow best practices for KFMS.

Tasks	Outcomes	Explanation
Verify compliance with functional requirements.	The functional requirements are well specified and met.	Examine the cart module's design to ensure that it satisfies important requirements such as item addition or removal, amount modification, and cart display.
Assess design for security standards.	Design contains the required security standards.	Ensures that security measures such as data encryption and user session management, are included in the cart module's architecture to protect client data and prevent unauthorised access.
Confirm User Experience (UX) conformity.	Plans meets to UX standard for usability and accessibility.	Verifies that the cart interface and user processes are built for simple navigation and responsiveness resulting in a consistent experience across several devices and user requirements.

2.5.1.2 Evaluate product for conformance

Description: This task involves evaluating the cart module to ensure that it works as planned and satisfies the specifications. This includes testing the module's functionality for example, adding items, changing amounts, and computing totals as well as assuring usability, security, and performance.

Tasks	Outcomes	Explanation
Validate Functionality against Specifications.	Product performs all specified cart functions.	Confirms that the cart module allows users to add, delete, and update products, as specified in the initial plan of KFS requirements.
Assess User Interface (UI) Consistency	UI aligns to design principles and provides usability.	Ensures that the cart module's interface is match with the approved design, including visual elements, navigation, and responsiveness across several devices.

2.5.1.3 Evaluate plans for acceptability

Description: To evaluate plans for acceptability, analyse the cart module's design and development plans to verify they meet user needs, stakeholder expectations, and industry standards. This involves determining if the planned features such as real-time updates, security standards, and responsiveness are appropriate for users and meet KFS objectives.

Tasks	Outcomes	Explanation
Review user needs and requirements.	Plans meet user expectations and requirements.	Confirms that the cart module plan has features like quick item addition or removal, clear pricing display, and simple checkout, making it user-friendly and practical for end users.
Validate compatibility with existing systems.	Plans integrate smoothly with other system components.	Ensures that the cart module may interact smoothly with existing systems such as stocks, payment gateways, and user accounts, resulting in a smooth end-to-end shopping experience.
Confirm feasibility within the project constraints.	Plans are achievable within scope, timeline, and budget.	Assesses the plan's feasibility by determining if the proposed cart features can be developed within the existing resources, timetable, and budget, lowering the possibility of delays or over cost.

2.5.1.4 Evaluate product life cycle support for conformance

Description: This process evaluates the cart module's lifecycle support, ensuring that plans for maintenance, upgrades, and future scalability are in line with standards. This involves ensuring that resources and documentation are in place to support the cart's maintenance and future upgrades.

Tasks	Outcomes	Explanation
Evaluate maintenance requirements.	The product has a clear plan for updates and fixes.	Maintains a set maintenance plan and allocates resources to handle bugs, security patches, and periodic updates in order to maintain the cart module functioning and secure over time.
Verify scalability for future growth.	The product can manage increased user and data loads.	Confirms that the cart module is designed to handle expansion, such as increasing item stock or user traffic, ensuring smooth operation as the KFMS and customer base develop.
Review the documentation for long-term support.	Comprehensive documentation is supplied to available for continuity.	Checks for completeness and accessibility of all technical documentation, including setup, usage, troubleshooting, and update instructions, allowing future teams to properly maintain and support the cart module.

2.5.1.5 Measure products

Description: To examine the cart module's quality and performance, measurements include tracking critical metrics such as loading speed, error rates, and transaction success rate. Regular monitoring and testing can assist evaluate whether the cart's module meet the required performance criteria.

Tasks	Outcomes	Explanation
Measure performance metrics.	Product meets the speed and efficiency requirements.	Evaluates the cart module's loading times, response times, and processing speed to ensure a smooth and efficient user experience with minimal latency during interactions.
Evaluate error rates.	Low error rates indicate reliability and robustness.	Monitors the frequency and types of errors such as unsuccessful item adds and checkout failures to detect possible problems and ensure cart functionality.
Track user interaction metrics.	User interactions align with usability goals.	Track data as click rates, drop-off points, and successful checkouts to evaluate usability and discover areas where users may encounter difficulties and leave or abandon the process.

Techniques to measure product assurance:

1. **Load Testing** – Average Response Time

Description: Measures how long the cart takes to respond under load, this helps determine if the cart can handle multiple users without delays.

Formula: Average Response Time = $\sum \frac{\text{Response Time For Each Request}}{\text{Total Number of Requests}}$

1. **Error Lodging** – Error Rate

Description: Calculate the frequency of errors, a lower error rate indicates better reliability of the Manage Cart Module.

Formula: Error Rate = $\frac{\text{Total Number of Errors}}{\text{Total Number of Transactions}} \times 100$

2. **Usability Testing** -Task Completion Rate

Description: Measures the percentage of users that can perform actions such as adding products or checking out, this represents how easy and useful the cart interface is for users.

Formula: Task Completion Rate = $\frac{\text{Number of Successful Tasks}}{\text{Total Number of Attempts}} \times 100$

Manage Review - Mohamad Hatta Rahman Bin Borhanuddin (CB21080)

Description: This activity enables customers to provide feedback on purchased items through detailed reviews, including the option to attach images. It supports users in updating or deleting their reviews, keeping feedback accurate and relevant. This module aims to enhance customer satisfaction and help future buyers make informed choices based on real customer experiences.

2.5.1.1 Usability

- **Evaluate plans for conformance:**
 - Review the development plan to ensure the review interface is user-friendly, with a simple layout for submitting, updating, or deleting reviews.
 - Make sure the instructions for attaching images are simple and clear.
- **Evaluate product for conformance:**
 - Test usability with representative users to determine how easy it is to leave reviews, update or deleting reviews, and attaching images.
 - Gather feedback on clarity and ease of use.
- **Evaluate plans for acceptability:**
 - Discuss with stakeholders about the usability features to acquire everyone on the same page about how easy and intuitive the review process should be.
- **Evaluate product life cycle support for conformance:**
 - Review documentation, to make sure they offer detailed instructions on how to use the review options.

- **Measure Products:**

- During usability testing, monitor task completion rates for adding, updating, and deleting reviews to make sure they are user-friendly.
- Gather user satisfaction ratings about the management and review submission processes' clarity.
- Determine any delays or pain spots by timing how long it takes users to finish tasks linked to maintaining and posting reviews.

- **Assess staff skill and knowledge:**

- Assess how well-versed the usability team is in testing methods like software for collection software or prototyping, as well as user-centered design ideas.
- Confirm staff skills in analyzing user feedback, identifying usability issues, and making actionable recommendations for enhancing user experience.

2.5.1.2. Reliability

- **Evaluate plans for conformance:**
 - Verify that the development plan has procedures for handling review data, such as making sure reviews are appropriately saved and displayed.
- **Evaluate product for conformance:**
 - Conduct stress testing to ensure the module handles a high volume of reviews without loss of data.
 - Examine the add, update, and delete review management functions' consistency to make sure they operate consistently in various scenarios.
- **Evaluate plans for acceptability:**
 - Discuss and confirm with stakeholders to make sure the module's reliability standards are up to par, particularly regarding data integrity and reliable execution.
- **Evaluate product life cycle support for conformance:**
 - Examine the documentation for details on how to handle common issues with the reliability of review operations, ensuring that users are aware of any restrictions or troubleshooting procedures.
- **Measure Products:**
 - Keep track of how many errors or issues have been reported in the Manage review module, especially those that have an impact on image uploads, review submission, or content presentation.
 - To ensure performance stability during periods of high usage, track the system's reaction time when users submit or load several reviews.

- Monitor the review module's reliability and uptime to make sure it continuously enables uninterrupted user interaction with reviews.
- **Assess staff skill and knowledge:**
 - Examine staff proficiency in interpreting test findings to guarantee consistency and data integrity as well as in reliability testing techniques including stress and load testing.
 - Verify the understanding of creating redundancy and error-handling procedures to preserve data integrity across different user operations (add, update, delete).

2.5.1.3. Security

- **Evaluate plans for conformance:**
 - Examine the development strategy to make sure it includes protections against potential vulnerabilities (such as XSS in reviews and picture upload validation) and secure input validation for all fields (text, images).
- **Evaluate product for conformance:**
 - Verify secure input handling by doing code reviews, making sure that reviews and image attachments are cleaned and saved safely.
 - Conduct vulnerability assessments for the process of review management.
- **Evaluate plans for acceptability:**
 - Discuss security measures with stakeholders to make sure they adhere to the necessary standards, especially regarding reviewing text and picture uploads.
- **Evaluate product life cycle support for conformance:**
 - Check the documentation for clear instructions on how users may submit reviews safely (e.g., picture size restrictions, content rules).
- **Measure Products:**
 - Keep track of how many security flaws in file uploads, content moderation, and user input validation were found during testing.
 - Monitor attempts to the upload offensive or dangerous information, such as illegal language or malicious links.
 - Calculate how frequently efforts are made to access review management capabilities without authorization (e.g., updating or deleting reviews without authority).

- **Assess staff skill and knowledge:**
 - Assess the team's expertise in secure coding procedures, especially in regard to user-generated content-related secure input validation, hashing, and sanitization methods.
 - Verify staff proficiency with tools and security vulnerability assessments to make sure they can successfully detect and reduce security threats.

Manage Stock - Nazrin Naim bin Ahmad Mahari (CB21072)

2.5.1.1 Product Assurance for Manage Stock

The Manage Stock module of the KOSSO Fishing Management System (KFMS) will go through detailed quality assurance steps to ensure it meets essential standards for security, reliability, and usability. This process focuses on delivering accurate, up-to-date stock information, making it easy for store staff to manage inventory, and ensuring the module works dependably for smooth operations.

2.5.1.2 *Evaluate Plans for Conformance*

This step reviews the design and requirements of the Manage Stock module to ensure they meet the necessary standards for security, reliability, and usability.

Quality Attribute	Application in Plans for Conformance
Security	Ensure the design includes secure access controls and user authentication to prevent unauthorized access to inventory data. The design should also handle data securely to protect sensitive information.
Reliability	Verify that the design includes backup processes and validation checks to ensure data remains accurate and is protected from errors. This helps maintain the integrity of inventory records.
Usability	Confirm that the interface is designed to be user-friendly, with simple workflows, clear labelling, and navigation that makes it easy for staff to add, update, and manage stock.

2.5.1.3 Evaluate Product for Conformance

In this stage, we test the Manage Stock module to make sure it meets the expected levels of security, reliability, and usability outlined in the design.

Quality Attribute	Application in Product for Conformance
Security	Run security tests, such as vulnerability scans and penetration testing, to identify and fix potential risks. Verify that access control work effectively and only authorized users can make changes to stock data.
Reliability	Perform functional and boundary testing to ensure stock levels update accurately. Test how the system handles invalid entries and simultaneous transactions to confirm stability and data accuracy.
Usability	Conduct usability testing to evaluate how easily users can complete tasks like updating stock. Gather feedback on navigation, labels, and ease of use to ensure the module is intuitive.

2.5.1.4 Evaluate Product for Acceptability

This step assesses whether the module meets user expectations for security, reliability, and usability.

Quality Attribute	Application in Product for Acceptability
Security	Make sure only authorized users can access and modify stock information. Check that sessions timeouts and logout functions work correctly to keep data secure.
Reliability	Confirm that stock levels are consistently accurate, even during high transaction loads. Ensure the system effectively handles any errors that might occur.
Usability	Conduct testing with end-users to see if they can easily navigate the module and use its features with minimal guidance. The goal is to confirm that the module is straightforward and efficient to use.

2.5.1.5 Evaluate Product Life Cycle Support for Conformance

This activity assesses whether adequate support processes are in place to maintain the quality of the Manage Stock module throughout its lifecycle. It ensures that the module remains secure, reliable, and usable even as updates and changes are implemented over time.

Quality Attribute	Application in Life Cycle Support
Security	Confirm that ongoing maintenance includes regular security updates, access control reviews, and vulnerability assessments to protect against new threats.
Reliability	Ensure that configuration management and backup processes are consistently applied to prevent data loss and maintain accurate stock information.
Usability	Verify that user feedback is incorporated into updates and provide ongoing support resources or training as needed to keep the module easy to use over time.

2.5.1.6 Measure Products

Define and monitor key metrics to ensure the module maintains standards for security, reliability, and usability throughout its lifecycle.

Quality Attribute	Measurement Metrics
Security	Track any unauthorized access attempts, monitor secure session handling, and verify logout and timeout effectiveness.
Reliability	Measure stock accuracy, uptime, and error rates to confirm consistent and reliable performance.
Usability	Collect data on task completion times and user satisfaction to assess ease of use and overall user experience.

2.5.2 Process assurance

Manage User & Analyse Sales - Abdul Razzaq bin Abdul Aziz (CB22056)

Process assurance activities guarantee that all development and operational procedures for the modules adhere to stated standards and best practices. Evaluating the life cycle, surroundings, subcontractor procedures, and performance measures contributes to high-quality outputs while ensuring that employees have the requisite skills and expertise. These assurance methods help to ensure that the Manage User and Analyse Sales modules meet their quality objectives and continue to improve.

2.5.2.1 Evaluate Life Cycle Processes for Conformance

Evaluating life cycle procedures in both the Manage User and Analyse Sales modules involves assessing implementation to the software development life cycle (SDLC). This includes ensuring that each stage, from requirements collecting to deployment, follows the project plan's specifications. This evaluation confirms that procedures are structured, regulated, and aligned with project timeframes, resulting in consistent and high-quality outcomes.

2.5.2.2 Evaluate Environments for Conformance

Evaluating environments focuses on exact configurations for both testing and production environments. The Manage User module provides assessments of security measures, data integrity, and role-based access controls to protect sensitive information and appropriately maintain user permissions. Then, the Analyse Sales module requires checks on data storage capacity and processing efficiency. These evaluations indicate that both settings enable secure, dependable, and effective operations across both modules.

2.5.2.3 Evaluate Subcontractor Processes for Conformance

In a project with a small internal team and a tight budget, the focus shifts from subcontractors to evaluating external tools and plugins. For both modules, this involves assessing

WordPress plugins and any other tools used to ensure they meet project quality standards. Within the Manage User and Analyse Sales modules, verifying how secure, compatible, and performant the plugins is necessary to be considered. This evaluation ensures that third-party components support the project's objectives while minimising security and functionality issues.

2.5.2.4 Measure Processes

Measuring processes is critical for determining and tracking efficiency and performance in module-specific tasks. The Manage User module includes monitoring the effectiveness of user access operations, ensuring that users are authenticated promptly and accurately. The Analyse Sales module requires data processing times to be monitored when generating reports, which ensures that reporting is completed on time and meets the performance criteria. These measures provide information on process reliability and prospective areas for improvement.

2.5.2.5 Assess Staff Skills and Knowledge

Finally, as the developer responsible for both modules, I must undertake a self-assessment of my project-related abilities and knowledge. This evaluation focuses on my past project handling secure data in the Manage User module and reporting accurate data in the Analyse Sales module. By identifying my strengths and areas for improvement, I can ensure that I have the required skills to create and manage high-quality modules. This self-assessment not only confirms my capacity to accomplish project objectives but also improves the overall quality of the work as a whole.

Manage Cart – Aina Jaleena binti Jusop (CB21089)

2.5.2.1 Evaluate life cycle processes for conformance

Description: This activity evaluates each aspect of the cart module's lifecycle, from design and development to testing, deployment, and maintenance, to ensure that process is consistent with stated standards and best practices. This involves validating that each stage follows to specified rules and quality standards, ensuring that cart module is created effectively.

Tasks	Outcomes	Explanation
Evaluate Development Process Adherence	Development follows standardised procedures.	Verifies that the development process for the cart module follows the established standards and industry best practices, reducing inconsistencies and ensure quality control all the way through the project lifecycle.
Validate the procedures for quality control and testing.	Quality standards are consistently applied	Ensures that testing protocols such as unit, integration, and user acceptability testing are in place and strictly followed, reducing defects and ensuring that the cart module meets quality standards.
Review the Update and Maintenance Processes.	Processes provide reliable long-term cart updates.	Evaluates the processes for maintaining and upgrading the cart module, ensuring that the procedures allow for smooth updates, patching, and scalability as the system changes over time.

2.5.2.2 Evaluate environments for conformance

Description: Evaluating environments for conformance involves verifying that development, testing, and production environments meet the necessary requirements for the cart module. This includes checking for compatibility with required tools, sufficient security measures, and adequate performance resources such as server capacity for smooth operation.

Tasks	Outcomes	Explanation
Verify the development of cart environment setup.	Development environment meets project requirements	Ensures that the development environment such as tools, libraries, and settings is appropriately configured to meet the cart module's technical standards and compatibility requirements.
Evaluate testing environment configuration	The testing environment simulates real-world conditions.	Confirms that the testing environment accurately reflects the production environment, including server settings, network configurations, and user rights, to provide realistic and reliable test results.
Check the staging environment for deployment preparation.	The staging environment supports pre-launch evaluations.	To avoid deployment concerns, ensure that the staging environment is designed to allow for final testing and validation prior to production, including data correctness and end-to-end functionality.
Review production environment compliance.	Production environment meets security and performance standards.	Verifies that the live environment meets critical compliance criteria for security, data privacy, and performance, to protecting user data.

2.5.2.3 Evaluate subcontractor processes for conformance

Description: This procedure ensures that all subcontractors and external suppliers participating in the cart module's development or support follow the agreed-upon standards and protocols. It includes checking for adherence to deadlines, quality standards, and security policies in order to ensure consistent cart module quality.

Tasks	Outcomes	Explanation
Review subcontractor selection and qualification.	Subcontractors meet required standards.	Verifies that chosen subcontractors have the necessary credentials, certificates, and experience to successfully contribute to the cart module development while adhering to KFMS requirements.
Assess subcontractors' quality control processes.	Consistent quality in deliverables from subcontractors.	Confirms that they have implemented quality control methods for the cart module, such as testing, defect tracking, and review techniques to fulfil KFMS quality requirements.
Monitor project timelines.	Subcontractor deliverables are timely and coordinated.	Ensures that subcontractors follow to agreed-upon dates reducing delays in module development and integration while maintaining an efficient KFMS timetable.
Verify compliance with security and privacy standards.	Subcontractors follow to security and data protection standards.	Ensures they follow data protection rules and secure coding methods, protecting user information throughout the development of KFMS process.

2.5.2.4 Measure processes

Description: Process efficacy is measured by assessing important parts of the cart module's development and testing procedures. Metrics like task completion time, defect rates, and code review efficiency are used to evaluate and optimise these processes, ensuring that they contribute to cart module quality and timeline.

Tasks	Outcomes	Explanation
Track the process cycle time.	The processes are efficient and finished on schedule.	Measures the time required for each process stage such as development, testing in the cart module, detecting any delays in order to increase efficiency and KFMS timeline compliance.
Evaluate defect removal efficiency	High defect detection before production.	Measures the effectiveness of testing methods by determining how many defects are found and fixed before release, ensuring high quality at launch.
Analyse rework rate.	Minimal rework required in process phases.	Calculates the percentage of operations that require rework, showing process efficiency and identifying steps that might be improved for a smoother workflow.

Techniques to measure process assurance:

1. Process Cycle Time Measurements

Description: This estimates the time required to complete each process stage, which helps in the identification of bottlenecks.

Formula: Process Cycle Time = End Time - Start Time

2. Defect Removal Efficiency (DRE)

Description: This metric measures how effectively testing processes catch defects before production.

Formula:
$$DRE = \frac{\text{Defects Found}}{\text{Total Defects (Found in Testing + Found in Production)}} \times 100$$

2.5.2.5 Assess staff skill and knowledge

Description: Assessing staff competence and knowledge include determining if team members working on the cart module has the appropriate skills and training to carry out jobs successfully. This comprises talent tests, certificates, and training sessions to improve expertise in key areas such as UI/UX design and e-commerce functionality.

Tasks	Outcomes	Explanation
Evaluate technical skills for cart module.	Staff have the technical competencies required	Ensures team members have the necessary technical capabilities such as front-end and back-end development, database administration to successfully manage cart module requirement and follow best practices.
Conduct knowledge assessments for process standards.	Staff comprehend the appropriate process standards and regulations.	Confirms that team members understand process assurance standards, security regulations, and quality expectations, allowing them to correctly follow and implement them in their positions.
Identify training needs for skills gaps.	Training requirements are assessed and planned.	Assesses areas where team members may require further training such as new technology, tools, or processes contribute fully to the cart module's success.

Manage Review - Mohamad Hatta Rahman bin Borhanuddin (CB21080)

2.5.2.1. Evaluate Life Cycle Processes for Conformance

- Review the Software Development Life Cycle (SDLC) for alignment with industry best practices, focusing on key areas for the Manage Review module:
 - **Content Moderation and Security:** To avoid vulnerabilities in image uploads, review moderation, and review submission, ensure that secure coding principles are used.
 - **User-Friendly Design:** Make advantage of user-centered design concepts to create a simple, user-friendly review interface which makes it simple for users to add, update, or delete reviews.
 - **Comprehensive Testing:** To find and fix possible problems, such as inappropriate content filtering, system load management, and the review process's ease, include moderation, performance, and usability testing into the SDLC.
- Opportunities to Integrate Quality Considerations Throughout SDLC Phases:
 - **Design:** Include a simple user interface for the review part along with content moderation features.
 - **Development:** Provide a user-friendly review form with secure input validation for reviews and picture attachments.
 - **Testing:** To guarantee seamless operation under high demand, carry out performance testing, usability testing (for simplicity of use) and security testing (for review and image validation).
 - **Deployment:** Create safe deployment procedures that include continuous system performance monitoring and content moderation.

2.5.2.2. Evaluate Environments for Conformance

- Assess Development and Testing Environments to confirm they support secure development, effective usability testing, and reliable performance:
 - **Security Testing:** Confirm that secure file storage setups, coding tools, and libraries are available for reviewing material and attachments.
 - **Usability Testing:** Verify that platforms that gather feedback from users on the review interface and screen recording software are accessible for usability testing.
 - **Performance Testing:** Ensure that you have access to load testing tools that replicate numerous users engaging with the review system, especially during busy times.

2.5.2.3. Evaluate Subcontractor Processes for Conformance

- If subcontractors are involved in developing the Manage Review module, evaluate their processes to ensure alignment with project quality standards:
 - **Secure Coding Practices:** Verify that subcontractors follow secure coding guidelines, particularly when managing media uploads and user-generated material.
 - **User-Centered Design:** Ensure that subcontractors adhere to user-centered design standards to establish a clear, accessible review submission and editing procedure.
 - **Robust Testing Procedures:** Confirm that subcontractors carry out the required load, security, and usability testing in order to ensure the module satisfies functional and quality requirements.

- **Documentation:** Keep records of any inefficiencies or opportunities for process improvement in the subcontractor's operations, especially in relation to security and content control.

2.5.2.4. Measure Processes

- Implement Metrics and KPIs to assess process effectiveness, with a focus on Security, Usability, and Reliability for the Manage Review module:

➤ **Security:**

- I. Keep track on the quantity and seriousness of security flaws found during testing, especially when it comes to managing user content.
- II. Keep track of the quantity and severity of vulnerabilities discovered during penetration testing and code review, along with the amount of time needed to fix each one.
- III. Evaluate to what extent the text and image input validation procedures work to avoid possible security risks like malicious file uploads and XSS attacks.

Input Validation Success Rate:

Success Rate =

$$\frac{\text{Number of malicious inputs blocked}}{\text{Total malicious inputs attempted}} \times 100$$

Indicates the effectiveness of validation measures.

➤ **Usability:**

- I. Monitor job completion rates for actions like adding, amending, and removing reviews, as well as user satisfaction scores from usability testing.
- II. Keep an eye on the frequency and turnaround time of any situations involving problems with data integrity, such as inconsistent or lost review data.

- III. Monitor stress test scenario success rates, including the module's performance during high usage, to guarantee dependable operation for all users.

Task Completion Rate (TCR):

TCR =

$$\frac{\text{Number of successful task completions}}{\text{Total tasks attempted}} \times 100$$

Shows the percentage of users completing tasks like adding or editing reviews.

User Satisfaction Score (USS):

USS =

$$\frac{\text{Total satisfaction ratings given by users}}{\text{Total users surveyed}}$$

Calculates average user satisfaction based on surveys (e.g., 1-5 rating scale).

➤ **Reliability:**

- I. Keep updated on response times, system uptime, and the frequency of any problems with the review feature (such as media upload or content moderation issues).
- II. Keep track of and record the effectiveness of usability testing sessions, including how long it takes users to finish particular tasks such adding, updating, and deleting reviews.
- III. Track the types and frequency of issues with usability that are reported, to identify patterns that may be used to iteratively improve the interface design.

Error Rate:

Error Rate =

$$\frac{\text{Number of errors}}{\text{Total user interactions}} \times 100$$

Tracks how often users experience issues like failed reviews or slow response times.

- **Data Analysis:** To find areas for improvement and make sure the Manage Review module is dependable, safe, and easy to use, gather and examine data from these metrics.

2.5.2.5. Assess Staff Skill and Knowledge

- Evaluate Team Skills and Knowledge related to the Manage Review module, with an emphasis on:
 - **Secure Coding Practices:** Evaluate knowledge of media handling, content moderation best practices, and secure input validation methods.
 - **User Interface Design Principles:** Verify the team's proficiency in creating user-friendly interfaces, emphasizing parts that make it simple to interact with review submission and management.
 - **Testing Methodologies:** Assess proficiency in relevant testing methodologies, such as load testing (for performance under high usage), usability testing (for user satisfaction), and security testing (for vulnerabilities).
- **Training Plans:** Create training plans based on the assessment to fill up any skill shortages in user-centered design, secure coding, and reliable testing techniques so the team can produce a Manage Review module of the highest possible standard.

Manage Stock - Nazrin Naim bin Ahmad Mahari (CB21072)

Ensure that all processes in developing and maintaining the Manage Stock module are well-documented, consistently followed, and effective. This includes verifying adherence to methodologies, alignment of testing environments with production, and adequacy of team skills.

2.5.2.1 Evaluate Life Cycle Processes for Conformance

i. Conduct Reviews

Perform regular reviews and audits to ensure Agile practices, quality standards, and feedback loops are followed.

ii. Maintain Documentation

Update process documentation as needed to reflect any changes or improvements

iii. Align with Goals

Ensure development processes support project goals for security, reliability, and usability

2.5.2.2 Evaluate Environments for Conformance

i. Match Testing to Production

Ensure the testing environment closely matches production in configurations, data handling, and setups

ii. Consistency in Settings

Keep database settings, access controls, and other configurations consistent between environments.

iii. Check Tool Compatibility

Verify that testing tools and frameworks align with the production setup for accurate testing.

2.5.2.3 Evaluate Subcontractor Processes for Conformance

i. Assess Outsourced Workflows

Confirm that suppliers meet project standards, timelines, and quality requirements.

ii. Ensure Integration

Verify compatibility between outsourced components and in-house workflows to prevent disruptions.

iii. Standardize Deliverables

Ensure that supplier outputs meet the same quality standards as in-house components.

2.5.2.4 Measure Processes

i. Track Key Metrics

Monitor defect resolution time, sprint completion rate, and testing frequency for process effectiveness

ii. Analyse Responsiveness

Use defect resolution time to gauge how quickly issues are addressed

iii. Monitor Progress

Track sprint completion and testing frequency to confirm consistent quality checks.

2.5.2.5 Assess staff skill and knowledge

Action	Description
Review Team Skills	Check current skills to ensure team members meet project needs, especially in quality assurance tools and method.
Spot Skill Gaps	Identify areas where training or development is needed to strengthen capabilities
Provide Training and Support	Offer targeted training and resources to fill skills gaps and boost team effectiveness

2.6 Additional Considerations

Manage User & Analyse Sales- Abdul Razzaq bin Abdul Aziz (CB22056)

In addition to the core quality assurance activities outlined earlier, several additional factors must be considered to ensure the successful implementation of the Manage User and Analyse Sales modules. These considerations include reviewing contractual obligations, measuring quality, handling waivers and deviations, managing task repetition, assessing risks associated with Software Quality Assurance (SQA), establishing a clear communications strategy, and addressing non-conformance issues. Each of these elements plays a critical role in maintaining project integrity and achieving the desired outcomes.

2.6.1 Contract Review

The contract review involves examining all project-related agreements to ensure that the Manage User and Analyse Sales modules meet specified terms and quality standards. For example, it ensures that requirements like secure data handling in the Manage User module and accurate, real-time reporting in the Analyse Sales module are clearly outlined and agreed upon. This process also verifies that timelines, deliverables, and quality expectations are well-defined and understood by all parties, helping to prevent misunderstandings and ensure accountability throughout the project's lifecycle.

2.6.2 Quality Measurement

Quality measurement is critical for evaluating the Manage User and Analyse Sales modules. By defining clear metrics and key performance indicators (KPIs), the modules' functionality, dependability, and user satisfaction can be assessed. For instance, in the Manage User module, KPIs such as average login time, successful login rate, and data retrieval speed can indicate system efficiency and user experience. In the Analyse Sales module, metrics like report generation time, data accuracy percentage, and visualization clarity help assess the accuracy and effectiveness of sales insights. This constant measurement enables rapid adjustments and continuous improvement, allowing the team to refine each module based on reliable, quantitative data.

2.6.3 Waivers and Deviations

Waivers and deviations, such as simplifying user data encryption to a lower security level due to time constraints, might be necessary to meet immediate project deadlines. In this case, the reduced encryption level would be a temporary measure for the initial release, with plans to upgrade security in the next release cycle. Despite the waiver, quality assurance measures like applying a secure hashing algorithm and conducting access control testing ensure that sensitive data remains protected and accessible only by authorized users. Approval among stakeholders is required, verifying that this adjustment meets minimum security standards without compromising data confidentiality or project integrity.

2.6.4 Task Repetition

Task repetition addresses the need for consistency and reliability in performing quality assurance activities. In the Manage User module, it ensures that security checks, like access control validations and data accuracy tests, are consistently applied to safeguard user information at every interaction. Likewise, for the Analyse Sales module, repetitive tasks such as data accuracy checks and report consistency reviews help maintain the reliability of sales insights by verifying metrics and visualizations each time they are generated. Standardizing these repetitive quality assurance activities minimizes errors and enhances the overall integrity and effectiveness of both modules.

2.6.5 Risk to Performing SQA

Assessing risks associated with Software Quality Assurance is essential to identifying potential threats to project quality, including issues like resource availability, technical challenges, and timeline adherence. For instance, limited resources could affect testing coverage in the Manage User module, while technical complexities in the Analyse Sales module might hinder accurate data reporting. By proactively identifying such risks, effective mitigation strategies can be developed, helping to minimize their impact on quality and maintain project standards.

2.6.6 Communications Strategy

A clear communications strategy is essential for keeping all stakeholders informed and engaged throughout the project. It should specify information-sharing methods, update frequency,

and communication channels. For instance, regular status updates on the Manage User and Analyse Sales modules could be provided through team meetings, ensuring that all parties remain aligned on progress and challenges. Effective communication fosters collaboration and enables timely resolution of quality concerns, contributing to smoother project execution and higher quality outcomes.

2.6.7 Non-conformance SQA

Non-conformance in Software Quality Assurance occurs when a module, such as Manage User or Analyse Sales, does not meet the established quality standards or requirements. For example, if the Manage User module fails to properly encrypt sensitive user data, it is crucial to identify and document this non-conformity promptly to prevent data breaches. Establishing a corrective action plan allows the team to address the issue effectively and ensures that lessons learned from this non-conformance are applied to future development activities. This process ultimately enhances overall quality assurance practices and fosters continuous improvement within the project.

Manage Cart – Aina Jaleena binti Jusop (CB21089)

2.6.1 Contract review

The contract review for the cart management module ensures that all project stakeholders understand the scope, deliverables, timelines, and quality standard specified in the agreement. This involves evaluating terms like as functionality, performance goals, and compliance requirements to avoid misunderstandings. Regular reviews are conducted to ensure that both the client and the project team are sticking to the agreed-upon standards and deliverables, ensuring consistency with the project's objectives.

2.6.2 Quality measurement

Quality measurement in the cart module focusses on defining quantitative measures for assessing functionality, usability, security, and performance. Key quality measures include response time, error rate, task completion rate, and security compliance. Regular measurement using testing and performance monitoring tools helps to ensure that the module meets the stated quality standard. In addition, quality audits are performed on a regular basis to ensure compliance with best practices and suggest areas for improvement.

2.6.3 Waivers and deviations

Waivers and deviations handle any allowed exceptions to the cart module's normal processes and standards. For example, if some feature requests are outside of the project's scope, a waiver may be requested to prevent affecting timelines or costs. Deviations may also be made to specified requirements where alternative solutions are considered acceptable. All exemptions and deviations are documented, approved by appropriate stakeholders, and periodically evaluated to determine their impact on project quality and compliance.

2.6.4 Task repetition

Task repetition is repeating crucial tasks throughout the cart module's lifecycle to assure consistency and quality. Security testing, usability assessments, and code reviews may be performed at various phases of development to ensure standards compliance and early detection

of flaws. Repetition is very helpful for checking features in reaction to updates or fixes, which helps to keep a module stable and reliable.

2.6.5 Risk to performing SQA

The risks associated with Software Quality Assurance (SQA) for the cart module include potential testing delays, resource constraints, and non-compliant standards. Other risks include difficulties with security requirements or sustaining performance during peak load conditions. To address these risks, contingency plans are created, which may include assigning more personnel, introducing automated testing, or conducting regular risk assessments to adjust SQA efforts effectively.

2.6.6 Communications strategy

The cart module's communication strategy establishes clear channels of communication among team members, stakeholders, and subcontractors. Regular updates, status reports, and review meetings are scheduled to discuss progress, address issues, and make decisions. Key stakeholders are kept informed about quality milestones and risks, and team members receive timely feedback and assistance. Effective communication enables cooperation, responsibility, and quick resolution of any issues that arise.

2.6.7 Non-conformance SQA

Non-conformance SQA resolves any situations in which the cart module does not match the specified quality requirements. This involves finding, documenting, and fixing issues such as functional gaps, safety risks, and performance limitations. When non-conformance is found, corrective measures are taken, and the module is reevaluated to assure compliance. An established method for managing non-conformance ensures that the cart module consistently meets quality and performance expectations.

Manage Review - Mohamad Hatta Rahman bin Borhanuddin (CB21080)

2.6.1 Contract review

When reviewing agreements related to the Manage Review module, it's essential to carefully assess all clauses that address the handling of user-generated content, such as product reviews and images. Special attention should be given to provisions on content moderation, intellectual property rights, data privacy, and compliance with relevant laws (e.g., GDPR, CCPA). For example, to ensure user privacy and content integrity, the contract should specify safeguards for content storage and access controls, defining protocols for storing, moderating, and displaying user reviews. These measures help protect both the rights of the users and the company and ensure that the module operates with clear accountability regarding content standards and user privacy.

2.6.2 Quality measurement

To assess the Manage Review module's efficacy and performance, provide quantifiable quality requirements in the contract. Establish KPIs (Key Performance Indicators) that correspond with company goals, such as system availability for review submission, review processing times, and content filtering accuracy. Maintaining high standards in content moderation and user experience, as well as monitoring service quality and identifying improvements, is made possible by clearly defining KPIs for uptime, moderation accuracy, and user happiness. These KPIs may be examined on a regular basis to make sure the Manage Review module is fulfilling predetermined standards for quality.

2.6.3 Waivers and deviations

While contracts outline guidelines and obligations, certain circumstances may require flexibility in managing reviews. It's important to include waiver and deviation clauses that specify when exceptions to content moderation or data storage policies may be permitted, along with the steps for requesting these exceptions. For example, if an unexpected volume of user reviews impacts

moderation response time, a deviation may be necessary to temporarily adjust service levels. Defining the criteria and approval process for such deviations enables flexibility while ensuring adherence to quality and compliance standards under controlled conditions.

2.6.4 Task repetition

To optimize operations within the Manage Review module, identify and eliminate unnecessary or duplicated tasks in content moderation and review processing. For example, if multiple departments are independently responsible for review moderation or response handling, consolidating these tasks into a centralized moderation system can improve efficiency and consistency. Reducing redundant tasks allows for streamlined workflows, lowers operational costs, and enhances the accuracy of content review processes, ultimately improving the module's productivity and reliability.

2.6.5 Risk to performing SQA

One of the key risks in performing Software Quality Assurance (SQA) activities for the Manage Review module is the challenge of ensuring thorough content moderation and data validation under time constraints. This module must filter user-generated content, such as text and images, to prevent inappropriate or harmful content, which may require additional resources and specialized tools. Limited time and resources for SQA could lead to inadequate content filtering, which can impact the reliability and reputation of the platform.

To mitigate this risk, the project team could collaborate with content moderation experts and adopt automated tools for initial content filtering, supplemented by human review as needed. This approach allows the team to balance resource limitations with the need for comprehensive review moderation, ensuring user safety and satisfaction while maintaining efficiency.

2.6.6 Communications strategy

The communications strategy for the Manage Review module prioritizes transparency and consistent information flow, especially regarding security, usability, and content integrity. Frequent updates ensure that stakeholders remain informed about actions taken to:

- **Ensure security** by safeguarding user data during review submission, updating, or deletion.
- **Enhance usability** by improving the review submission interface, simplifying review editing and deletion, and providing clear instructions.
- **Maintain content quality** by filtering inappropriate content, ensuring authenticity, and preventing spam.

Security Initiatives for Stakeholders:

- Implementing secure protocols for uploading and storing review images and text.
- Ensuring that only authorized users can modify or delete reviews.
- Conducting regular security audits to detect and address vulnerabilities related to content handling.

Usability Initiatives for Stakeholders:

- Designing an intuitive, accessible interface for submitting and managing reviews.
- Providing straightforward instructions and responsive feedback to guide users.
- Implementing an easy-to-use reporting feature for flagging inappropriate reviews.

Content Integrity Initiatives for Stakeholders:

- Regular content moderation practices to maintain review authenticity.
- Implementing anti-spam measures to reduce misleading or irrelevant reviews.
- Using machine learning or automated tools to assist in identifying inappropriate or harmful content.

2.6.7 Non-conformance SQA

The non-conformance process for the Manage Review module ensures that any deviation from quality standards in content quality, usability, or security is identified and addressed promptly.

1. Identification of Non-Conformance

- **Risk Assessments:** Regular assessments help identify potential issues with content moderation, security, or usability.
- **Regular Team Meetings:** Development and testing teams are encouraged to report any quality deviations related to review handling, moderation, or interface usability.

2. Specific Criteria for Non-Conformance

- **Reliability**
 - Instances of inappropriate or harmful content passing through moderation.
 - Low user satisfaction scores on content quality due to unfiltered spam or irrelevant reviews.
- **Usability:**
 - Confusing workflows for submitting, editing, or deleting reviews.
 - Lack of accessibility features that limit users with disabilities from engaging with review functionalities.
- **Security:**
 - Unauthorized changes to or deletions of user reviews.
 - Weak data handling or storage protocols for user-uploaded images or texts in reviews.

3. Resolution through Corrective Actions

- **Process Improvements:** Refining content moderation processes to filter inappropriate content more effectively.
- **Code Updates:** Addressing security vulnerabilities, ensuring robust access controls, and updating moderation protocols.
- **Interface Enhancements:** Simplifying review submission, editing, and deletion interfaces to improve user experience.

- **Training and Tools:** Providing team members with training in content moderation and security best practices, including usage of automated tools for content filtering.

Manage Stock - Nazrin Naim bin Ahmad Mahari (CB21072)

The following additional considerations address essential areas that support the quality assurance efforts for the Manage Stock module. These points ensure thorough planning, communication, and risk management to maintain the module's quality throughout its lifecycle.

2.6.1 Contract review

i. Identify Quality Requirements

Ensure the contract specifies quality standards, performance expectations and acceptance criteria, covering attributes like security, reliability, and usability

ii. Verify Scope and Deliverables

Confirm that the contract clearly outlines the scope of work and expected quality-related deliverables to avoid scope misunderstanding.

iii. Assess Timelines for Quality Activities

Check that timelines allow for thorough testing and quality verification, with realistic milestones for key quality activities.

iv. Clarify Roles and Responsibilities

Ensure the contract details responsibilities for quality tasks, including testing, reviews, and approvals, for both parties involved.

2.6.2 Quality measurement

Quality Attribute	Metric	Measurement
Security	Incident Rate	Track the number and severity of security incidents or unauthorized access attempts. Analyze log files and monitor for irregular activity to ensure data protection
	Access Control Effectiveness	Test authentication and authorization functions to confirm that only authorized users can access sensitive areas. Conduct periodic security audits and vulnerability scans.
Reliability	Defect Rate	Calculate the percentage of defects found during testing and post-deployment. Log issues identified in each testing cycle to assess stability and performance consistency.
	Uptime/Downtime	Monitor system uptime to ensure minimal downtime. Use monitoring tools to track availability over time, aiming for high reliability and reduced service disruptions.
Usability	User Satisfaction Score	Conduct surveys and user testing to gather feedback on ease of use, interface clarity, and overall satisfaction. Analyze results to identify areas for improving user experience.
	Task Completion Time	Measure the average time users take to complete core tasks. Track during usability testing to ensure tasks are completed efficiently without causing frustration.

2.6.3 Waivers and deviations

i. Document Approved Exceptions

Record all approved exceptions to quality standards that result from unique project constraints to ensure traceability.

ii. Review and approve with Stakeholder

Conduct a formal review and approval process with relevant stakeholders to maintain transparency and alignment on each deviation.

iii. Maintain Waiver Records

Keep a detailed log of all waivers to track and justify deviations from standard practices, supporting accountability and future reference

2.6.4 Task repetition

i. Regular Code Reviews

Frequency: After each development sprint or major update.

Purpose: To identify code issues early, ensure adherence to coding standards, and maintain code quality. Regular code reviews help prevent errors from accumulating, making later testing more effective.

ii. Usability Checks After Updates

Frequency: Following any updates that impact user interface or workflow.

Purpose: To ensure the module remains user-friendly and aligned with user expectations, especially after modifications. Repeating usability checks helps confirm that changes do not negatively affect user experience.

iii. Documentation Reviews

Frequency: At the end of each project phase or after significant changes to requirements or functionalities.

Purpose: To maintain accurate, up-to-date documentation that reflects the current state of the module. Repeating documentation reviews ensures that any changes are recorded and that team members have access to reliable information

2.6.5 Risk to performing SQA

i. Changing Requirements from Stakeholders

Stakeholders may request changes to the Manage Stock module's features or functionality as development progresses. Frequent changes could require significant rework of test cases and additional rounds of testing, which may impact project timelines and increase costs.

ii. Time Constrains Due to Project Deadlines

Tight deadlines for delivering the KFMS project could restrict the time allocated for quality assurance of the Manage Stock module. Reduced time may result in skipped or rushed testing, increasing the risk of undetected bugs or usability issues.

iii. Technical Challenges with Module Integration

Integrating the Manage Stock module with other parts of the KFMS might introduce unforeseen technical challenges. These issues could require extra time and expertise to resolve, potentially slowing down the SQA process and delaying overall project progress.

2.6.6 Communications strategy

Activity	Content
Regular Status Updates	Weekly updates through written reports or emails to summarize completed SQA activities, current quality status, key metrics, and any issues encountered. This ensures the development team, SQA team, and stakeholders stay informed of ongoing quality activities.
Monthly Progress Meetings	Monthly virtual or in-person meetings to review SQA progress, recent test results, defect resolution, upcoming SQA tasks, and any challenges faced.
Quality Review Sessions	Held at key project milestones, these sessions involve a detailed review of SQA results, testing summaries, and defect status.
Final Project Quality Report	Summarizing all SQA activities, final quality metrics, defect resolutions, and an overall quality assessment.

2.6.7 Non-conformance SQA

i. Identify and Document Non-Conformance

- a. Identify instances where the module fails to meet established quality standards or specific project requirements.
- b. Document each instance of non-conformance, including details about this issue, affected components, and potential impact on the module's functionality or usability

ii. Analyse Root Cause

- a. Conduct a root cause analysis to determine the underlying reasons for non-conformance.
- b. Identify contributing factors, such as design flaws, testing gaps, or process issues, to ensure a clear understanding of the source of the problem.

iii. Record and Review

- a. Maintain detailed records of all non-conformance instances, root cause analyses, corrective actions, and verification results.
- b. Regularly review these records to identify any patterns or recurring issues that may need further action or preventive measures.

iv. Verify Effectiveness

- a. Once corrective actions are completed, verify their effectiveness by retesting or re-evaluating the affected areas.
- b. Document the results of this verification to confirm that the non-conformance has been resolved and quality standards are met.

2.7 SQA Records (Optional)

Manage User & Analyse Sales - Abdul Razzaq bin Abdul Aziz (CB22056)

While the documentation for the project may be limited to the project proposal, Software Quality Assurance Plan (SQAP), and project design, these documents are essential for establishing a robust framework for quality assurance throughout the development of the Manage User and Analyse Sales modules. The project proposal outlines the objectives, scope, and deliverables, serving as a foundational reference for quality expectations. The SQAP details the strategies, methodologies, and performance metrics that will guide the quality assurance activities, ensuring consistency and reliability in assessing the software's functionality. Additionally, the project design document specifies the architecture and processes involved in the development, which aids in aligning quality standards with technical implementation. By effectively utilizing these key documents, the team can maintain transparency, accountability, and a clear understanding of quality requirements, facilitating a structured approach to quality assurance throughout the project lifecycle.

Manage Cart – Aina Jaleena binti Jusop (CB21089)

2.7.1 Analyse, Identify, Collect, File, Maintain and Dispose

This procedure ensures that all quality assurance activities, testing results, and necessary data for the cart module are maintained effectively, precisely, and in accordance with organisational standards throughout the module's lifecycle. This method to record management contributes to the cart module's SQA process by ensuring that quality-related information is reliably documented, securely maintained, and available as needed for compliance, improvement, and accountability.

Analyse	The cart module's focusses on discovering essential quality and performance parameters including user experience, reaction speed, error rates, and security compliance. The SQA team examines the module's objectives and usage scenarios such as adding or removing products, managing stock in the cart to decide which records must be tracked.
Identify	The cart module's records include test case documentation, performance reports, defect logs for difficulties with cart functionality such as item insertion or removal failures, and user feedback logs. Each record type facilitates quality monitoring for the cart's functioning, stability, and security, ensuring that all aspects meet defined standards.
Collect	Records are gathered continually during the cart module's development and testing phases. For example, test results are collected throughout unit, integration, and user acceptability testing, gathering information on how well the cart performs in various situations. Additionally, defect reports and security assessments are gathered to document any identified issues.
File	Records for the cart module are organised methodically in a secure, single project management repository. Records are labelled with IDs specific to the cart module, such as "CartTestID" or "DefectLog_CartFunctionality," to ensuring quick reference and retrieval.
Maintain	The cart module records are actively maintained, with updates to reflect the most recent test results, defect resolutions, and performance assessments. Version control ensures that earlier versions of documents are archived while the most recent updates are easily accessible for review.
Dispose	Once the cart module has been completed and deployed, records are evaluated for their retention value. Essential data, such as final performance metrics and defect resolution logs, are kept for future reference or audit reasons. Temporary records that no longer meet compliance or reference requirements are safely disposed of in accordance with data management policies.

Manage Review - Mohamad Hatta Rahman bin Borhanuddin (CB21080)

2.7.1 Analyse, Identify, Collect, File, Maintain and Dispose

1. **Analyze and Identify:** A thorough examination will be conducted to determine the critical documentation required for the Manage Review module. Essential records will include:
 - Content Moderation Guidelines: Documents outlining the criteria and protocols for filtering user reviews to ensure appropriateness and relevance.
 - Test Plans and Cases: Detailed test scenarios and results for review submission, editing, deletion, and content filtering functionalities.
 - Security Audits: Records of security assessments verifying that user data (especially image uploads and review content) is handled securely.
 - User Feedback Reports: Summaries of usability testing and user feedback to continuously improve review functionality.
2. **Collect:** Throughout the module's development lifecycle, relevant SQAP records will be systematically collected at each phase. This ensures that the documentation captures each stage of development, testing, and iteration, maintaining a consistent record of quality and compliance.
3. **File:** All collected SQAP records will be organized in a central, structured repository accessible to the project team. This central storage will act as a single source of truth for the module's quality documentation, including test cases, moderation guidelines, and feedback records. Each document will be tagged by version, phase, and type, making it easy for team members to retrieve and use.
4. **Maintain:** To uphold the relevance and accuracy of the Manage Review documentation, regular audits will be conducted. This involves updating records to reflect new findings or updates in moderation requirements, security protocols, or user experience

improvements. If discrepancies are found during maintenance reviews, immediate updates will be made to retain data integrity and usability.

5. **Dispose:** As the project progresses, a formal disposal protocol will govern the retention and deletion of SQAP records. This protocol will include criteria for identifying obsolete or redundant records, ensuring they are disposed of securely in alignment with organizational and data privacy policies. Any documents no longer necessary for the module's maintenance will be safely deleted to manage storage and information security responsibly.

Manage Stock - Nazrin Naim bin Ahmad Mahari (CB21072)

2.7.1 Analyse, Identify, Collect, File, Maintain, and Dispose

Step	Description
Analyse	Decide which quality records are essential, like test results, defect logs, and meeting notes.
Identify	Recognize all key documents and data needed to fully track quality activities.
Collect	Continuously gather records during testing, reviews, and quality tasks to capture a complete history.
File	Store records in a central, secure location, clearly labeled for easy access.
Maintain	Regularly review and update records to reflect the latest project and quality actions.
Dispose	Remove records that are no longer needed per retention policies, ensuring secure disposal.

2.7.2 Availability of Records

Aspect	Description
Centralized Storage	Keep all SQA records in a secure, central repository that authorized team members can access.
Access Control	Limit access to SQA records to authorized personnel to protect data.

3. Software Quality Assurance Design (SQAD)

3.1 Abstract

The Kosso Fishing Management System (KFMS) is an online platform that aims to improve the operations of the Kosso Fishing Store (KFS) by addressing the inefficiencies and constraints of its present manual management processes. KFMS aims to facilitate and improve the operations of KFS by simplifying user account management, facilitating the shopping and checkout process, improving stock tracking, incorporating customer feedback and providing accurate sales tracking and analysis. Manual approaches of managing Kosso Fishing Store business are prone to human error, delays, and inefficiencies, particularly in terms of stock levels, order processing, and customer data management. Replacing these manual procedures with KFMS will increase operational efficiency, improve the customer experience, and enable better decision-making for the store's owner and staff.

The focus of this project will be on implementing software quality attributes within KFMS's five primary functions which are user management, shopping process optimization, stock tracking, customer feedback integration, and sales analysis. The expected outcome is an effective and user-friendly system that enhances these functions, ultimately supporting the sustainable growth of KFS.

3.2 Introduction

Quality design in software development involves creating a system that performs as intended, efficiently, and accurately. This means carefully planning and structuring each part of the system to ensure it operates seamlessly, meets user requirements, and functions consistently under various conditions. For the Kosso Fishing Management System, quality design is essential to building a solution that can handle the specific needs of the fishing industry, from data processing to user interaction, ensuring each component works in harmony to deliver a smooth experience for all users.

The Manage User and Analyse Sales modules require careful attention during design to meet their unique functional demands. For Manage User, the focus will be on implementing structured workflows that allow for smooth user account management and maintain secure and accurate user data. In the Analyse Sales module, the design will centre on accurate data tracking and analysis to provide clear insights for business decisions. Ensuring these modules are designed with precision will contribute to an effective and user-friendly Kosso Fishing Management System.

In the Manage Stock module, quality design is focused on accurately tracking inventory levels and ensuring that stock data remains current and reliable. This module requires the system to handle inventory updates efficiently, provide clear visibility into stock status, and support smooth stock adjustments as sales and returns occur. A well-designed Manage Stock module will help prevent stockouts and overstocking, thus supporting smooth operations for the Kosso Fishing Management System.

For the Manage Cart module, quality design is centred on creating a user-friendly and responsive shopping experience. This module needs to allow users to easily add, modify, or remove items from their cart, calculate totals accurately, and transition smoothly to checkout. Ensuring that the Manage Cart module is intuitive and fast enhances the customer experience, making it straightforward for users to complete transactions efficiently.

In the Manage Review module, quality design emphasizes clear, secure feedback processes where users can leave, edit, and view product reviews. This module should handle reviews in a way that ensures they are accessible, authentic, and easy to navigate. With a focus on user engagement and transparency, the Manage Review module enables customers to make informed decisions based on feedback from other buyers.

In conclusion, the quality design of the Kosso Fishing Management System across modules, including Manage User, Analyse Sales, Manage Stock, Manage Cart, and Manage Review, ensures a comprehensive and cohesive platform. Each module has been carefully designed to meet specific functional requirements and support seamless interactions, which contributes to an effective and user-friendly experience. By focusing on quality throughout the design phase, the

system is better equipped to meet both business objectives and user expectations, providing a reliable foundation for long-term success.

3.3 Background of Study

In the era of digital transformation, e-commerce has become an essential avenue for retailers seeking to expand their reach and enhance customer convenience. KOSSO Fishing Store (KFS), a dedicated supplier of fishing equipment, currently operates without an online presence, limiting its customer base to those who can access its physical location. This absence of an e-commerce platform prevents KFS from tapping into the potential of the expanding online market, where consumers increasingly seek convenient, round-the-clock access to specialized products and services.

While traditional in-store shopping offers certain benefits, it can also result in operational inefficiencies, particularly in terms of managing inventory and sales during off-peak seasons. KFS faces notable seasonal fluctuations, such as reduced stock turnover during the rainy season when fewer people go fishing. A comprehensive e-commerce system, specifically designed to serve fishing enthusiasts, could address these challenges by allowing KFS to reach customers in unaffected regions, balancing stock levels and enhancing profitability across all seasons.

To develop a solution that meets these modern retail needs, the KOSSO Fishing Management System (KFMS) will focus on user-friendly product browsing, secure transaction capabilities, and streamlined inventory management. In this design approach, usability and accessibility are paramount, enabling customers to easily search, filter, and review products before purchase. Data security protocols, including encrypted transactions and role-based access control, will safeguard both customer and store data, building customer trust in the platform.

The implementation of robust Software Quality Assurance (SQA) practices is critical in ensuring KFMS meets quality standards at every stage of its lifecycle. By incorporating thorough requirement analysis, detailed test planning, and continuous integration, the SQA plan will maintain performance, reliability, and user satisfaction. This approach, combining a quality-driven design with comprehensive quality assurance, will empower KFS to build a strong digital presence, improving operational efficiency and creating a seamless shopping experience that resonates with fishing enthusiasts locally and nationally. This project ultimately supports the evolution of retail into a more accessible, resilient, and customer-focused model.

3.4 Introduction to Quality Attributes

Manage User & Analyze Sales- Abdul Razzaq bin Abdul Aziz (CB22056)

Quality attributes are essential to ensure the Kosso Fishing Management System (KFMS) operates securely, accurately, and reliably. With a focus on the Manage User and Analyze Sales modules, the system prioritizes Confidentiality, Integrity, and Reliability to protect sensitive user data, maintain the accuracy of user and sales-related operations, and ensure consistent performance. These attributes are critical for gaining user trust, safeguarding information, and supporting smooth functionality across these modules.

Confidentiality

Confidentiality is a critical quality attribute for protecting sensitive information within the KFMS, such as user data and transaction details, from unauthorized access. By implementing strict access control measures, such as role-based access control (RBAC) and data encryption, KFMS ensures that only authorized users can access sensitive information. This attribute guarantees the privacy and security of customer data, which is fundamental to maintaining trust in the platform.

Integrity

Integrity ensures that the data within the KFMS remains accurate, consistent, and trustworthy over time. This attribute is achieved through mechanisms like data validation, error detection systems, and audit trails that monitor changes to sensitive data. By enforcing integrity, the system prevents unauthorized modifications, ensuring that users can trust the data presented by the platform for decision-making, such as product stock levels, sales reports, and customer information.

Reliability

Reliability is the attribute that ensures the KFMS system consistently performs as expected, with minimal downtime and without errors. This is achieved through a combination of robust error handling, failover strategies, and regular system monitoring to identify and resolve potential issues before they affect the user experience. A reliable system enhances customer satisfaction by offering uninterrupted access to services such as browsing, purchasing, and managing user profiles.

Manage Cart – Aina Jaleena binti Jusop (CB21089)

Quality attributes are the key characteristics that determine the overall quality and functionality of a software system. These attributes are essential to assess the software's performance, user experience, and its ability to meet the needs of the stakeholders. In software design and development, quality attributes guide decisions and help ensure the system's success from a user and technical perspective. Different software features may emphasize different qualities depending on the requirements and use cases. For the Kosso Fishing Management System (KFMS), which includes a cart management module, the focus is on attributes such as **usability**, **maintainability** and **reliability**.

The **usability** attribute is particularly important for the cart module, since it allows customers to simply add, modify, and review items in their cart. An intuitive interface and seamless interaction are key to a positive user experience, which directly impacts customer satisfaction and sales (Vijay, 2024). **Maintainability** refers to the ease with which developers can update and modify the cart module. This includes making changes to product listings, pricing rules, and promotions with minimal downtime or issues. A maintainable system ensures that new features or bug fixes can be fixed quickly and efficiently (Priyanka, 2024).

Lastly, **reliability** is another critical quality for the cart module. This quality ensures that the system performs as expected in a variety of situations, such as when customers return to their cart after a period or when there are changes in stock or pricing. The cart should always appropriately represent the selected products and their amounts.

In this quality design document, these key quality attributes which are usability, maintainability and reliability in the context of **Manage Cart** module for KFMS. These attributes will ensure that the cart module not only meets the expectations of the users but also provides a secure and maintainable foundation for the development and operational success.

Manage Review - Mohamad Hatta bin Borhanuddin (CB21080)

The Manage Review module in the KOSSO Fishing Store (KFS) project is designed to allow customers to provide feedback on products, ensuring a transparent and reliable review system. To maintain the system's efficiency and effectiveness, three quality attributes have been identified: Usability, Reliability, and Security.

Usability

This quality attribute ensures that the review submission process is intuitive and easy for all users, regardless of their technical expertise. By focusing on clear instructions, a simple interface, and smooth navigation, usability aims to enhance the user experience and encourage active participation in leaving reviews.

Reliability

Given the potential volume of reviews, especially during peak sales periods, the system must function without errors or delays. Reliability ensures that the review system operates consistently, with minimal downtime and accurate data storage, allowing users to submit their feedback seamlessly, even under high traffic conditions.

Scalability

Ensuring the Manage Review module can handle increased user activity and data growth is essential for maintaining performance and reliability. Scalability mechanisms, such as load balancing, database optimization, and efficient caching strategies, enable the system to manage higher volumes of review submissions, retrievals, and concurrent users without performance degradation. This ensures the module remains responsive and reliable as the KOSSO Fishing Store grows and attracts more users, providing a seamless experience for all stakeholders.

Manage Stock - Nazrin Naim bin Ahmad Mahari (CB21072)

The Manage Stock modules in the KOSSO Fishing Management System (KFMS) requires attention to specific quality attributes to ensure smooth and accurate stock management. Focusing on quality attributes such as security, reliability and usability will create a robust system that supports accurate inventory handling, protects data, and provides ease of use for all administrative users. Each attribute is important in fulfilling the module's purpose and in meeting the expectations of the users and stakeholders.

Security

Security in software systems involves measures to prevent unauthorized access to the system and to protect data integrity. In the manage stock module, security safeguards inventory data by ensuring that only authorized users can view or modify stock levels and related information. This is important because inventory management systems contain sensitive information that directly impacts business operation. Proper security measures minimize risks such as unauthorized data changes, which could lead to stock tracking errors or financial losses.

Reliability

Reliability in software ensures that the system consistently performs its intended functions without failure. In the manage stock module, reliability means accurately updating stock data in real-time and making it consistently available across all system interfaces without delays or errors. This is crucial for avoiding issues such as stock inconsistencies, overselling or underselling. Inaccurate stock data can lead to loss of customer trust, restock disruption and operational inefficiencies, making reliability a key factor in the management module effectiveness.

Usability

Usability is a quality attribute that ensures a system is easy to use, enabling users to achieve their goal effectively and efficiently. In the manage stock module, usability focuses on allowing staff to

view, add, update and remove stock entries quickly with minimal steps. This is essential for improving operational efficiency and reducing the training time for new users. A user-friendly interface not only minimizes errors during stock updates but also makes inventory management more accurate and straightforward for staff.

3.5 Quality Attributes Scenario

Manage User Analyze Sales - Abdul Razzaq bin Abdul Aziz (CB22056)

Confidentiality

An unauthorized individual tries to access confidential user information within the Manage User module by manipulating the URL and attempting to reach the profile page directly. The system detects that this individual is not logged in and lacks authorization to view the requested information page. As a result, access is immediately denied, preventing exposure of sensitive data, and any further attempts are blocked based on predefined access rules stored in the database.

Source: Unauthorized individual.

Stimulus: Attempt to access confidential user data via URL manipulation.

Artifact: Manage User module.

Environment: Regular operational hours.

Response: Deny access for the user who is not logged in or lacks authorization.

Response measure: Immediate access denial based on access rules, ensuring that unauthorized attempts are blocked within 5 seconds.

Integrity

Customers (end-users) update their profile information on the edit user information page, entering a new address in the required field. The system automatically checks the entered address to ensure it meets the minimum character requirement. Since the entered address does not meet the criteria, the system immediately prompts the customer to re-enter the correct address. The system accepts the input once the customer provides a valid address, ensuring all user data adheres to established validation rules.

Source: Customer (end-user).

Stimulus: Customer interacts with the edit user information page.

Artifact: Edit user information interface.

Environment: Regular operational hours.

Response: The system accepts the input once the customer provides a valid address

Response measure: Validation results are processed within 2 seconds.

Reliability

An admin accesses the Analyse Sales module to generate a monthly sales report. The system automatically retrieves and processes the sales data from the database, generating the report without delay. Even during peak operational hours, the system remains functional, ensuring the report is available whenever the admin needs it. This ensures accurate and timely decision-making without interruptions.

Source: Admin.

Stimulus: Request for monthly sales report.

Artifact: Analyze Sales module.

Environment: Peak operational hours.

Response: The system reliably generates and makes the report available to the admin.

Response measure: Ensure the report is generated within 10 seconds with 99% uptime during peak loads.

Manage Cart – Aina Jaleena binti Jusop (CB21089)

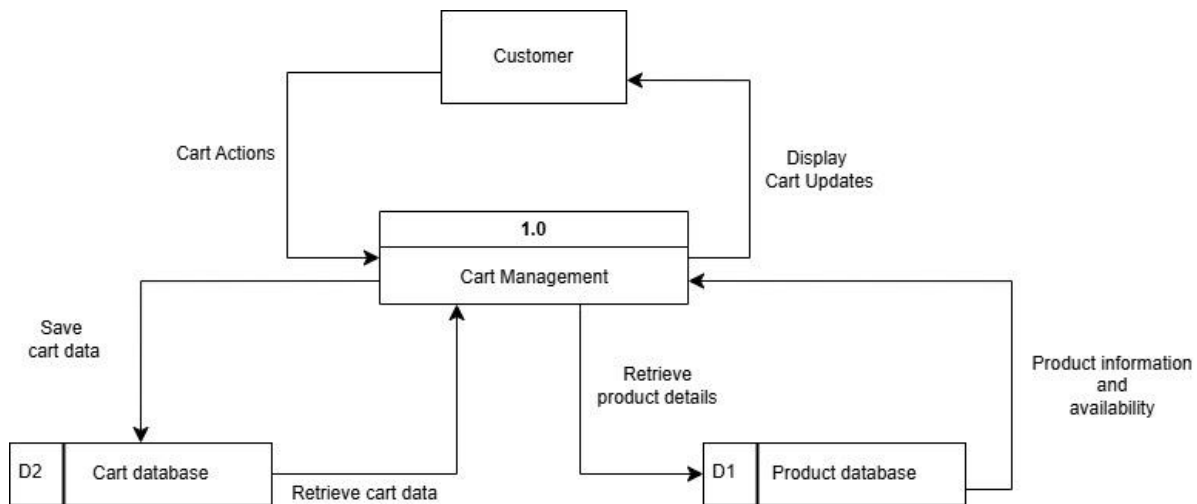


Figure 1: Data Flow Diagram for Manage Cart Module

Usability

A customer browses the KFMS to purchase fishing equipment. After selecting a few items, the customer navigates to their cart to review the items, edit quantities, removes an item they no longer want, and proceeds to checkout.

Source: A customer using the Kosso Fishing Management System (KFMS).

Stimulus: Customer adds items to their cart, views the cart, updates item quantities, and removes unwanted items.

Artifact: The cart interface within the KFMS platform.

Environment: Normal operation using web page.

Response: The system allows customers to simply adjust their carts, instantly updating the quantities and pricing while ensuring that changes are saved and reflected correctly across sessions.

Response measure:

- Changes in the cart such as removing an item or updating quantity are reflected in real-time.
- The customer is able to complete the process of updating the cart within 2 minutes without any issues.

Maintainability

A developer needs to update the price rules in the cart module due to a seasonal promotion. After applying the modifications, they test the cart functionality to ensure that the updates are applied successfully without affecting overall system's general operation.

Source: A developer maintaining and updating the code in the cart module of the KFMS platform.

Stimulus: The developer modifies the pricing rules and applies a seasonal discount to specific items in the cart.

Artifact: The cart's code and configuration for price calculations.

Environment: The system is accessed in a development or staging environment, using development tools to implement and test changes.

Response: The system should implement the new pricing rules correctly, displaying updated prices in the cart, while ensuring that no other cart features such as item quantity updates and checkout process are disrupted.

Response measure:

- The pricing changes are properly applied within the cart and reflected in less than 10 minutes of development time.
- The cart continues to function correctly without errors after the changes, with no negative impact on other system functionalities.

Reliability

A customer adds multiple items to their cart, updates the quantities, and then returns to the cart after a few hours to ensure that the modifications were properly recorded and that the cart still represents the right items and quantity.

Source: A customer using the Kosso Fishing Management System (KFMS) to manage their shopping cart.

Stimulus: The customer adds items to the cart, updates the quantities, and leaves the site for a while, returning later to check if the cart information is correct.

Artifact: The cart interface and underlying database system in KFMS.

Environment: Normal operation using web page.

Response: The system reliably saves the cart data, ensuring that the items and updated quantities remain in the cart even after the session ends or the customer navigates away from the page.

Response measure:

- The cart data such as items, quantities, and prices are saved correctly and can be retrieved accurately when returning to the site.
- In 99% of test cases, the cart displays accurate information without any inconsistencies or errors.
- User feedback shows that 95% of customers trust the system to keep their cart contents correct throughout many sessions.

Manage Review - Mohamad Hatta Rahman bin Borhanuddin (CB21080)

Usability

Usability ensures that customers can submit, view, and edit their reviews with ease, promoting a positive user experience.

A new user named Nazrin recently bought a fishing rod and wanted to leave a review about his experience. Since it was his first time using the review system, he wasn't sure how it worked. The Manage Review module should make it easy for him by providing clear steps and a simple interface. For example, when Nazrin opens the review page, he sees a clean form with labels like "Rate this product" and "Write your review," along with a star rating option and a text box. After submitting his review, a confirmation message pops up, and his review shows up immediately under the product. To ensure the system works well for everyone, its usability can be checked by looking at how often users successfully submit reviews and collecting feedback on how easy it is to use.

Source: New user (Nazrin)

Stimulus: Attempts to leave a review for the first time

Artifact: Manage Review Module

Environment: First-time user unfamiliar with the system

Response: The system provides a clean layout with labeled fields, a star rating system, and a text box. Upon submission, a success message appears, and the review is instantly displayed under the product.

Response Measure: Track the percentage of successful review submissions and collect user feedback on the interface's clarity and simplicity.

Reliability

Reliability ensures that the Manage Review module consistently performs its intended functions, even under high usage.

Imagine during a big sale, hundreds of people are leaving reviews all at once. The Manage Review module should still run smoothly without slowing down or crashing. It should handle every review submission properly, saving the data to the database and showing the reviews right away. To make sure it's reliable, the system's performance can be tracked during busy times by checking uptime, successful review submissions, and how quickly it responds. Any errors or delays should be minimal or nonexistent.

Source: Large group of users

Stimulus: Hundreds of customers submit reviews simultaneously

Artifact: Manage Review Module

Environment: High user load during peak activity such as sale event

Response: The system handles the load seamlessly, ensuring all reviews are processed and saved in the database without delays or errors. Reviews are displayed instantly under the product.

Response Measure: Monitor system uptime, successful review submissions, and response times during peak activity. Analyze logs for errors or performance issues.

Scalability

The Manage Review module is designed to efficiently handle increasing volumes of customer feedback by dynamically allocating resources, such as database connections and server capacity, during periods of high activity. This ensures that users can seamlessly submit and view reviews, even as the number of users and reviews grows, without experiencing delays or system slowdowns.

During peak times, such as special sales events or holiday promotions, the Manage Review module may experience a surge in traffic as more customers leave reviews. The system should be able to scale automatically to accommodate the increased load, ensuring that users can still

submit, update, or view reviews without experiencing delays or downtime. This can be tested by simulating high traffic scenarios and monitoring the system's response, ensuring that it maintains performance levels and doesn't degrade under heavy usage. Scalability features like load balancing and resource allocation should be validated to confirm their effectiveness.

Source: High traffic or sudden surge in users

Stimulus: Increased volume of review submissions during peak periods (e.g., promotions, sales events)

Environment: Multiple simultaneous review submissions from users

Response: The system automatically scales to handle the increased load, ensuring fast submission, retrieval, and processing of reviews without delays or downtime. Load balancing and resource allocation mechanisms are engaged to maintain performance.

Response Measure: Monitor system performance during peak traffic, including submission success rates, response times, and any downtime. Test the system's scalability under simulated high traffic conditions to verify its ability to handle heavy loads. Track user experience and response times to ensure the system remains stable and efficient.

Manage Stock - Nazrin Naim bin Ahmad Mahari (CB21072)

The manage stock module is an important part of the system that helps manage inventory smoothly while meeting key quality standards. The main quality attributes are security, reliability and usability are essential for creating a system that is safe, reliable, and easy to use.

Security

During off-peak hours, when staff are not active, someone who is not authorized tries to access the stock database by guessing usernames and passwords. They attempt different combinations to exploit weak security in the login system. If they succeed, they could make unauthorized changes to stock data, causing errors or even deleting important records. To prevent this, the system tracks login attempts and detects suspicious activity after three failed tries. It quickly locks the account; blocks access from the user's IP address and sends a notification to the system administrator for further action.

Source: Unauthorized user

Stimulus: Multiple failed login attempts

Artifact: Stock database and authentication system

Environment: Off-peak hours when staff not active

Response: The system locks the account after three failed attempts, blocks the source IP address, and sends a notification to the administrator

Response Measure: Unauthorized access is completely prevented, and the alert is sent to the administrator within 5 seconds of detection

Reliability

On a busy day, the staff needs to update stock levels for several products using the manage stock module. While adjusting the quantity of items, they also add new stock for item and make changes to other items. This sudden increase in activity could cause delays or errors if the system isn't reliable. To handle this, the system processes each update one at a time, making sure no changes overwrite or conflict with others. All updates are shown instantly across the system, including on inventory records and staff reports.

Source: Store staff

Stimulus: Multiple stock updates for different products at the same time

Artifact: Stock database and inventory records

Environment: During a busy day with high stock management activity

Response: The system processes update as atomic transaction, ensuring that no changes overwrite or conflict with others. All updates are shown in real-time across the system.

Response Measure: Updates are processed within 2 seconds, and no errors or conflicts occur

Usability

A new staff member, who isn't familiar with using the manage stock module, tries to update product quantities for the first time. The system guides them with clear instructions, including labelled fields for entering product codes, stock levels, and descriptions. If they make a mistake, like entering a negative number, the system shows a simple error message explaining the problem and how to fix it. Features like auto-complete for product names and helpful tips make the process easier. This ensures the new staff member can successfully update stock without errors.

Source: New staff member

Stimulus: Attempts to update stock but enter an invalid quantity.

Artifact: Manage stock module interface

Environment: During normal operation

Response: The system displays a clear error message, provides tips, and prevents the invalid update

Response Measure: The task is successfully completed within 5 minutes, and no incorrect data is saved

3.6 Tactics

Manage User & Analyze Sales- Abdul Razzaq bin Abdul Aziz (CB22056)

Confidentiality

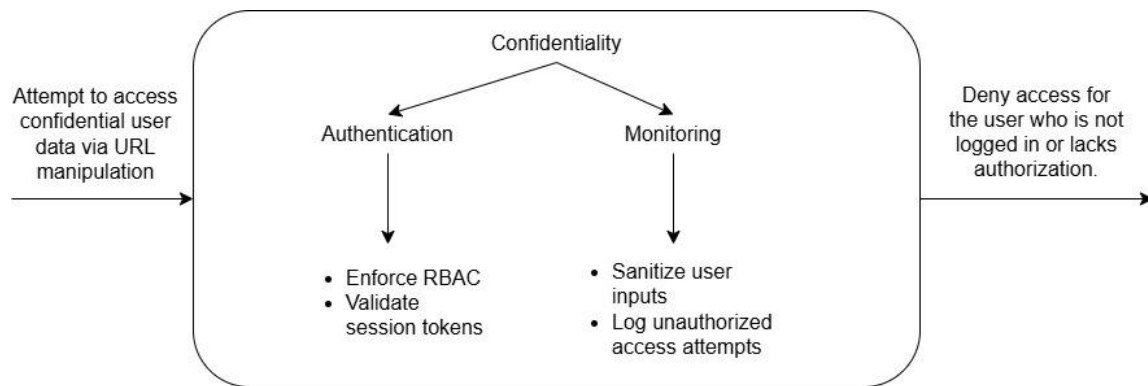


Figure 5.1 Tactics for Unauthorized Access

1. Authentication

- **Enforce RBAC:** Implement role-based access control (RBAC) to ensure that only authorized users can access sensitive data. This ensures permissions are assigned based on user roles or attributes, restricting access to confidential information.
- **Validate Session Tokens:** Use secure session tokens and verify their validity on each request. Expired or tampered tokens must be invalidated immediately to prevent unauthorized access.

2. Input Validation and Monitoring

- **Sanitize User Inputs:** Sanitize and validate all inputs before processing to prevent malicious attacks like SQL injection or cross-site scripting (XSS). This ensures that data entering the system is clean and safe.
- **Log Unauthorized Access Attempts:** Continuously monitor and log unauthorized access attempts, such as URL manipulation or brute-force attacks. Real-time logging enables administrators to respond quickly to suspicious activities.

Integrity

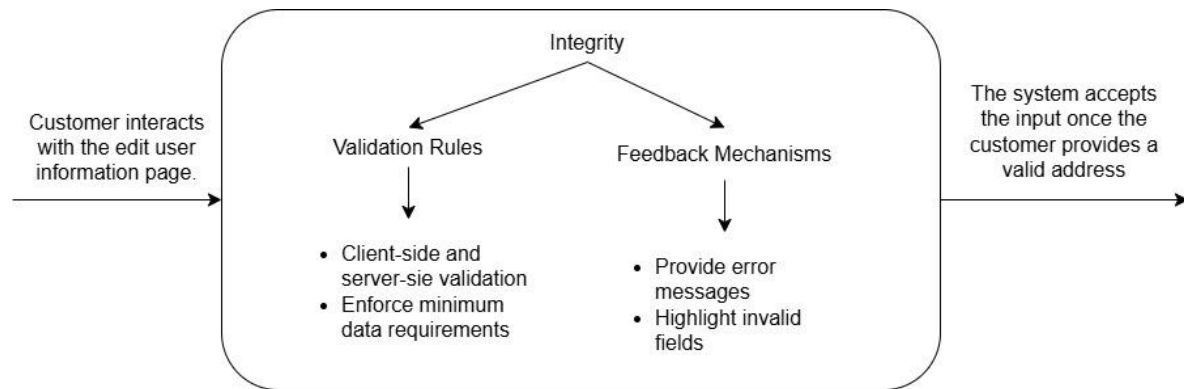


Figure 5.2 Tactics for Data Validation

1. Validation Rules

- **Client-Side and Server-Side Validation:** Perform input validation on both the client and server sides. Client-side validation ensures quick feedback to users, while server-side validation provides a final check to prevent invalid data from being stored.
- **Enforce Minimum Data Requirements:** Define strict validation rules, such as minimum character counts for addresses or specific formats for email addresses, ensuring the data is complete and adheres to system standards.

2. Feedback Mechanisms

- **Provide Error Messages:** Display clear and actionable error messages when validation fails. For example, "Address must be at least 10 characters long."
- **Highlight Invalid Fields:** Visually mark the fields with validation errors like, red borders to guide users toward corrections, improving the user experience.

Reliability

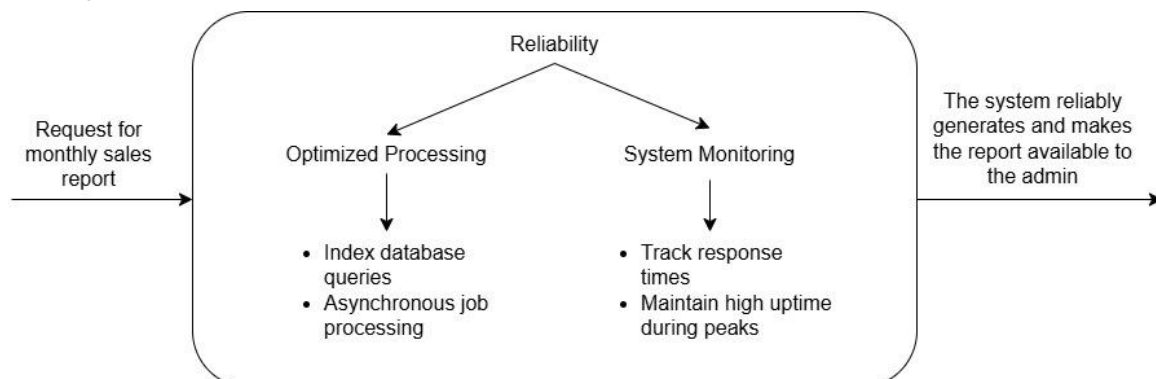


Figure 5.3 Tactics for Report Reliability

1. Optimized Processing

- **Index Database Queries:** Use indexing in the database to improve the performance of queries, especially for large datasets. This ensures timely data retrieval even during high traffic.
- **Asynchronous Job Processing:** Offload heavy tasks, such as report generation, to background processes. This ensures the system remains responsive for other users while the task is completed in the background.

2. System Monitoring

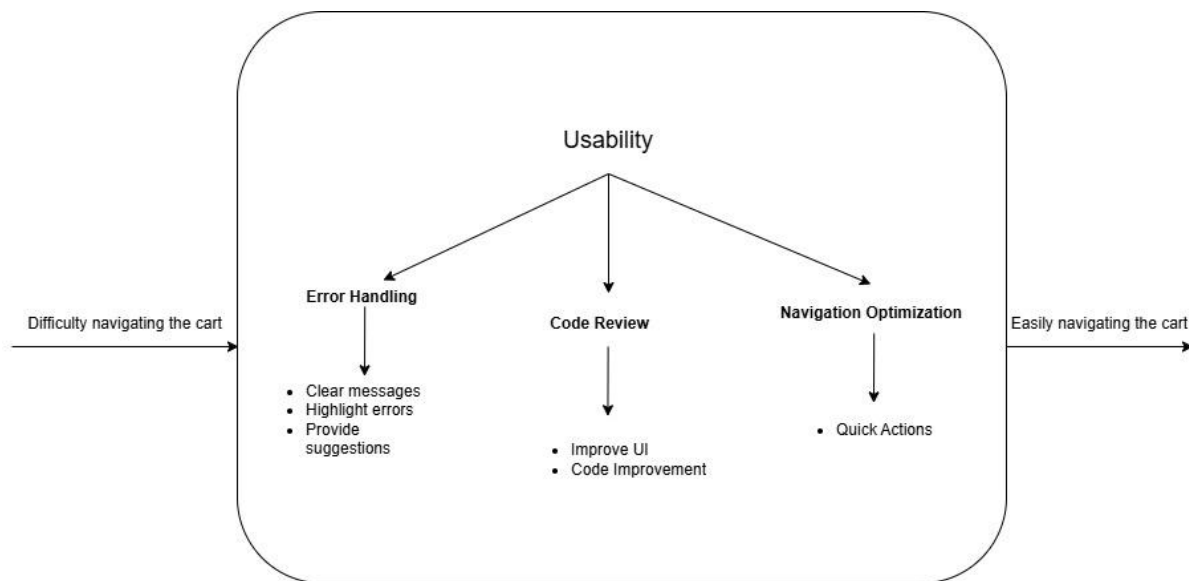
- **Track Response Times:** Continuously monitor the system's response time to identify bottlenecks or slowdowns in real time. Alerts should be triggered for delays beyond acceptable limits.

Maintain High Uptime During Peaks: Implement load balancing, caching, and failover systems to ensure the system achieves 99% uptime even during peak operational hours.

Manage Cart – Aina Jaleena binti Jusop (CB21089)

Usability

Difficulty in navigating the cart.



1) Error Handling

➤ Clear messages

- Provide clear and concise feedback when a user adds, removes, or updates an item in the cart. If an error occurs such as an item is out of stock, display a message that clearly state the issue and suggests possible solutions, such as checking other product options.

➤ Highlight errors

- If a user attempts to update the quantity of an unavailable product or makes an invalid change, highlight the problematic field or item to help the user quickly identify and correct the issue.

➤ Provide Suggestions

- When an error occurs, provide actionable alternatives such as recommending similar products when an item is out of stock.

2) Code Review

➤ Improve UI

- Regularly review and refine the cart interface to ensure it is user-friendly. Use easily recognizable icons and make sure that interactive features such as buttons are properly positioned. The layout should be consistent and straightforward.

➤ Code Improvement

- Regularly monitor and review the cart module's code to identify areas for performance enhancement, bug fixes, or accessibility improvements. This ensures the cart's efficiency and functionality as it evolves.

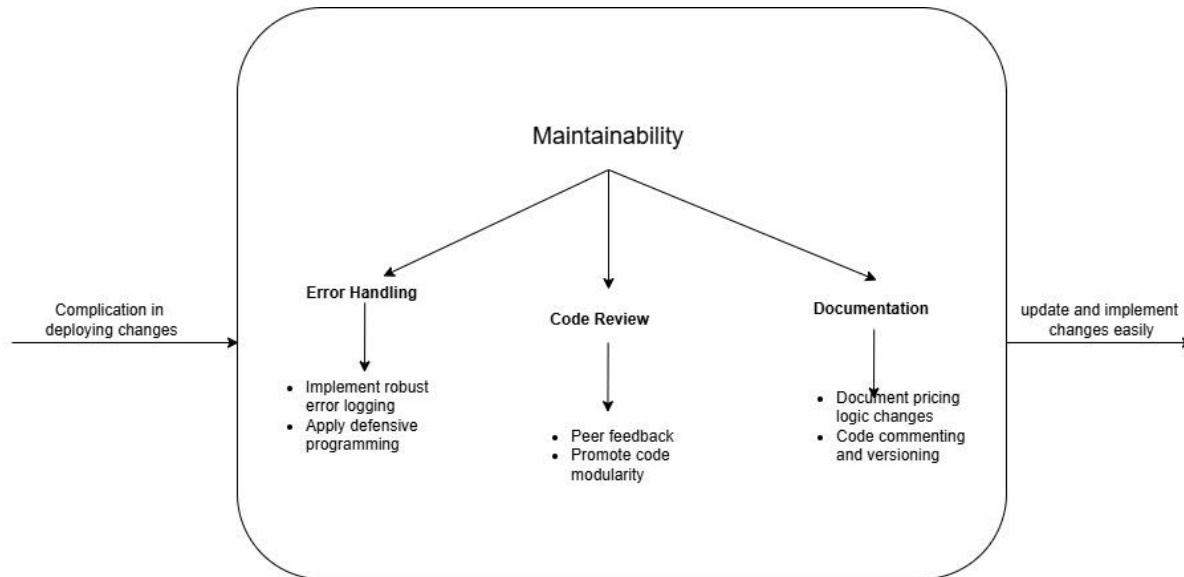
3) Navigation Optimization

➤ Quick Actions

- Implement quick actions, such as one-click removal of items or quantity updates, to improve user interaction with the cart.

Maintainability

Complication in deploying changes.



1) Error Handling

➤ Implement robust error logging

- Implement a robust error logging system to track and record any issues related to the new pricing rules or cart functionalities. This will allow that developers can quickly trace and resolve errors that occur during updates. Clear error messages and logs must be standardized to facilitate future troubleshooting.

➤ Apply defensive programming

- Use defensive programming techniques to ensure that the cart logic handles unexpected inputs or situations such as if a promotion code fails, a clear message is displayed, and no wrong calculations are performed.

2) Code Review

➤ Peer Feedback

- Conduct thorough peer reviews of the cart module's code before and after making changes. Having multiple developers analyse the modifications means that possible faults or improvements are detected early resulting in better code quality and system stability.

➤ Promote Code Modularity

- Ensure that the cart logic is modular so that components such as price rules and discounts can be updated independently without affecting the rest of the system.

3) Documentation

➤ Document Pricing Logic Changes

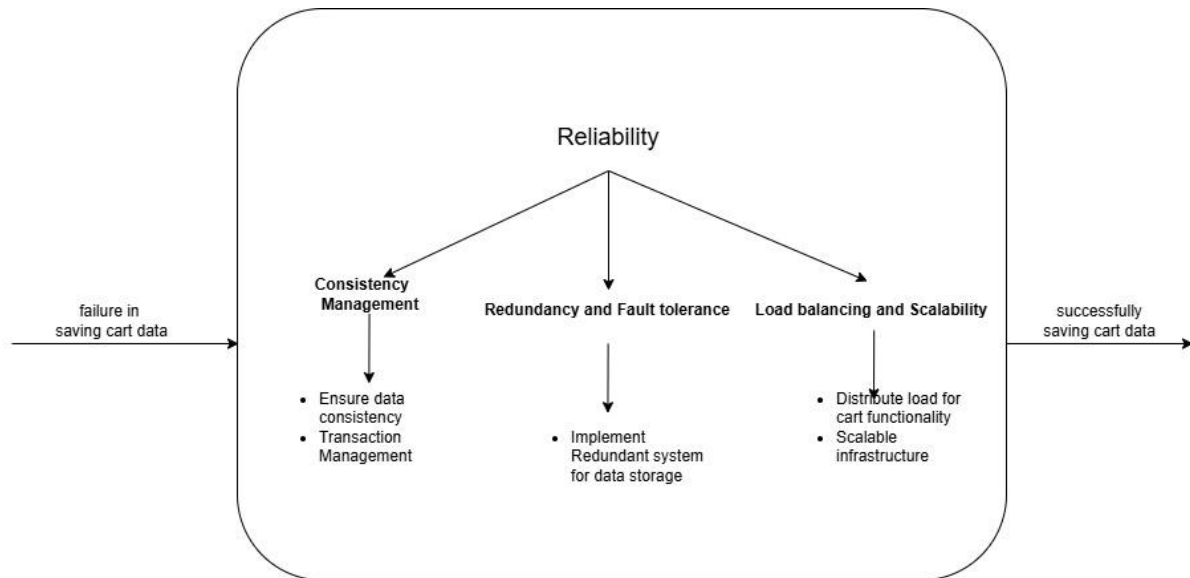
- Keep detailed documentation of the seasonal pricing logic and the changes made to the cart. This will make it easier for future developers to understand the reasoning behind modifications and the parts of code that were changed, resulting in faster future updates or corrections.

➤ Code Commenting and Versioning

- Properly comment code changes and maintain version control so that any modification to the price rules is well-documented and can be reviewed if necessary. This ensures that updates are clear and understandable.

Reliability

Failure in saving cart data



1) Consistency Management

➤ Ensure data consistency

- Implementing techniques to keep the data in the cart remains consistent between user sessions. When the customer returns to the cart, the system should automatically verify that any changes such as added items and updated quantities have been correctly stored and retrieved.

➤ Transaction Management

- Implement proper transaction management for cart updates, ensuring that any changes performed during a session such as adding products or updating quantities are properly committed to the database without partial updates.

2) Redundancy and Fault Tolerance

➤ Implement Redundant Systems for Data Storage

- Use redundant systems to ensure that cart data is backed up in several locations. If one system fails, the backup will keep the cart available and dependable until the customer returns. This technique helps prevent data loss and ensures the cart data is always available.

3) Load Balancing and Scalability

➤ Distribute Load for Cart Functionality

- Use load balancing techniques to distribute traffic and ensure that cart-related processes such as adding items or updating quantities perform reliably, even under high user loads. This ensures that customers can always interact with their cart without delays or errors, regardless of traffic volume.

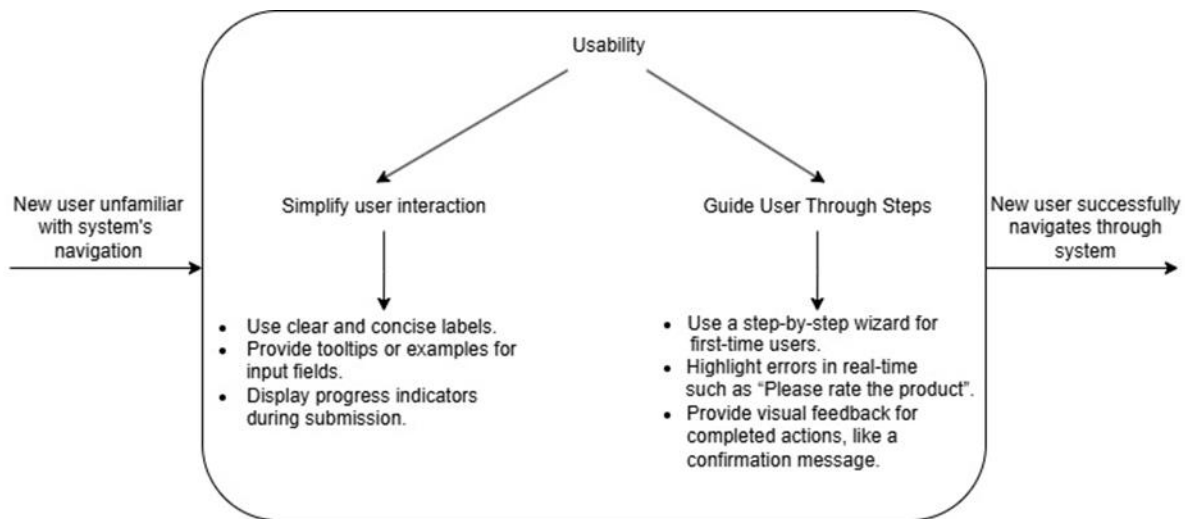
➤ Scalable Infrastructure

- Ensure that the infrastructure can expand as needed to manage an increase in traffic or data load, especially during high-traffic periods like promotions or seasonal sales. This scalability ensures that the cart stays stable, even as the number of users and transactions increases.

Manage Review - Mohamad Hatta Rahman bin Borhanuddin (CB21080)

The section will describe the tactics for each quality attribute. Tactics are design decisions that could affect quality attributes' stimulus responses, tactics can be made to improve the system in different attributes and ways.

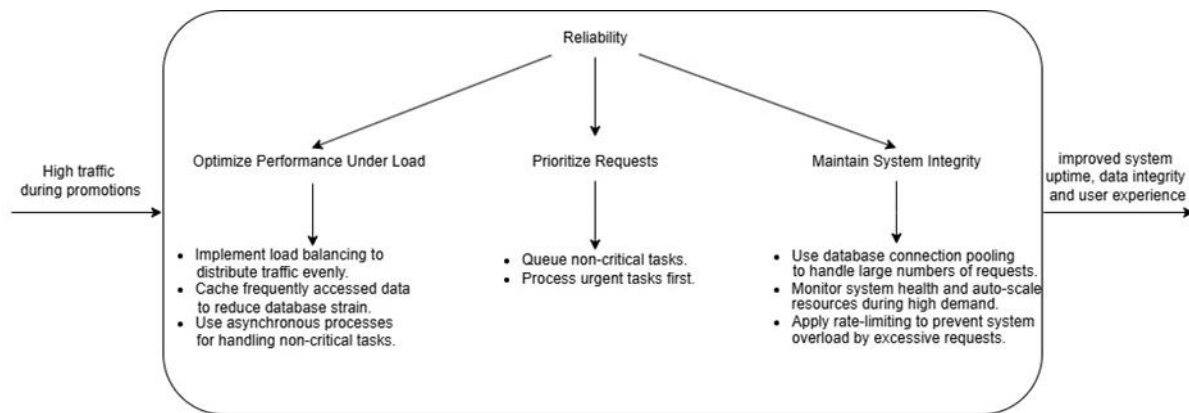
Usability



Tactics	Elaboration
Simplify user interaction	Use clear and concise labels. Developers ensure that form labels, buttons, and instructions are straightforward, reducing confusion for users. Provide tooltips or examples for input fields. The system offers small pop-ups or placeholder examples to guide users in entering the correct data in input fields. Display progress indicators during submission. A progress bar or spinner is displayed during the submission process to reassure users that their action is being processed.

Guide User Through Steps	Use a Step-by-Step Wizard for First-Time Users The system breaks down the process into smaller steps, guiding users through each stage with clear instructions. Highlight Errors in Real-Time such Developers ensure the system detects and displays errors immediately, such as reminding users to rate the product before submission. Provide Visual Feedback for Completed Actions A confirmation message or visual indicator, like a checkmark, reassures users that their action was successful.
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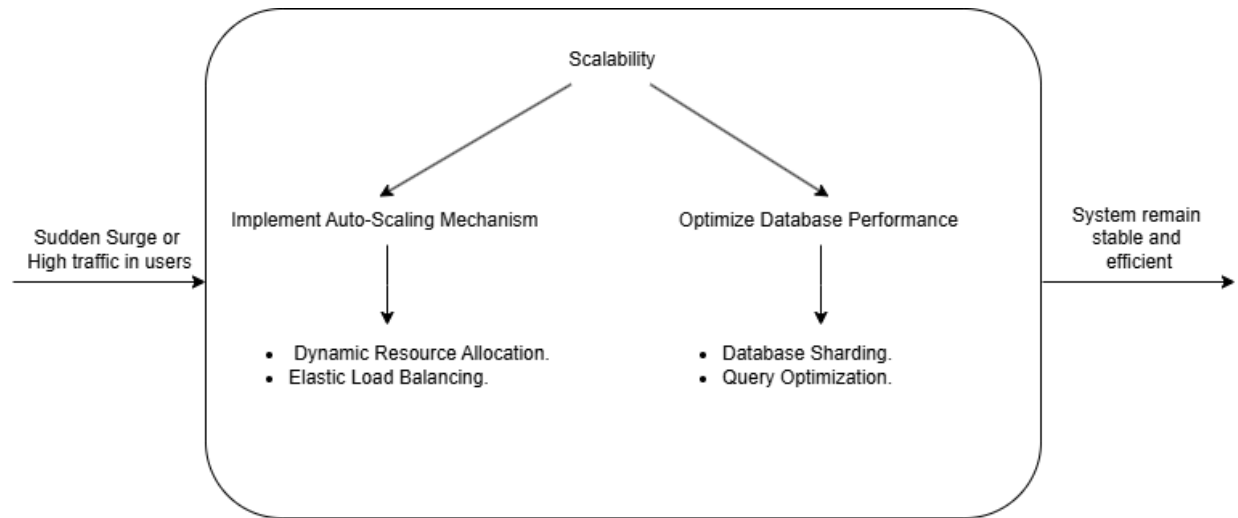
Reliability



Tactics	Elaboration
Optimize performance under load	Implement Load Balancing to Distribute Traffic Evenly The system uses load balancers to route incoming traffic across multiple servers, ensuring stable performance during high demand. Cache Frequently Accessed Data to Reduce Database Strain Developers store commonly requested data in memory, minimizing repeated queries to the database and speeding up response times. Use Asynchronous Processes for Handling Non-Critical Tasks Non-essential operations, like sending confirmation emails, are handled in the background, freeing up resources for critical tasks.
Prioritize requests	Queue Non-Critical Tasks The system places less urgent tasks, like background updates, into a queue, ensuring critical operations are processed first.

	Process Urgent Tasks First Developers implement a priority system that prioritizes essential requests, like review submissions, over lower-priority actions during peak usage.
Maintain system integrity	Use Database Connection Pooling The system reuses existing database connections to efficiently handle multiple simultaneous requests, reducing latency and resource usage. Monitor System Health and Auto-Scale Resources Real-time monitoring tools detect high demand and trigger the addition of extra server resources to maintain performance. Apply Rate-Limiting The system restricts the number of requests a user or IP address can send in a given time frame, preventing overload and abuse.

Scalability



Tactics	Elaboration
Implement Auto-Scaling Mechanism	Dynamic Resource Allocation The system automatically adjusts resources such as server capacity, memory, and processing power based on traffic volume. During peak usage periods, additional resources are provisioned to prevent slowdowns, ensuring reviews are processed promptly without performance degradation. Elastic Load Balancing Implement load balancing to distribute incoming traffic across multiple servers. This ensures that no single server is overwhelmed, improving response times and system reliability, especially during high traffic or when multiple users submit reviews simultaneously.
Optimize Database Performance	Database Sharding Split the database into smaller, more manageable parts (shards) to ensure faster read and write operations. This improves the system's ability to handle a large number of review submissions, ensuring quick access and updates during high demand.

	Query Optimization Optimize database queries by indexing frequently accessed fields, reducing query execution time. This helps maintain fast response times when users are submitting or viewing reviews, ensuring smooth performance even when the system is under heavy load.
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Security

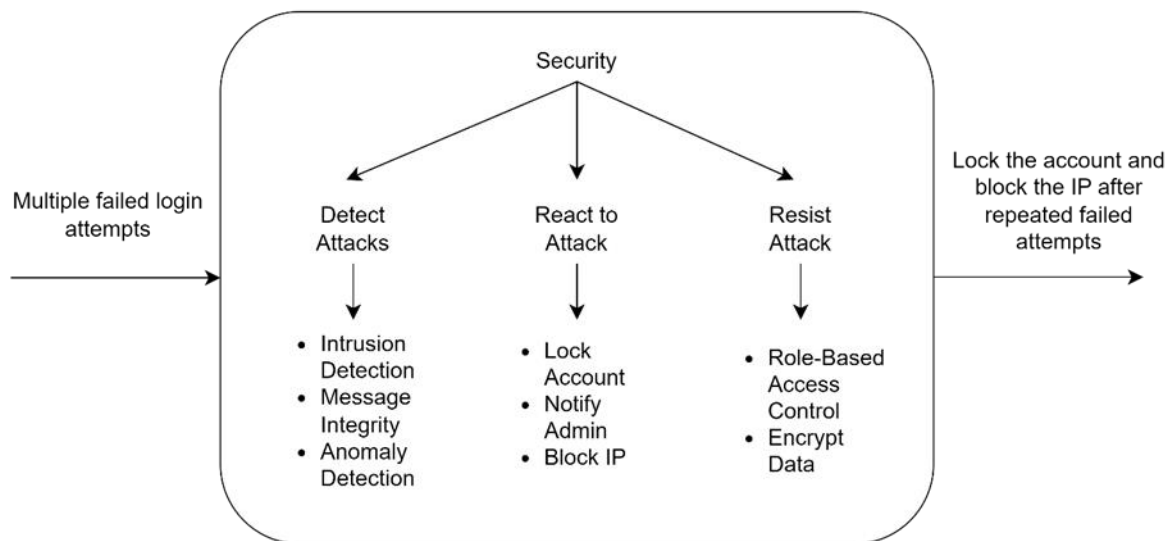


Figure 5.1 Tactics for Security

Detect Attacks

- **Intrusion Detection:** The system watches login attempts to spot unusual activities, such as too many failed attempts login or access from unknown location. This helps to find attackers early.
- **Message Integrity:** Ensures that the data sent between users and the system is not changed or tampered with, using secure methods like Transport Layer Security (TLS) encryption.
- **Anomaly Detection:** Checks if user behavior is unusual, like logging in at odd hours or from unknown devices. This helps to detect threats.

React to Attack

- **Lock Account:** Automatically locks a user account if they fail to log in multiple times, stopping brute force attacks.
- **Notify Administrator:** Send alerts to the admin as soon as suspicious behaviour is detected so they can act quickly
- **Block IP Address:** Block IP addresses that seem malicious or keep trying to access the system without permission.

Resist Attack

- **Role-Based Access Control (RBAC):** Limits what users can access based on their roles. For example, only admins can change sensitive stock data.
- **Encrypt Data:** Protects important data like stock records and passwords by encrypting it so it can't be read even if stolen.

Reliability

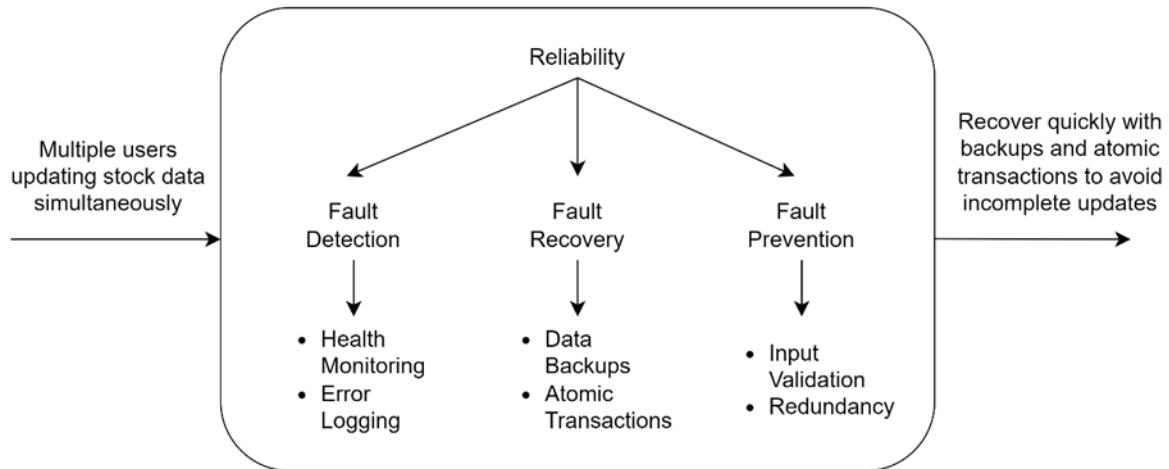


Figure 5.2 Tactics for Reliability

Fault Detection

- **Health Monitoring:** Continuously checks the system's health, like how quickly it responds or how much load the database is handling, to find problems early.
- **Error Logging:** Keeps records of errors, like failed stock updates, so they can be reviewed and fixed later.

Fault Recovery

- **Data Backups:** Regularly saves copies of stock data, so if the system crashes, everything can be restored quickly.
- **Atomic Transactions:** Makes sure that stock updates are either fully saved or not saved at all, preventing half-finished changes or errors.

Fault Prevention:

- **Input Validation:** Checks that data entered by users is correct before saving it. For example, it won't allow negative stock quantities or invalid product codes.
- **Redundancy:** Keeps backup systems ready, so if one system fails, another can take over without downtime.

Usability

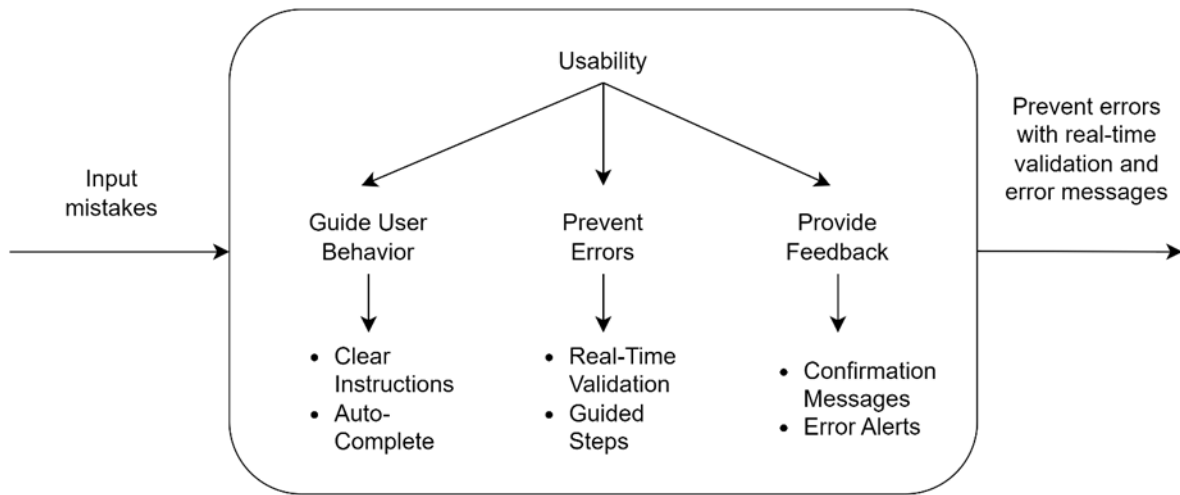


Figure 5.3 Tactics for Usability

Guide User Behaviour

- **Clear Instructions:** Provides simple instructions, tooltips, and labels to guide users through tasks like adding or updating stock.
- **Auto Complete:** Suggests product names or categories as users type to save time and reduce mistakes.

Prevent Errors

- **Real-Time Validation:** Shows error messages right away if the user enters incorrect data, like a negative quantity, so it can be fixed immediately.
- **Guided Steps:** Breaks tasks into smaller steps, like entering a product code first and then its quantity, to make the process easier.

Provide Feedback

- **Confirmation Messages:** Shows a success message when a task, like updating stock, is completed.
- **Error Alerts:** Displays simple, clear error messages if something goes wrong, explaining what happened and how to fix it.

3.7 Conclusion

In conclusion, this quality design document is created to ensure the KOSSO Fishing Management System (KFMS) meets high-quality standards. By integrating detailed quality attribute scenarios, effective tactics, and proactive risk management strategies, the design provides a strong foundation for tackling challenges and maintaining consistent system performance. Adopting industry best practices and aligning with established standards further ensures that the system is both efficient and adaptable to future needs. The focus on user satisfaction and operational efficiency ensures the system remains intuitive, dependable, and straightforward to manage. Overall, this document acts as a comprehensive guide for the team to build a strong and reliable system. It effectively meets the store's current needs while laying the groundwork for ongoing enhancements and future growth.

3.8 References

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4. Implementation

4.1 Introduction

The implementation of the online store for KOSSO Fishing Store involved the development of various modules to deliver a user-friendly and feature-rich e-commerce platform. The website was built using WordPress, leveraging its extensive plugin ecosystem and customization capabilities to achieve most of the desired functionalities.

Each module was designed and developed to meet the requirements set during the planning phase. Core features like the Landing Page, Login Page, Product List Page, and Cart Page were implemented successfully, delivering the expected outcomes with a seamless user experience. These modules effectively utilized WordPress themes and plugins, such as Elementor for page design and WooCommerce for e-commerce functionalities.

However, certain advanced features, such as dynamic stock management, customized sales analytics, and detailed review editing, were constrained by plugin limitations. In these cases, plugins like WooCommerce Stock Manager and WP Customer Reviews were used to provide a practical solution, even if they didn't fully align with the original requirements.

In summary, while some modules achieved the expected outcomes, others were adapted using available plugins to deliver a functional and efficient solution. This approach allowed the team to balance development efficiency with the project's goals, ensuring the completion of the online store within the given timeframe.

4.2 Methodology

The chosen methodology method for the development of Kosso Fishing Management System is Agile methodology.



Phase	Description
Requirements Gathering	In this phase, stakeholder needs are identified and documented. Our team collaborates closely with the owner of Kosso Fishing Store to understand the system requirements, which includes Manage User, Manage Stock, Analyze Sales, Manage Cart and Manage Review. Activities: Conduct interviews and brainstorming sessions with stakeholders and the owner of KFS to collect requirements. Outcome: Prioritized the list of functional and non-functional requirements that will guide the development of KFMS.
Design	This phase focusses on developing the User Interface (UI) on building a dynamic and adaptable framework to ensure the system meets both business and user needs effectively. This phase involves close collaboration among stakeholders, including the system owner, developers, and end users, to outline the interface, and functionalities. This ensures the design remains user-friendly, scalable, and capable of evolving as new ideas, requirements, or issues are identified during the project. By prioritizing a flexible

	design approach, KFMS can deliver an effective system for managing operations in KFS. Activities: Create user interface that emphasizes usability and user experience (UX). Outcome: Detailed system design documents and UI prototypes.
Implementation	During the implementation phase, the Agile development team works in short iterations called sprints, which generally run for two to four weeks. Each sprint begins with a planning session in which activities are prioritized and divided into smaller, easier to manage parts based on the design phase. Activities: - Development on 5 primary modules which are Manage User, Manage Stock, Analyze Sales, Manage Cart and Manage Review. - Implement the related quality attributes based on each module. - Regular code reviews and meetings to address development issues promptly.
Testing	Testing is performed iteratively to ensure the system meets quality standards and user requirements. Each sprint includes both functional and non-functional testing in order to identify and resolve issues as early as possible. Activities: - Execute a set of test cases at the end of each iteration to identify defects, inconsistencies, or performance issues. - Perform manual testing to validate system requirements such as user interaction, data quality, and functionalities.
Deployment	Once the system has passed testing, it is deployed to the production environment for usage by stakeholders. The deployment is carried out in a planned way to minimize disruption to operations. Activities: - Prepare the server or hosting environment for deployment. - Install and configure the system for real-world use. - Provide user training and documentation to ensure smooth adoption of KFMS. Outcome: KFMS is accessible to users, improving store operations and customer experience.
Review	The review phase is an essential aspect of the Agile methodology, conducted at the conclusion of each sprint. During this phase, the

	team evaluates the sprint's outcomes, discussing successes, challenges, and areas for improvement. allowing team members to provide feedback and share insights. For the Kosso Fishing Management System (KFMS. The lessons learned during this phase drive ongoing improvements, ensuring the system remains adaptive and responsive to evolving requirements. Activities: Conduct regular meetings with stakeholders to gather feedback and analyze system usage and identify any potential issues or gaps. Outcome: An optimized Kosso Fishing Management System that is constantly adapting to the needs of stakeholders, to ensure long-term use and performance.
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4.3 Tools

Development Tools

Development tools are used for creating, managing, and maintaining software, websites, or applications. They support activities like coding, debugging, version control, and deployment.

1. WordPress
2. GitHub

Document Tools

Document tools are used for creating, editing, formatting, and managing text-based documents or presentations.

1. Microsoft Word

Communication/Storage Tools

These tools facilitate communication, collaboration, and file storage or sharing, often via the cloud.

1. Google Drive

Design Tools

Design tools are used for creating visual assets, diagrams, user interface designs, or prototypes, aiding in the design phase of a project.

1. Draw.io
2. Figma

Database Tools

Database tools are used for managing, querying, and maintaining structured data stored in relational or non-relational databases.

1. MySQL

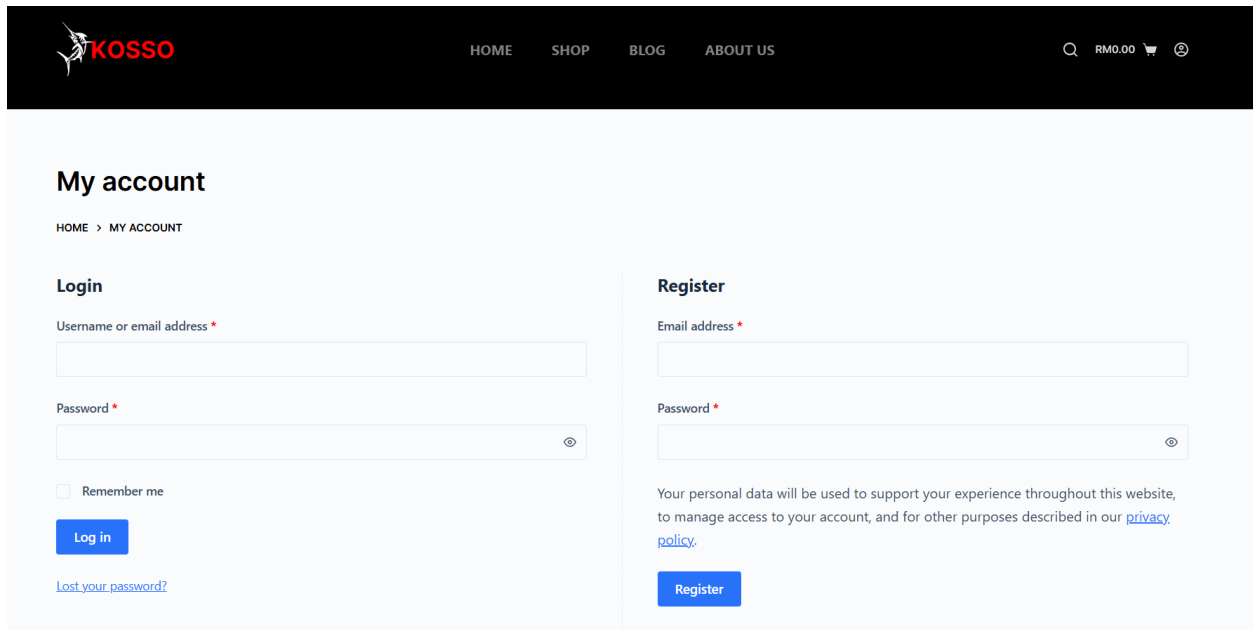
Tools	Description
WordPress	Adaptable content management platform, powers the KFMS's frontend, making it simple to build, organize, and update content. It keeps navigation smooth and delivers a user-friendly experience for everyone interacting with the system.
Microsoft word	The KFMS team relies on Microsoft Word to keep project documentation in check. It's handy for drafting and organizing everything from requirements to reports and meeting notes, helping to stay clear on details and track progress easily.
GitHub	A platform for version control with Git is used in developing the KFMS to store source code, track changes, and collaborate with the development team.
Google Drive	For the KFMS, we keep everything on Google Drive. It's the central hub for storing, sharing, and updating files, so everyone has easy access and stays aligned.
Draw.io	The diagramming and flowcharting tool helps create different types of diagrams. For the KFMS development, we used these tools to design a database structure that organizes our system's data and to create a use case diagram that maps out user relationships within each module.
Figma	A design tool built for collaboration, plays a key role in shaping the KFMS's UI. It lets the team bring interface ideas to life, fine-tune design details, and work together in real-time, helping create a polished and intuitive user experience.
SQL	A programming language used for managing and manipulating relational databases. We use Structured Query Language (SQL) to handle database queries, requests and validation of KFMS.

4.4 System Result

Manage User & Analyse Sales - Abdul Razzaq bin Abdul Aziz (CB22056)

Manage User

- Confidentiality



The screenshot displays the 'My account' section of the KOSSO website. The header is black with the KOSSO logo on the left and navigation links (HOME, SHOP, BLOG, ABOUT US) in the center. On the right, there are icons for search, cart (RM0.00), and user profile. The main content area is white and titled 'My account' with a breadcrumb 'HOME > MY ACCOUNT'. It is divided into two columns: 'Login' and 'Register'. The 'Login' form includes fields for 'Username or email address' and 'Password', a 'Remember me' checkbox, a 'Log in' button, and a 'Lost your password?' link. The 'Register' form includes fields for 'Email address' and 'Password', a privacy policy notice, and a 'Register' button.

Figure 4.4.1 (a)

The Manage User UI for the login page is designed with **Confidentiality** as a key focus, ensuring that users' sensitive information remains secure throughout the authentication process. The login page features strong encryption methods for password storage and transmission, protecting user credentials from unauthorized access. Additionally, the registration page ensures that new users provide valid and secure information, maintaining data confidentiality right from the start. This system aims to deliver a seamless yet highly secure user experience and protect user data at every step.

The screenshot shows the 'My account' section of a website. At the top, there is a navigation bar with the 'KOSSO' logo and links for HOME, SHOP, BLOG, and ABOUT US. A search icon and a shopping cart icon with 'RM0.00' are also present. Below the navigation bar, the 'My account' title is followed by a breadcrumb 'HOME > MY ACCOUNT'. The page is divided into two columns. The left column is for 'Login' and contains a 'Username or email address' field with 'superadmin' entered, a 'Password' field with masked characters, a 'Remember me' checkbox, a 'Log in' button, and a 'Lost your password?' link. The right column is for 'Register' and contains an 'Email address' field, a 'Password' field, a privacy policy notice, and a 'Register' button.

Figure 4.4.1 (b)

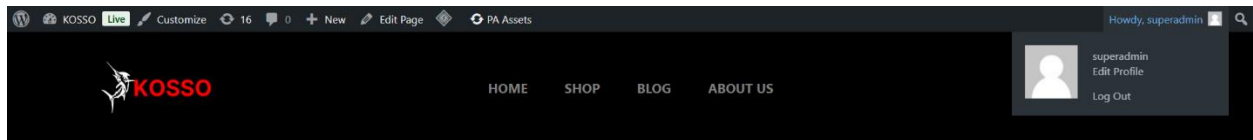


Figure 4.4.1 (c)

Figure 4.4.1 (b) and Figure 4.4.1 (c) shows the **Admin Login** page of the Manage User UI is designed with **Confidentiality** as a top priority, ensuring that only authorized personnel can access the admin portal. This ensures that only verified admins can access critical system settings and user data, protecting the system from unauthorized access while always maintaining confidentiality.

- Integrity

KOSSO HOME SHOP BLOG ABOUT US RM0.00

My account

HOME > MY ACCOUNT

Unknown email address. Check again or try your username.

Login

Username or email address *

razzaq@gmail.com

Password *

☐ Remember me

Log in

Register

Email address *

Password *

Your personal data will be used to support your experience throughout this website, to manage access to your account, and for other purposes described in our [privacy policy](#).

Figure 4.4.1 (d)

The **User Login** page focuses on **Integrity** by making sure that the user's data is correct and consistent. If a user enters an "**Unknown Email Address**", the system checks the email against the records in the database to make sure it matches a registered account. This helps to avoid mistakes and ensures that the right email is being used.

- Reliability

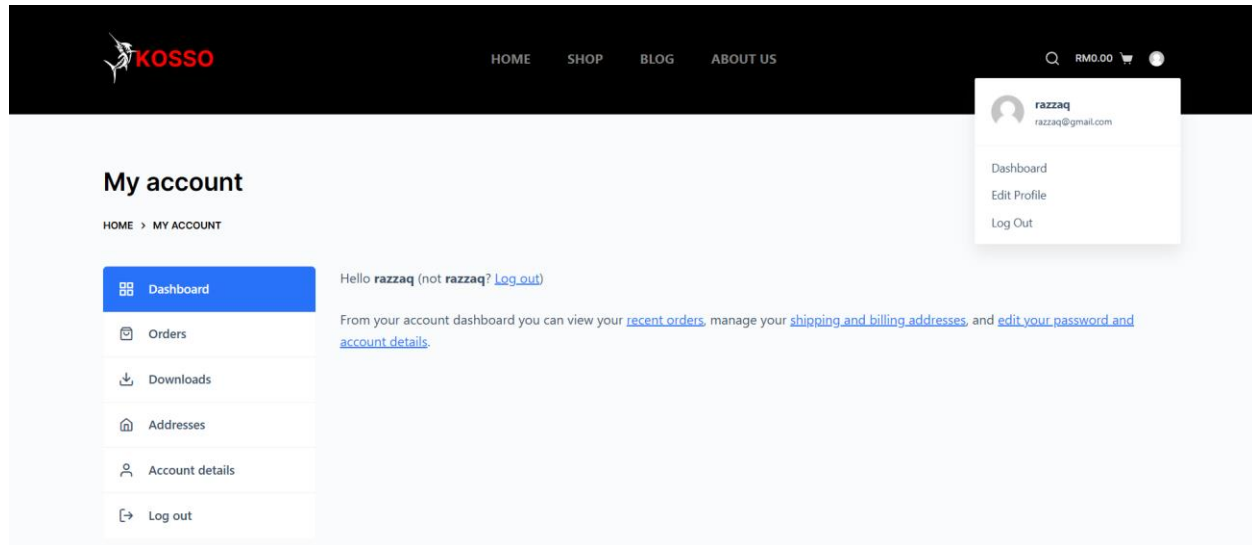


Figure 4.4.1 (e)

When a user successfully logs in, the **User Login** page provides a reliable experience by quickly and securely letting the user into their account. The system checks that the email and password are correct, and then smoothly takes the user to their dashboard without any delays. The login process is designed to be stable, ensuring that it works consistently without errors. Even during busy times, the system handles logins without any problems. If something goes wrong, the system is set up to recover quickly, so users can always get into their accounts easily and without frustration. This makes sure that the login experience is reliable and smooth whenever user use the KFMS.

Analyse Sales

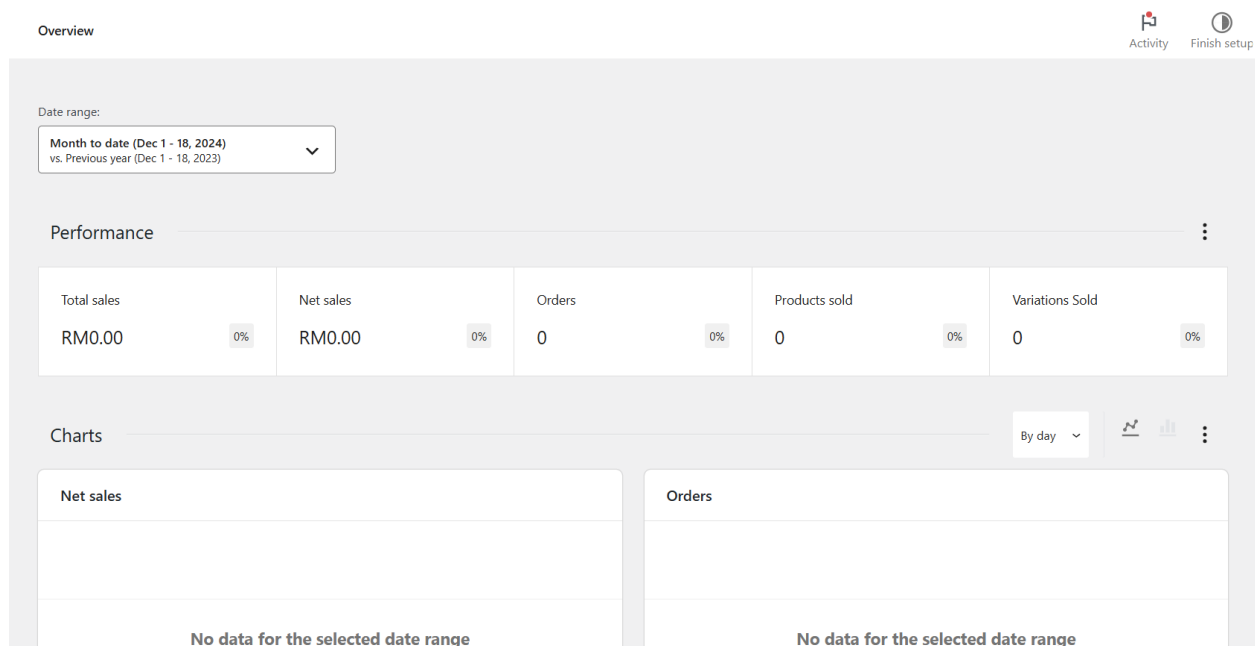
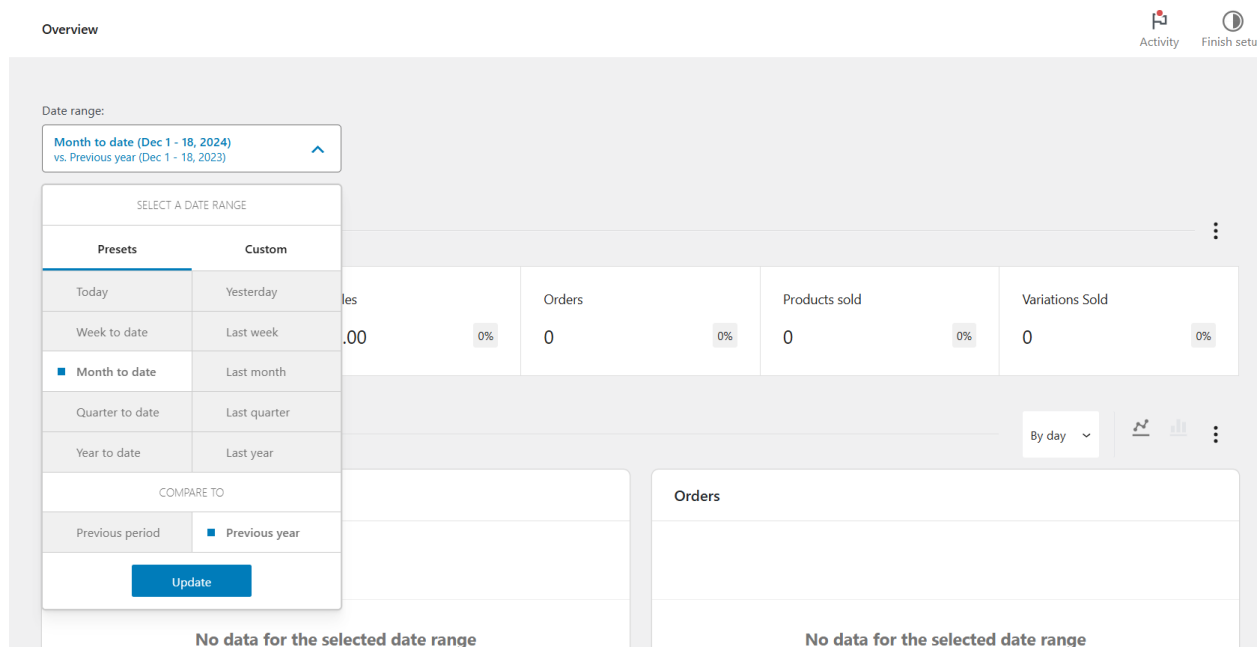
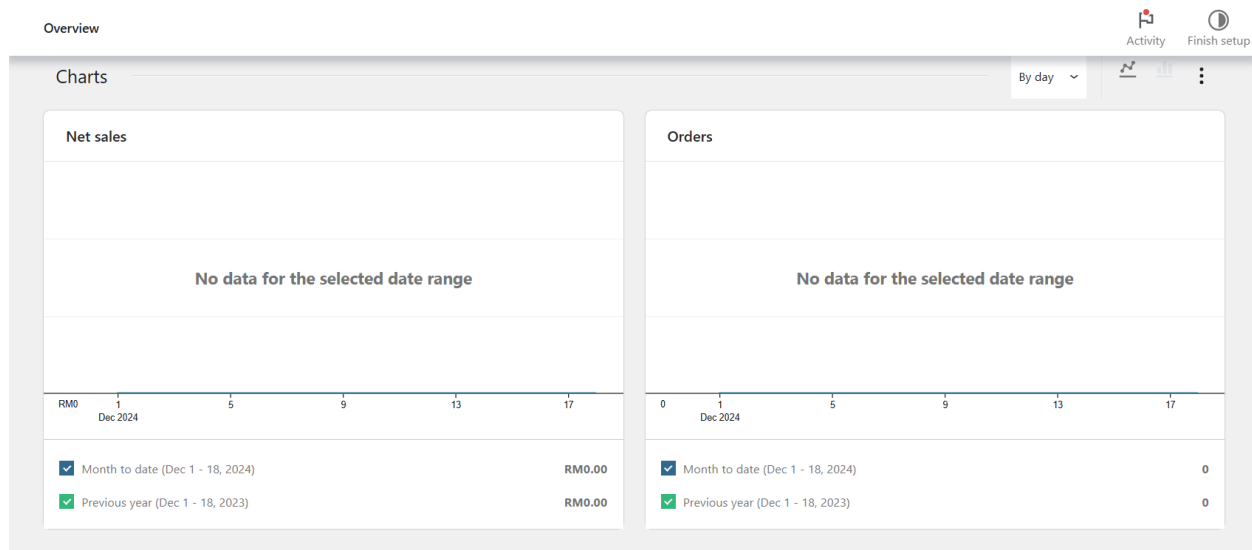


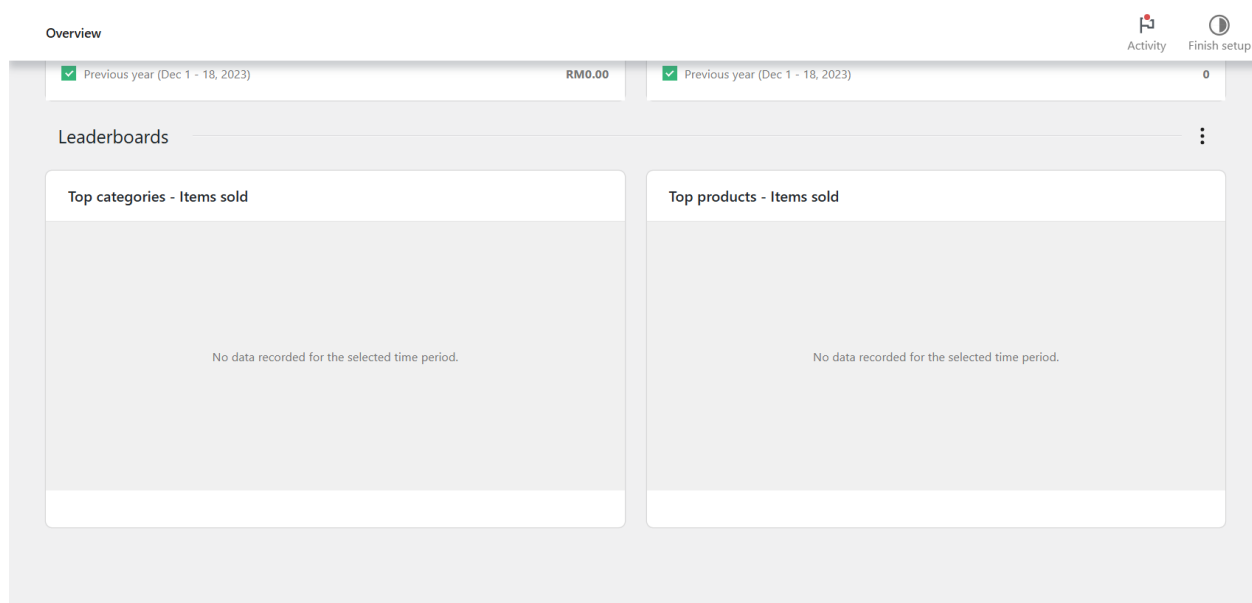
Figure above presents the detailed sales analysis for KFMS, offering a comprehensive view of key performance metrics. It includes the total sales, net sales, number of orders, products sold, and variations sold, providing users with a complete overview of the store's sales performance. The page also features a sales chart that visually represents sales trends over a specified period, making it easier for administrators to track fluctuations in sales and identify patterns. With this detailed analysis, users can gain valuable insights into the store's performance, aiding in decision-making and strategic planning. The design is intuitive, ensuring that all information is easily accessible and understandable.



The next figure displays the options for selecting the date and month to analyze sales. Users can choose specific date ranges or select particular months to view sales data, allowing for more granular analysis. This feature enables administrators to track sales performance over different time periods, helping to identify trends, seasonal fluctuations, and other key insights. The intuitive interface ensures that users can easily filter the data by date or month, making it simple to perform customized analyses and make informed decisions based on time-specific sales patterns.



The next figure illustrates the graph chart for sales, providing a visual representation of sales data over a specified period. The chart displays trends in total sales, highlighting peaks and troughs to help users quickly identify patterns in sales performance. By using dynamic axes, the chart allows users to track sales fluctuations over time, making it easier to spot seasonal trends, high-sales periods, and potential areas for improvement. This graphical representation enhances the user experience by presenting complex data in an easily interpretable format, empowering administrators to make informed decisions based on visual insights.



The next figure displays the leaderboard for top categories and top products sold, providing a ranking of the best-performing items. The leaderboard highlights the top categories based on sales volume, allowing administrators to quickly identify which product categories are driving the most revenue. Additionally, it showcases the top individual products sold, offering insights into which specific items are most popular among customers. This feature helps users pinpoint high-demand products and categories, enabling them to optimize inventory and marketing strategies accordingly. The leaderboard format ensures that these key insights are presented clearly and prominently for easy analysis.

Manage Cart – Aina Jaleena binti Jusop (CB21089)

- Usability

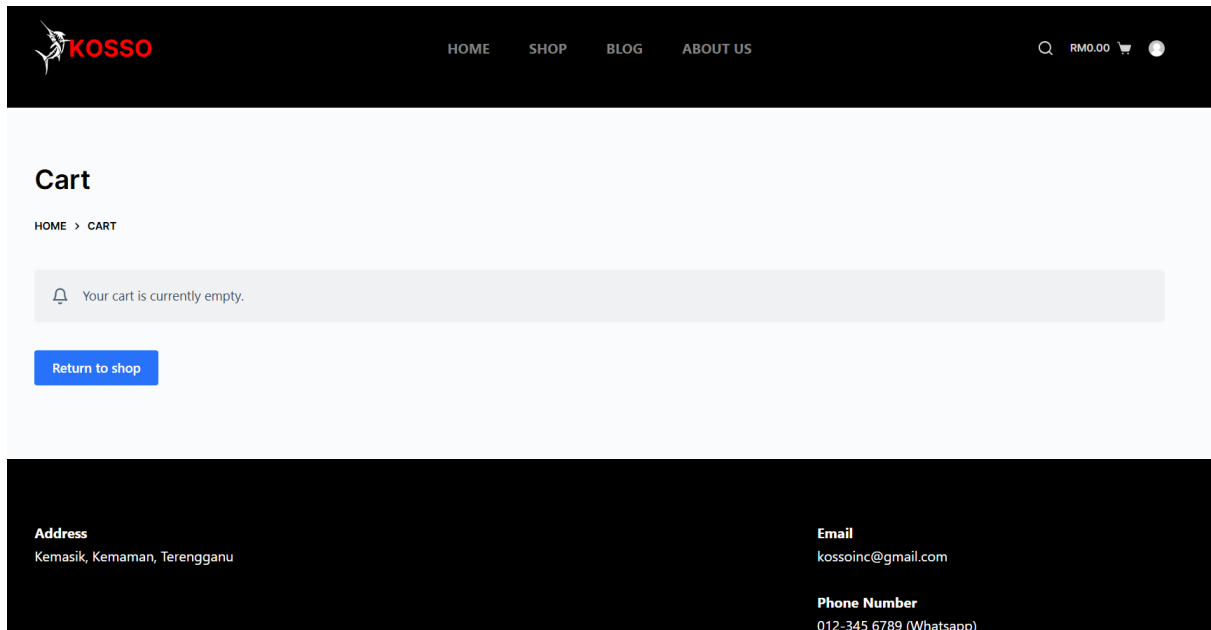


Figure above shows the Add to Cart Page for KFMS. The process is designed to be simple and user-friendly, ensuring users can easily view and manage their selected items. In this case, the cart is currently empty, as indicated by clear message, “Your cart is currently empty”. Next, a “Return to shop” button is displayed below the message, allowing users to seamlessly navigate back to the product page to continue shopping. The layout is minimalistic to avoid distractions focusing on smooth user experience. Once products are added to the cart, the page is expected to display the summary of the total amount of products. The simple and clean design ensures the users of KFMS can interact with the cart functionality without confusion.

Checkout

HOME > CHECKOUT

Billing details

First name *
Aina

Last name *
Jaleena

Company name (optional)

Country / Region *
Malaysia

Street address *
no 123 jalan tasik

Apartment, suite, unit, etc. (optional)

Town / City *
seremban

State / County *

Your order

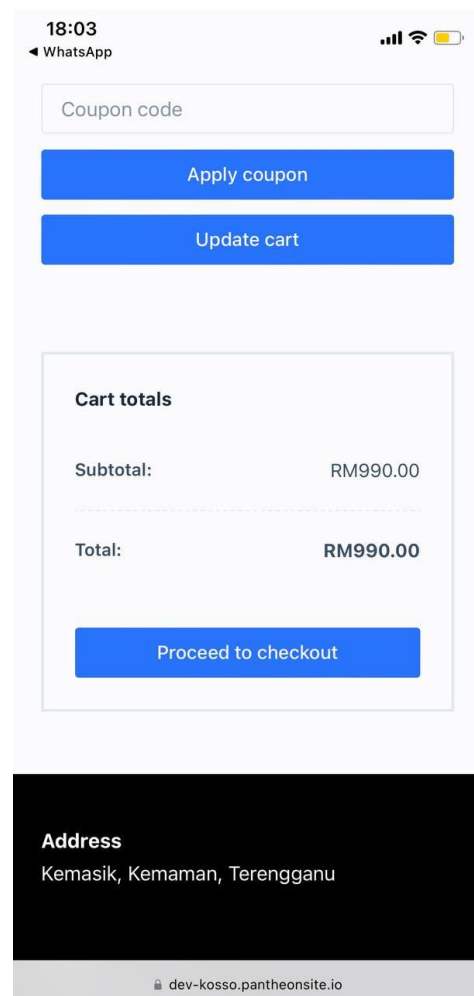
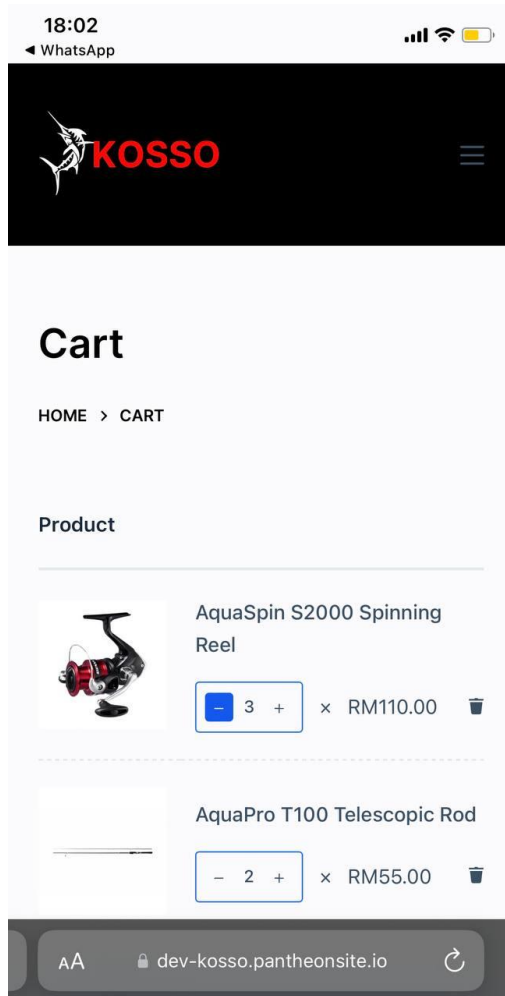
Product	Subtotal
AquaPro T100 Telescopic Rod x 1	RM55.00
AquaSpin S2000 Spinning Reel x 1	RM110.00
Subtotal	RM165.00
Total	RM165.00

Sorry, it seems that there are no available payment methods. Please contact us if you require assistance or wish to make alternate arrangements.

Your personal data will be used to process your order, support your experience.

The figure above shows the Checkout section where user needs to fill in their billing address before proceeding to check out the products from their cart. The interface is simple and easy to use, with clear labels for each field and a logical layout to guide users. Drop-down menus for selecting states or countries, help users enter their details quickly and accurately. Clear instructions and messages make the process smooth, while confirmation messages let users know their billing details were successfully submitted. This ensures a convenient and user-friendly checkout experience.

- Maintainability



The **Responsive Cart Design** helps make the system easy to maintain by ensuring the cart works well on different devices like **mobile**, tablet, and desktop. On smaller screens, like mobile phones, the cart adjusts by stacking elements vertically, so users can easily navigate and interact with the cart. On larger screens, such as tablets and desktops, the cart spreads out to use the extra space, showing products in a wider layout. This flexibility makes it easier to update the cart's design in the future since the same design can adjust to different screen sizes automatically. By keeping the cart simple and consistent across devices, it becomes easier to fix problems, add new features, and keep everything working smoothly for all users.

- Reliability

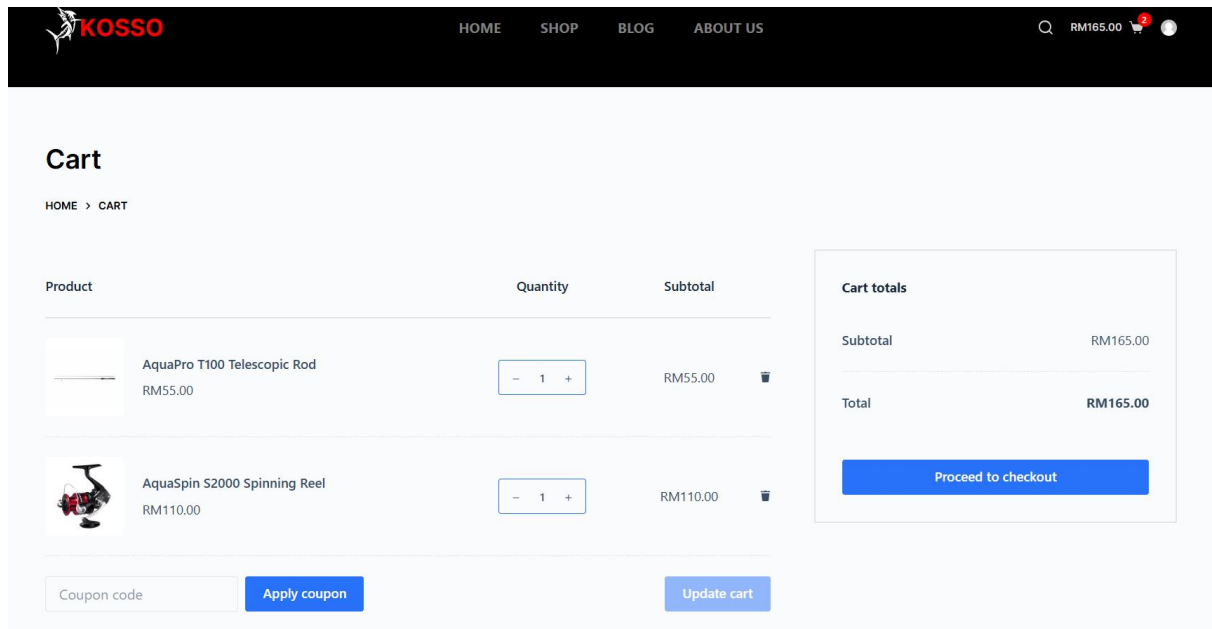


Figure above displays the Cart Page for KFMS that is designed with reliability of users. The page ensures that users can **rely** on the total amount to be **correctly calculated** based on the selected product quantity and any applied discounts or updates. Key information such as the **product name, price, quantity, and subtotal** are displayed clearly, and the system automatically updates the **cart total** whenever changes are made. This reliable calculation process provides users with confidence that the total amount shown is accurate and free from errors.



HOME SHOP BLOG ABOUT US

 RM165.00 

Cart

HOME > CART

Product	Quantity	Subtotal
 AquaPro T100 Telescopic Rod RM55.00	- 2 +	RM55.00 
 AquaSpin S2000 Spinning Reel RM110.00	- 3 +	RM110.00 

Apply coupon Update cart

Cart totals

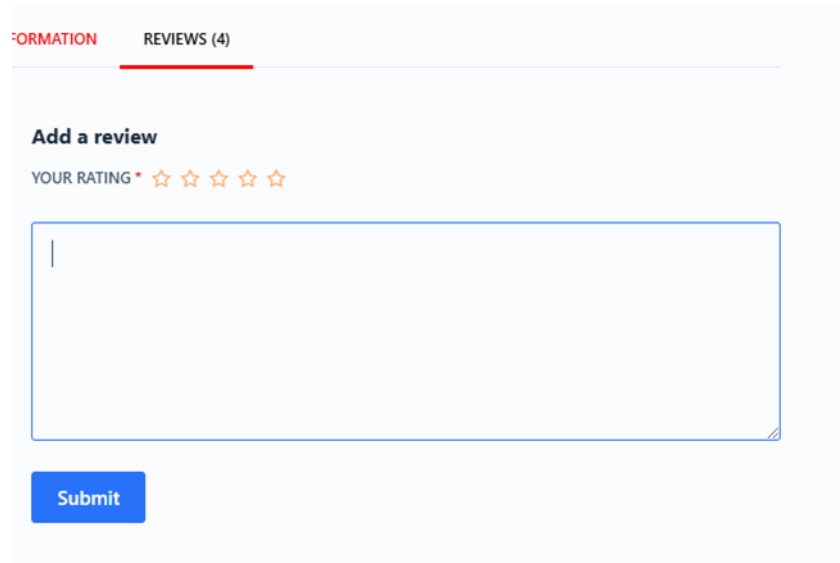
Subtotal	RM165.00
Total	RM165.00

Proceed to checkout

The "**Update Cart**" button is available when user adds the quantity of products to make sure any changes, like updating the quantity or applying a coupon, are correctly shown in the cart. The "**Proceed to Checkout**" button provides an easier way for users to continue with their purchase.

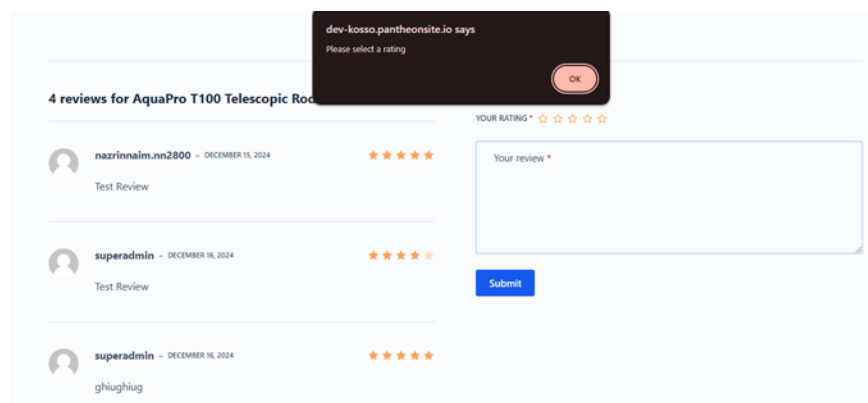
Manage Review - Mohamad Hatta Rahman bin Borhanuddin (CB21080)

Usability



The screenshot shows a web interface for adding a review. At the top, there are two tabs: 'INFORMATION' and 'REVIEWS (4)', with 'REVIEWS (4)' being the active tab. Below the tabs, the heading 'Add a review' is displayed. Underneath, there is a section for 'YOUR RATING' with five orange star icons. Below the rating stars is a large, empty text input box for the review content. At the bottom of the form is a blue 'Submit' button.

Figure above illustrates the Add a Review page for KFMS. The design is intentionally streamlined and user-centric, ensuring that customers can effortlessly share their feedback on the products they've purchased. At the moment, the review container is vacant, providing users with a clear space to interact. Users can easily click on the star button to assign a rating to the product, followed by entering a detailed review in the designated text box. The interface is designed with clarity and simplicity in mind, allowing users to provide their feedback with minimal effort, enhancing the overall user experience and engagement on the platform.



The screenshot displays a product review page for 'AquaPro T100 Telescopic Rod'. At the top, there is a notification banner from 'dev-kosso.pantheonsite.io' asking the user to select a rating, with an 'OK' button. Below the notification, the product name is followed by '4 reviews for AquaPro T100 Telescopic Rod'. The reviews section shows three entries, each with a user profile picture, name, date, and rating. The first review is by 'nazrinnaim.nn2800' on December 15, 2024, with a 5-star rating and the text 'Test Review'. The second review is by 'superadmin' on December 16, 2024, with a 5-star rating and the text 'Test Review'. The third review is also by 'superadmin' on December 16, 2024, with a 5-star rating and the text 'ghiughug'. To the right of the reviews, there is a 'YOUR RATING' section with five orange star icons. Below the stars is a large, empty text input box for the user's review. At the bottom of this section is a blue 'Submit' button.

Figure above demonstrates the prompt that appears when a user attempts to submit a review without selecting a star rating or entering any text. The design is simple yet clear, providing an immediate reminder to the user that both a rating and a written review are required before submission. This prompt is strategically placed to ensure that the user is aware of the necessary actions, thus preventing incomplete submissions and enhancing the overall functionality of the review process. The intention behind this design is to encourage users to provide thorough and meaningful feedback, ensuring a more complete review experience for both customers and store administrators.

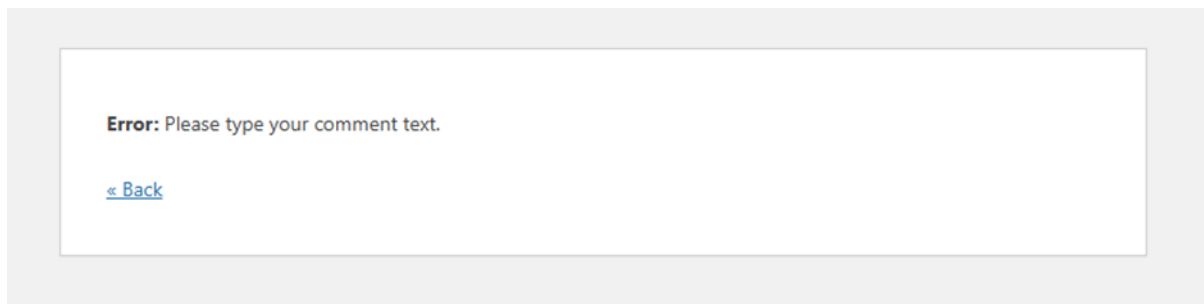
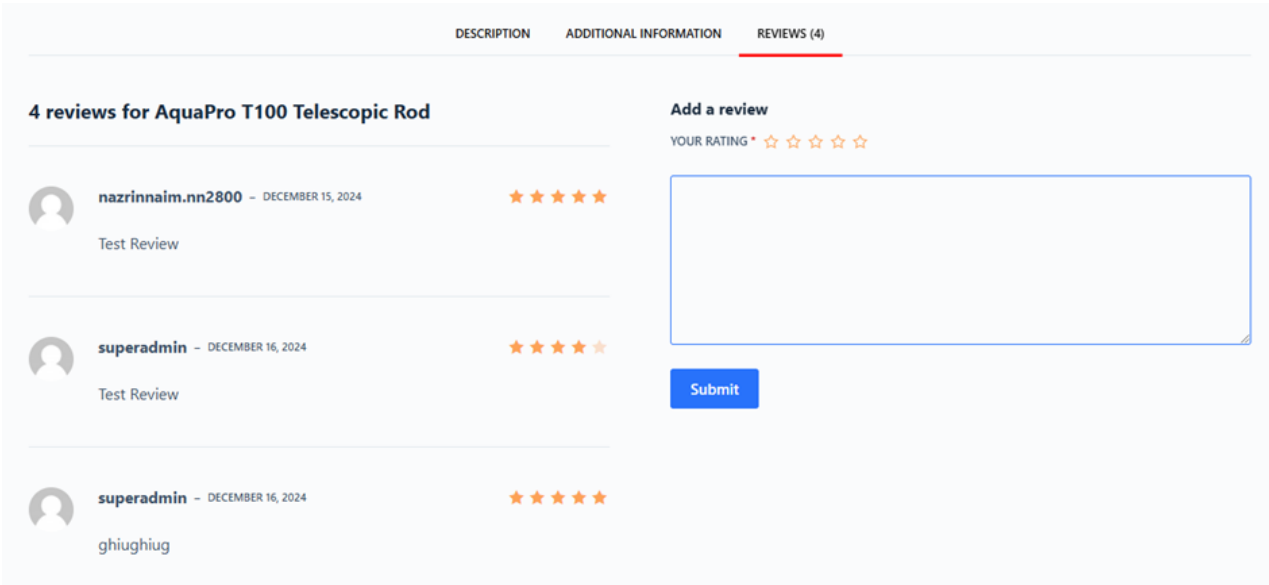
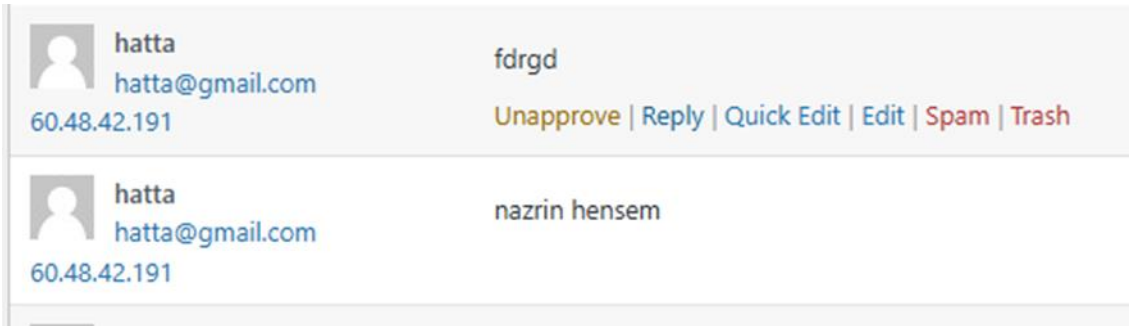


Figure above illustrates the error message displayed when a user attempts to submit a review without selecting a star rating or entering text. If the user fills in one of the required fields but leaves the other empty such as submitting a review with only text or a star rating, the system redirects them to a dedicated error page. This page clearly communicates the issue, informing the user that both a star rating and written feedback are necessary for the review to be submitted successfully. The error message is designed to be direct yet supportive, guiding the user to correct the omission and complete their review submission properly. By ensuring that all necessary fields are completed, this approach aims to maintain the quality and consistency of the reviews posted on the platform.

Reliability



The figure above highlights the reliability of the system, demonstrating how it updates seamlessly as soon as the user clicks the "Submit" button for their review. This instant response ensures that the submitted review is processed and displayed without delay, providing users with immediate confirmation of their action. The rapid update reflects the system's efficiency and stability, fostering a smooth and dependable user experience. By maintaining this level of responsiveness, the platform enhances user trust and satisfaction, showcasing its commitment to reliable and high-performance functionality.



The figure above illustrates the reliability of the system by showcasing its capability to handle simultaneous operations efficiently. While a user submits a review, the admin is granted full access to manage existing reviews in real-time. This includes the ability to read, edit, update,

delete, and unapprove reviews seamlessly. The system ensures that these administrative actions are performed without any interruptions or conflicts, even as new reviews are being added by users. This robust functionality demonstrates the system's ability to maintain stability and reliability under concurrent usage, ensuring a smooth and dependable experience for both users and administrators.

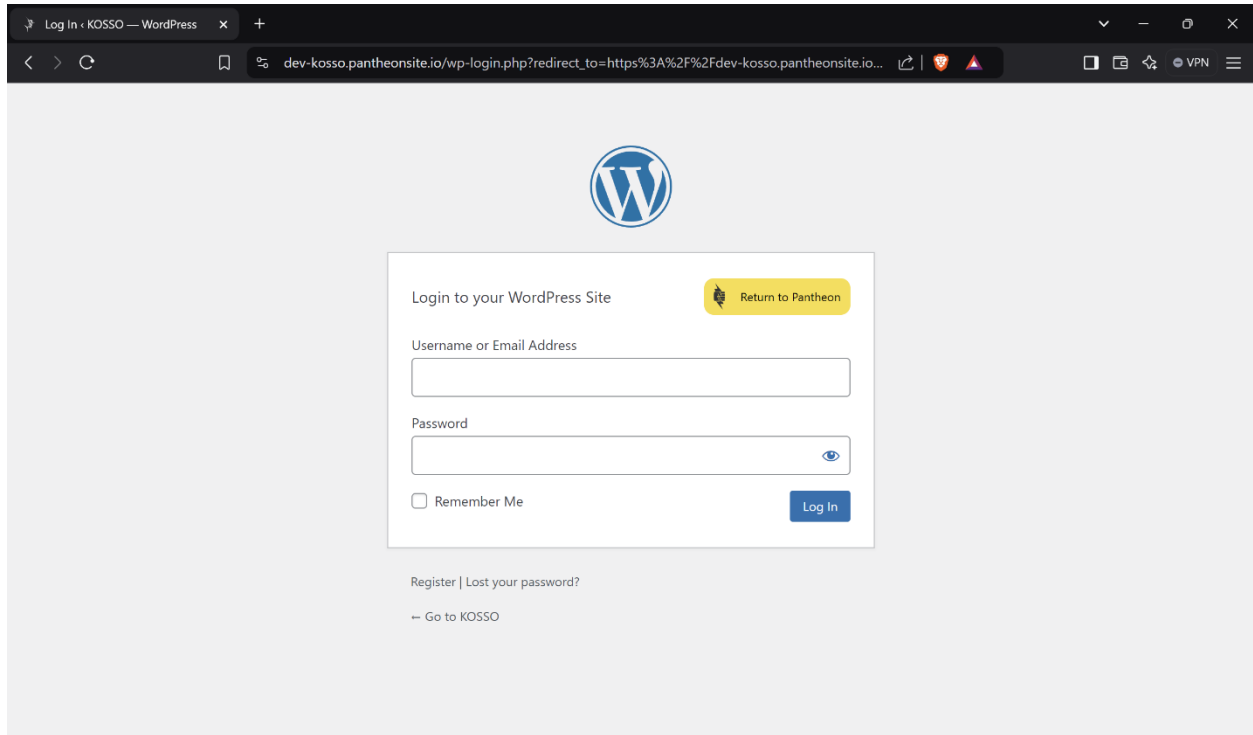
Scalability

Reviews				
Add Comment				
	hatta hatta@gmail.com 60.48.42.191	fdrgd		
	hatta hatta@gmail.com 60.48.42.191	nazrin hensem		
	superadmin dev-kosso.pantheonsite.io superadmin@gmail.com 127.0.0.1	ghiughiug		
	superadmin dev-kosso.pantheonsite.io superadmin@gmail.com 127.0.0.1	Test Review		
	nazrinnaim.nn2800 nazrinnaim.nn2800@gmail.co m 127.0.0.1	Test Review		

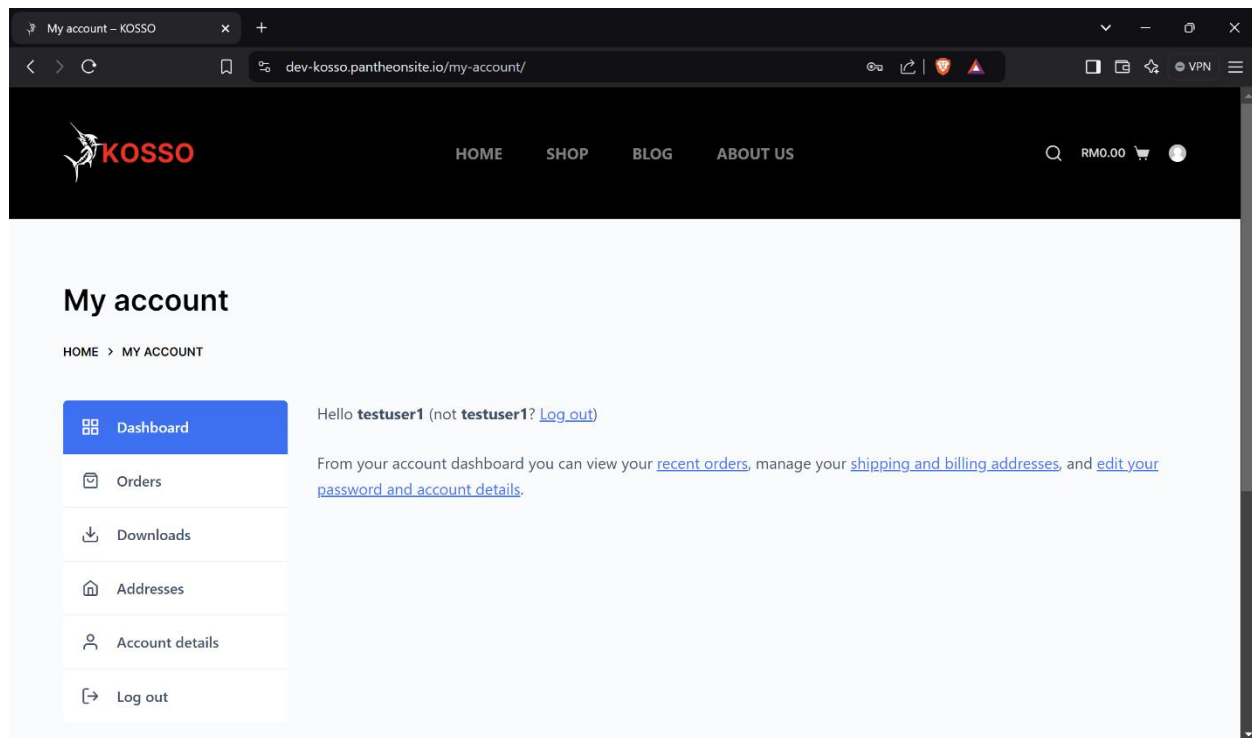
The figure above highlights the scalability of the system by displaying the reviews stored within it. The system is designed to handle a growing volume of reviews efficiently, ensuring that as more users engage and submit feedback, the performance remains consistent. This scalability ensures that the system can accommodate an increasing number of reviews without compromising load times or functionality. The ability to store, organize, and retrieve reviews seamlessly, regardless of the system's size or user activity, demonstrates its preparedness for future growth and higher demand.

Manage Stock - Nazrin Naim bin Ahmad Mahari (CB21072)

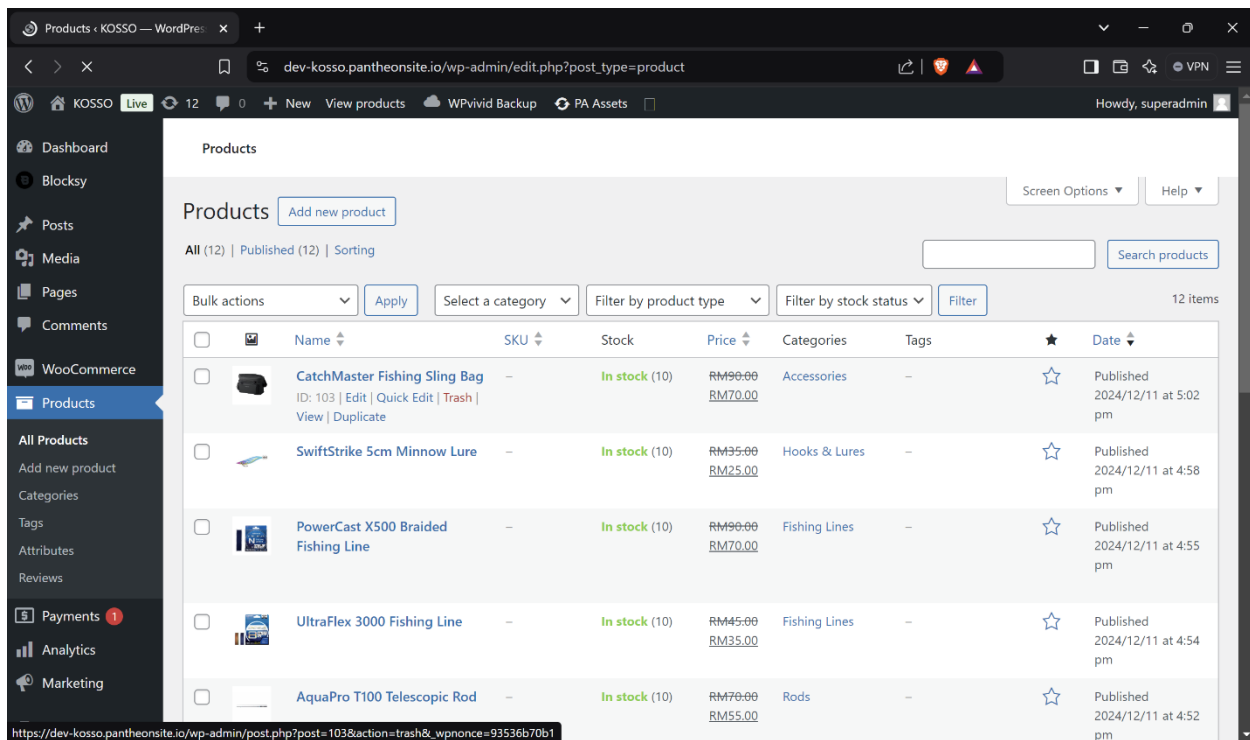
- Security



The figure above shows the main login page, which is primarily used by admins as it redirects them to the admin dashboard. The login page ensures security through role-based access control (RBAC) by redirecting users based on their roles after authentication. When a user logs in, their credentials are verified, and their role is identified. Admins are redirected to the admin dashboard, while customers are sent to the My Account page. This separation ensures that only authorized users can access the admin dashboard, preventing unauthorized access.



The figure above shows the My Account page, which is where users or customers are redirected if they log in through the same login page used by admins. While the login page is mainly designed for admins to access the admin dashboard, if a user or customer logs in, they will be redirected to the My Account page instead. This ensures users are directed to the correct page based on their role.



The figure above shows the Products page from the admin dashboard, which is exclusively accessible to admins. This page allows admins to manage product details, such as adding, updating, or deleting items. It is restricted to prevent unauthorized users, like customers, from accessing or modifying product information, ensuring secure and proper management of the store's inventory.

- Usability

The image displays two screenshots of the WooCommerce 'Add New Product' interface, illustrating its usability for entering product details.

Top Screenshot: Initial Product Form

- Header:** 'Add New' button and user profile 'Howdy, superadmin'.
- Left Sidebar:** Navigation menu with options like Dashboard, Blocks, Posts, Media, Pages, Comments, WooCommerce, Products, All Products, Add new product, Categories, Tags, Attributes, Reviews, Payments, Analytics, Marketing, Elementor, and Templates.
- Main Content Area:**
 - Add new product:** A large text input field for the product name.
 - Product description:** A rich text editor with a toolbar (Paragraph, Bold, Italic, List, Link, etc.) and a 'Visual' tab.
 - Product image:** A field with a 'Set product image' button.
 - Product gallery:** A field with an 'Add product gallery images' button.
 - Product categories:** A field for selecting categories.
- Right Sidebar:**
 - Publish:** Includes 'Save Draft', 'Preview', 'Status: Draft', 'Visibility: Public', 'Publish immediately', and 'Catalog visibility: Shop and search results'.

Bottom Screenshot: Product Data Section

- Product data:** A dropdown menu set to 'Simple product' with checkboxes for 'Virtual' and 'Downloadable'.
- General Tab:**
 - Regular price (RM):** A text input field.
 - Sale price (RM):** A text input field with a 'Schedule' link below it.
- Product short description:** A rich text editor with a toolbar and a 'Visual' tab.
- Right Sidebar:**
 - Product tags:** A field with an 'Add' button and a link to 'Choose from the most used tags'.
 - LiteSpeed Options:** Includes checkboxes for 'Disable Cache', 'Disable Image Lazyload', and 'Disable VPI'.

Figure above shows The Manage Stock module is designed with a focus on Usability to make it easy for admins to input product details efficiently and accurately. The user interface includes clear, labelled fields for essential product information such as product name, price, and description. A preview option allows admins to review the details before saving, ensuring accuracy. The layout is intuitive, with a clean design that organizes input fields logically, helping admins complete tasks quickly without confusion. By prioritizing usability, the module ensures that managing stock is straightforward and user-friendly, even for those with minimal technical experience.

- Reliability

The top screenshot shows the 'Products' page in the WooCommerce admin. It displays a list of 12 items, including 'CatchMaster Fishing Sling Bag', 'SwiftStrike 5cm Minnow Lure', 'PowerCast X500 Braided Fishing Line', 'UltraFlex 3000 Fishing Line', 'AquaPro T100 Telescopic Rod', and 'PowerCast C300 Casting Rod'. Each product entry includes a checkbox, a product image, the product name, SKU, stock status (e.g., 'In stock (10)'), price (e.g., 'RM90.00'), category (e.g., 'Accessories'), tags, and a star icon. The bottom screenshot shows a similar view for fishing reels, including 'StormFlex S220 Spinning Rod', 'AquaSpin S2000 Spinning Reel', 'StreamForce C300 Casting Reel', 'PowerCast Z400 Casting Reel', 'ProCast X500 Casting Reel', and 'SpinMax 3000 Spinning Reel'. Both interfaces feature a sidebar with navigation options like 'Dashboard', 'Blocksy', 'Posts', 'Media', 'Pages', 'Comments', 'WooCommerce', 'Products', 'All Products', 'Add new product', 'Categories', 'Tags', 'Attributes', 'Reviews', 'Payments', 'Analytics', 'Marketing', 'Elementor', and 'Templates'.

Figure above shows a consistent experience for admins to view and edit stock details. The stock listing page displays all products in an organized table format, showing key details like product name, category, price, quantity, and last updated date. The interface is designed to load quickly and handle large datasets without crashing or lagging, ensuring a smooth experience even with frequent updates. Admins can trust the system to display accurate and up-to-date information. When editing stock details, the system uses safeguards like confirmation prompts and real-time updates to prevent accidental changes or data loss. This design ensures that the module consistently supports accurate and secure stock management for the admin.

5. Testing/Validation

5.1 Introduction

Testing and validation are important steps in the software development process to make sure the KOSSO Fishing Store system works properly and meets user expectations. These steps help identify and fix issues like bugs, performance problems, and security risks before the system goes live. This ensures the system is reliable and easy to use for everyone involved. Both automated and manual testing methods are used to check how well the system performs in different situations. Tests like unit testing, integration testing, system testing, and user acceptance testing are carried out to make sure all features work as expected and provide a smooth experience for users. Thorough testing not only prepares the system for real-world use but also gives users and stakeholders confidence in its performance. By delivering a well-tested system, the KOSSO Fishing Store can provide a reliable and user-friendly experience, helping to ensure its success and keep customers satisfied.

5.2 Test Cases

Manage User - Abdul Razzaq bin Abdul Aziz (CB22056)

- Confidentiality

No	Test Steps	Expected Results	Actual Result	
			Pass	Fail
1	Attempt to access sensitive user data without logging in.	Access is denied, and the system redirects to the login page or displays a message stating that authentication is required.	✓	
2	Log in with a role that has restricted access (e.g., a non-admin role) and attempt to view sensitive user details.	The system denies access, displaying a message that the user lacks the necessary permissions.	✓	
3	Log in with an admin role and access sensitive user details.	The system grants access, displaying the relevant data as per the user's permissions.	✓	
4	Perform a brute-force attack by entering multiple incorrect login attempts.	The system locks the account after a predefined number of failed attempts and logs the activity for administrative review.		X
5	Attempt unauthorized actions such as accessing admin-only features or manipulating user roles.	The system logs the unauthorized attempt with details such as the user's IP address,		X

		timestamp, and attempted action, and sends an alert to the administrator.		
6	Simulate simultaneous logins from two different devices using the same user credentials	The system either logs out the first session or alerts the user of the concurrent access, depending on the configured policy.		X
7	Submit a user form with invalid data, such as a username with special characters or an email without a valid format.	The system prevents submission, highlighting the invalid fields with specific error messages (e.g., "Invalid email format").	✓	

- Integrity

No	Test Steps	Expected Results	Actual Result	
			Pass	Fail
1	Attempt to submit a form with missing required fields, such as leaving the "email" and "username" fields blank	The system prevents form submission and displays an error message, such as "Email is required" or "Username cannot be empty."	✓	
2	Enter an invalid email format (e.g., "userexample.com") and attempt to save the user profile.	The system rejects the input, highlighting the email field with a red border and displaying an error message like "Please enter a valid email address."	✓	
3	Submit a form with multiple invalid fields, such as an invalid email and a missing address.	The system displays a summary of errors with actionable messages for each invalid field and highlights the fields with red borders	✓	
4	Enter valid inputs in all fields and submit the form.	The system accepts the inputs, saves the data, and displays a success message like "User profile updated successfully."	✓	
5	Enter a username shorter than the required minimum character count (e.g., "usr" when the minimum is 5 characters).	The system rejects the input, highlights the username field, and displays an error message like "Username must be at least 5 characters long."	✓	

6	Attempt to save a profile with a duplicate email already used by another user.	The system prevents submission, highlights the email field, and displays the message "This email is already in use. Please enter a different email."	✓	
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- Reliability

No	Test Steps	Expected Results	Actual Result	
			Pass	Fail
1	Perform a search for a specific user using a common field (e.g., email or username)	The system quickly locates the user profile and displays the results, confirming that database queries are optimized for performance.	✓	
2	Trigger a heavy task like generating a detailed activity report for multiple users while performing other regular tasks, such as adding or updating user profiles.	The system offloads the report generation to a background process, remains responsive for regular tasks, and notifies the user when the report is ready.		X
3	Attempt to load a list of user profiles with sorting and filtering options (e.g., by role or registration date).	Sorting and filtering operations are performed quickly without noticeable lag, even with a large dataset.	✓	

4	Introduce multiple simultaneous requests to view or edit user profiles to simulate high traffic.	The system handles concurrent requests efficiently, maintaining a consistent response time without errors or crashes.	✓	
5	Simulate a high-traffic scenario where multiple users access the manage user page simultaneously.	The system maintains uptime and continues functioning without degradation, supported by load balancing and caching mechanisms.		X
6	Simulate a server failure during operations on the manage user page.	The failover system activates, redirecting the workload to backup servers, ensuring uninterrupted access to the page.	✓	
7	Clear the cache and attempt to reload the manage user page under normal traffic conditions.	The system quickly regenerates the cache and serves the page without noticeable delay, ensuring minimal disruption.	✓	

Manage Cart – Aina Jaleena binti Jusop (CB21089)

- Usability

No	Test Steps	Expected Results	Actual Result	
			Pass	Fail
1	Add a product to the cart by clicking on the “Add to Cart” button.	The product is successfully added to the cart, and a confirmation message appears (“View Cart”)	✓	
2	View the cart page after adding products	The cart page displays a clear summary of the added products, including product name, quantity, price, and total cost.	✓	
3	Increase the product quantity using the quantity selector (use "+" or typing a number).	The quantity updates correctly, and the total price adjusts automatically without delays.	✓	
4	Decrease the product quantity using the quantity selector or "-" button.	The product quantity decreases appropriately, and the total price updates instantly. If quantity is reduced to zero, an option to remove the product appears.	✓	
5	Remove a product from the cart using the “Remove” button.	The product is removed instantly, and the cart updates with the correct total. A message appears confirming the product	✓	

		removal (e.g., “Product removed from cart”).		
6	Verify the cart for multiple products.	The cart displays all products in an organized list with clear details (product image, name, price, quantity, and subtotal for each item).	✓	
7	Verify the “Proceed to Checkout” button functionality.	Clicking the button navigates the user to the checkout page without errors.	✓	
8	Validate the cart for empty-cart scenarios.	If no products are in the cart, the page displays a clear message (“Your cart is empty”) and suggests continuing to shop.	✓	
9	Test the “Update Cart” button functionality after modifying quantities.	Clicking “Update Cart” reflects all changes (e.g., quantity updates and recalculated totals).	✓	
10	Verify error recovery if the page reloads unexpectedly.	The cart retains all added products and details even after refreshing the page.	✓	
11	Test the loading time of the cart page.	The cart page loads within a reasonable time (under 3 seconds), even when multiple products are added.	✓	

Quality Metrics

Usability Testing -Task Completion Rate

$$\text{Task Completion Rate} = \frac{\text{Number of Successful Tasks}}{\text{Total Number of Attempts}} \times 100$$

$$= \frac{11}{11} \times 100$$

$$= 100 \%$$

The **Task Completion Rate** is 100%. This indicates that users were able to successfully complete all assigned tasks during usability testing for Manage Cart module.

- Maintainability

No	Test Steps	Expected Results	Actual Result	
			Pass	Fail
1	Simulate adding a new feature to the cart page (“Apply Discount Code” button).	The new feature integrates seamlessly with minimal changes to existing code and no adverse effects.	✓	
2	Test the time required to fix a minor bug in the cart (e.g., incorrect total price calculation)	The bug is resolved quickly due to clear code structure, well-documented logic, and minimal interdependencies.		✗
3	Check how quickly a feature can be rolled back (revert “Cart Auto-Update Quantity” feature).	Features can be rolled back or reverted with minimal effort and no significant disruptions.	✓	
4	Add logging for cart actions (add, update, remove) and check for accuracy and readability.	Logs are correctly generated, capturing accurate information about cart actions for debugging purposes.		✗
5	Test the process of deploying a cart-related update (updating the “Remove Product” function).	The update deploys smoothly with no adverse impact on other functionalities of the cart page.	✓	

Quality Metrics

Load Testing – Average Response Time

Description: Measures how long the cart takes to respond under load, this helps determine if the cart can handle multiple users without delays.

Formula: Average Response Time = $\sum \frac{\text{Response Time For Each Request}}{\text{Total Number of Requests}}$

Request 1: **2.0 seconds**

Request 2: **2.5 seconds**

Request 3: **1.8 seconds**

Request 4: **2.2 seconds**

1. Sum of Response Times:

2.0 seconds + 2.5 seconds + 1.8 seconds + 2.2 seconds = 8.5 seconds

2. Total Number of Requests:

4 Requests

3. Average Response Time = $\sum \frac{\text{Response Time For Each Request}}{\text{Total Number of Requests}}$

$$= \frac{8.5}{4}$$

= 2.1 seconds

Therefore, on average, the Manage Cart Module responds within 2.1 seconds under load.

- Reliability

No	Test Steps	Expected Results	Actual Result	
			Pass	Fail
1	Add multiple products (5 items) and verify all are listed in the cart.	All 5 products appear correctly in the cart with the correct quantity and details.	✓	
2	Refresh the page after adding products to the cart.	The cart retains the added products and quantity without loss of data.	✓	
3	Perform multiple cart actions (add, remove, update) repeatedly (10 times).	The cart remains stable, and all actions (add, remove, update) work correctly without system crashes.	✓	
4	Add a large number of products to the cart	The system handles a large cart size efficiently without errors or performance issues.		✗
5	Log out and log back in after adding products to the cart.	The cart retains the previously added products and their quantities after logging back in.	✓	
6	Perform checkout from the cart with correct product and quantity details.	The system successfully processes the checkout, and no errors occur during the order submission.	✓	
7	Verify cart data is retained after closing and reopening the browser.	Verify cart data is retained after closing and reopening the browser.	✓	

Error Lodging – Error Rate

Description: Calculate the frequency of errors, a lower error rate indicates better reliability of the Manage Cart Module.

Formula: Error Rate = $\frac{\text{Total Number of Errors}}{\text{Total Number of Transactions}} \times 100$

$$= \frac{1}{100} \times 100$$

$$= 1\%$$

A **lower error rate** that is close to 0% reflects higher reliability of the Manage Cart module. Therefore, the Error Rate for Error Lodging is **1%** that indicates the higher reliability of the Manage Cart Module since it is closer to 0%.

Manage Review - Mohamad Hatta Rahman bin Borhanuddin (CB21080)

- Scalability

No	Test Steps	Expected Results	Actual Result	
			Pass	Fail
1	High traffic scenario by having 100+ users submitting text-only reviews simultaneously.	The system should automatically scale resources (e.g., servers, processing power) to handle the increased load, ensuring that all reviews are submitted without delays or timeouts.	✓	
2	Submit multiple reviews from different users at the same time.	The system should efficiently handle multiple submissions without performance degradation or errors, processing all reviews within a reasonable timeframe.	✓	
3	Perform multiple actions like review submissions, updates, and deletions across different user accounts simultaneously.	The system should handle concurrent actions effectively, ensuring that all submissions, updates, and deletions are processed accurately and promptly without significant delays.	✓	
4	Test the system under varying traffic conditions (low, moderate, and high loads) using only text-based reviews.	Under low to moderate traffic, the system should perform normally with no performance issues. During high traffic, the system should scale resources as necessary to maintain responsiveness and reliability.	✓	
5	Trigger a surge in traffic (e.g., 500+ concurrent users submitting text-only reviews).	The system should scale automatically to accommodate the surge in traffic. It should continue to process reviews without delays, and there should be no system downtime or major slowdowns.	✓	

Quality Metrics

Throughput (Transaction Rate)

Throughput =

$$\text{Throughput} = \frac{\text{Number of Reviews Submitted}}{\text{Time Taken}}$$

$$\text{Throughput} = \frac{5}{2} = 2.5 \text{ reviews per second}$$

$$= 2.5 \text{ reviews per second}$$

The **Throughput Transaction Rate** is 2.5 reviews per seconds. This indicates that scalable since the time taken is short during scalability testing for Manage Review module.

- **Usability**

No	Test Steps	Expected Results	Actual Result	
			Pass	Fail
1	The user submits a text review with a star rating and clicks “Submit Review.”	The system saves the review and shows the review on the product page.	✓	
2	The user navigates to the product page to view existing reviews.	The reviews are displayed in chronological order with clear labels and formatting for text and star ratings.	✓	
3	The admin clicks on a “quick edit” “spam” or “trash” to edit or delete it.	The admin is directed to a form or action confirmation page to edit or delete the review.	✓	
4	The user submits a review without entering any text.	The system prompts the user to complete the required fields before submission.	✓	
5	The user submits a review without selecting a star rating.	The system prompts the user to complete the required fields before submission.	✓	
6	The user submits a review without entering both text and selecting a star rating.	The system prompts the user to complete the required fields before submission.	✓	
7	The review submission page is responsive on both desktop and mobile devices.	The page adjusts seamlessly to different screen sizes, ensuring a smooth user experience.	✓	

Quality Metrics

Usability Testing -Task Completion Rate

TCR =

$$\frac{\text{Number of successful task completions}}{\text{Total tasks attempted}} \times 100$$

$$= \frac{7}{7} \times 100$$

$$= 100\%$$

The **Task Completion Rate** is 100%. This indicates that users were able to successfully complete all assigned tasks during usability testing for Manage Review module.

User Satisfaction Score (USS):

Calculates average user satisfaction based on surveys (e.g., 1-5 rating scale).

USS =

$$\frac{\text{Total satisfaction ratings given by users}}{\text{Total users surveyed}}$$

$$= \frac{19}{4}$$

$$= \frac{4.79}{5} \times 100$$

$$= 95.8\%$$

The **Task Completion Rate** is 95.8%. This indicates that the score satisfaction of the product during usability testing for Manage Cart module.

- Reliability

No	Test Steps	Expected Results	Actual Result	
			Pass	Fail
1	Multiple users submit reviews simultaneously during a high-traffic event.	The system processes all submissions without delays or errors, and reviews appear on the product pages.	✓	
2	The system continues to allow review submissions while performing a database backup.	Users can still submit reviews without any interruptions or data loss.		✗
3	The admin edits or deletes a review while users are submitting new reviews.	The system handles all actions concurrently without errors or delays.	✓	
4	The review feature remains functional after the system undergoes minor updates or patches.	Users and admins can submit, view, and manage reviews without encountering any disruptions.		✗
5	The system retries saving a review if the initial attempt fails due to a temporary server issue.	The review is saved successfully after the retry, with no user intervention required.	✓	

Quality Metrics

Error Rate:

Tracks how often users experience issues like failed reviews or slow response times.

Error Rate =

$$\frac{\text{Number of errors}}{\text{Total user interactions}} \times 100$$

$$= \frac{1}{100} \times 100$$

=1%

A **lower error rate** that is close to 0% reflects higher reliability of the Manage Review module. Therefore, the Error Rate for Error Lodging is **1%** that indicates the higher reliability of the Manage Review Module since it is closer to 0%.

Manage Stock - Nazrin Naim bin Ahmad Mahari (CB21072)

Security

No	Test Steps	Expected Results	Actual Result	
			Pass	Fail
1	The administrator enters valid credentials (username and password) and accesses the Manage Stock module.	The system grants access to the module and displays stock details.	✓	
2	The administrator enters invalid credentials and tries to access the Manage Stock module.	The system denies access and displays an error message like "Invalid login credentials."	✓	
3	A non-administrator user attempts to access the Manage Stock module via a direct URL.	The system redirects the user to the login page or My Account page if account already registered.	✓	
4	The administrator tries to access the Manage Stock module after their session has expired.	The system redirects the administrator to the login page and prevents access to the module.	✓	
5	A brute-force attack is simulated, with repeated incorrect login attempts to access the Manage Stock module.	The system locks the account after a set number of failed attempts and notifies the administrator via email.		X
6	The administrator enters valid credentials (username and password) and accesses the Manage Stock module.	The system grants access to the module and displays stock details.	✓	

Quality Metrics

Security Metric:

$$\text{Security Metric} = \frac{\text{Number of Vulnerabilities Found}}{\text{Total Number of Vulnerabilities Tested}} \times 100$$

$$\text{Security Metric} = \frac{4}{10} \times 100 = 40\% \text{ (Security Coverage)}$$

= 40%

The **Security Metric** is 40% for the coverage. This indicates that the system is secure for Security testing for Manage Stock module.

Usability

No	Test Steps	Expected Results	Actual Result	
			Pass	Fail
1	The administrator clicks the "Add Product" button without filling in mandatory fields (e.g., product name or price).	The system highlights the missing fields and displays an error message such as "Product name is required."	✓	
2	The administrator inputs a negative value for stock quantity when adding a product.	The system displays an error message like "Stock quantity cannot be negative" and prevents saving.	✓	
3	The administrator updates a product's information, and the system provides clear success feedback (e.g., "Product updated successfully").	The system updates the product details and displays a confirmation message.	✓	
4	The administrator attempts to search for a product using a partial keyword (e.g., typing "fish" instead of "fishing rod").	The system provides a list of matching products based on the partial keyword.	✓	
5	The administrator uses the module on a mobile device or a smaller screen.	The module adapts to the screen size, ensuring all functionalities are accessible and buttons are easy to click.		X
6	The system times out due to inactivity, and the administrator attempts to continue adding or updating stock.	The system displays a session timeout notification and redirects to the login page, preserving unsaved changes if possible.		X

Quality Metrics

Task Completion Rate:

$$\text{Task Completion Rate} = \frac{\text{Number of Tasks Completed Successfully}}{\text{Total Number of Tasks}} \times 100$$

$$\text{Task Completion Rate} = \frac{8}{10} \times 100 = 80\%$$

= 80%

The **Task Completion Rate is 80%**. This indicates that the system is Usable for Usability testing for Manage Stock module.

- Reliability

No	Test Steps	Expected Results	Actual Result	
			Pass	Fail
1	The administrator adjusts stock levels for a product, and another administrator views the same product details simultaneously.	The system displays the updated stock levels in real-time for both administrators.	✓	
2	Multiple administrators update stock information for different products simultaneously.	The system handles the updates without crashing or overwriting data incorrectly.	✓	
3	The system experiences a temporary network failure while updating a product's stock.	The system retries the update or notifies the administrator of the failure without losing data integrity.		X
4	The system experiences a sudden power outage or unexpected shutdown while the administrator is updating product information.	The system ensures data integrity by saving the last stable state or displaying a message to recheck the updated details.	✓	
5	The module processes a bulk update of stock for 100+ products simultaneously.	The system completes the bulk update successfully without errors or delays.		X

Quality Metrics

Failure Rate:

$$\text{Failure Rate} = \frac{\text{Number of Failures}}{\text{Total Operating Time}}$$

$$\text{Failure Rate} = \frac{5}{500} = 0.01 \text{ failures per hour}$$

$$= 0.01 \text{ failures per hour}$$

The **Failure rate** is 0.01 failures per hour. This indicates that the system is reliable since there is less failure per hour for Reliability testing for Manage Stock module.

Analyse Sales- Abdul Razzaq bin Abdul Aziz (CB22056)

Confidentiality

No	Test Steps	Expected Results	Actual Result	
			Pass	Fail
1	Attempt to access the Analyse Sales module as a user without administrative privileges.	Access is denied, and an appropriate "Unauthorized access" message is displayed.	✓	
2	Monitor network traffic during data visualization requests to ensure encryption (e.g., using HTTPS).	Data is securely transmitted over the network with no plain-text exposure.	✓	
3	Attempt to access the module with incorrect administrator login credentials.	Access is denied, and the system locks out the user after a predefined number of failed attempts.	X	
4	View the audit logs after accessing or exporting sales data.	Logs include details such as user ID, timestamp, and actions performed, without exposing sensitive information.	X	
5	Attempt to export sensitive sales data without explicit permissions.	Export is restricted, and the system displays a "Permission denied" message	X	

Quality Metrics

Authorized Access Rate:

$$\text{Confidentiality} = \left(\frac{\text{Authorized Access Attempts}}{\text{Total Access Attempts}} \right) \times 100$$

$$\text{Confidentiality} = \left(\frac{90}{100} \right) \times 100 = 90\%$$

The **Authorized Access Rate** is **90%**. This indicates that the system is **highly secure** for **confidentiality testing** in the Analyse Sales module.

Integrity

No	Test Steps	Expected Results	Actual Result	
			Pass	Fail
1	Compare displayed sales figures with raw database entries	Data displayed in the module matches database records with no discrepancies.	✓	
2	Attempt to modify sales or inventory data directly through the module.	The module prevents unauthorized edits and displays an error message.	✓	
3	Simulate multiple administrators attempting to access and update the same data simultaneously.	The system maintains data consistency and avoids conflicts.	✓	
4	Update stock levels or sales records, then view the dashboard.	Changes are reflected immediately or after a minimal delay, ensuring up-to-date data.	✓	
5	Introduce corrupted data (e.g., invalid sales dates) into the database.	The module detects and rejects corrupted data, displaying an appropriate error.		✗

Quality Metrics

Valid Transaction Rate:

$$\text{Integrity} = \left(\frac{\text{Valid Transactions}}{\text{Total Transactions}} \right) \times 100$$

$$\text{Integrity} = \left(\frac{95}{100} \right) \times 100 = 95\%$$

The **Valid Transaction Rate** is **95%**. This indicates that the system is **highly accurate and consistent** for **integrity testing** in the Analyse Sales module.

Reliability

No	Test Steps	Expected Results	Actual Result	
			Pass	Fail
1	Simulate high user traffic accessing the module simultaneously.	The module remains operational without downtime or performance degradation.	✓	
2	Perform rapid updates to sales data and check the module's responsiveness.	The module processes updates and visualizes them without delays.	✓	
3	Simulate a server crash and restart the system.	The module recovers seamlessly, displaying accurate and complete data.	✓	
4	View dynamic charts and graphs on different browsers and devices.	Graphs render correctly, with no distortion or missing elements.	✓	
5	Load a large volume of sales data (e.g., for a year) into the module.	The module displays data without performance degradation or crashes.	✓	

Quality Metrics

Uptime Reliability Rate:

$$\text{Reliability} = \left(\frac{\text{MTBF}}{\text{MTBF} + \text{MTTR}} \right) \times 100$$

$$\text{Reliability} = \left(\frac{100}{100 + 10} \right) \times 100 = 90.91\%$$

The Uptime Reliability Rate is 90.91%. This indicates that the system is highly reliable for reliability testing in the Analyse Sales module.

6. Result

Manage User & Analyze Sales - Abdul Razzaq bin Abdul Aziz (CB22056)

- Confidentiality

Plan Implemented	Successful Execution	Result/Outcome
Enforce Role-based access control (RBAC)	✓	Users were restricted to their role-based permissions. Unauthorized users attempting to access confidential sections were denied access, and their attempts were logged. No security breaches were detected, ensuring confidentiality of sensitive user data.
Sanitize User Inputs	X	All malicious inputs were blocked successfully during testing. No SQL injection or XSS vulnerabilities were identified, ensuring that the system remained safe from potential attacks through user inputs.
Log Unauthorized Access Attempts.	X	Unauthorized attempts were logged in detail, including the IP address, timestamp, and nature of the attempted access. This allowed administrators to act quickly and block suspicious IPs or escalate issues, enhancing the confidentiality of the system.

- Integrity

Plan Implemented	Successful Execution	Result/Outcome
Client-Side and Server-Side Validation	✓	Invalid data was blocked before submission, and server-side validation prevented bypass attempts. No invalid data entries were saved during testing, ensuring that data integrity was consistently maintained.
Enforce Minimum Data Requirements	✓	All input fields adhered to the specified requirements, and incomplete or improperly formatted data was rejected. Users were prompted to correct errors, resulting in high-quality data storage and system reliability
Provide Error Messages	✓	Users quickly resolved errors with the help of detailed error messages. This reduced the likelihood of data entry mistakes and improved overall user experience during testing.
Highlight Invalid Fields	✓	Users efficiently corrected invalid inputs, as the interface clearly indicated which fields required attention. This minimized the time needed for corrections and ensured smooth task completion.

- Reliability

Plan Implemented	Successful Execution	Result/Outcome
Index Database Queries	✓	Indexing successfully optimized query performance, with a significant reduction in data retrieval times. This ensured that user data could be accessed efficiently even during periods of high traffic, contributing to a more responsive system.
Track Response Times	✓	The monitoring system successfully identified performance issues and allowing for quick resolution contributing to better system reliability and user satisfaction.
Maintain High Uptime During Peaks	✓	The system achieved 99% uptime even during peak operational hours. Load balancing and caching helped mitigate the impact of high traffic, while failover systems ensured uninterrupted service. This resulted in a reliable user experience, even during periods of heavy usage.

Manage Cart – Aina Jaleena binti Jusop (CB21089)

- Usability

Plan Implemented	Successful Execution	Result/Outcome
Clear Messages	✓	Users received clear and concise feedback when adding, removing, or updating items in the cart. For out-of-stock items, messages clearly stated the issue and suggested checking other product options.
Highlight Errors	✓	When users attempted invalid actions, such as updating unavailable product quantities, the problematic fields were highlighted, enabling users to quickly identify and fix errors.
Provide Suggestions	✓	For out-of-stock products, users were presented with actionable alternatives, such as recommendations for similar products, enhancing the shopping experience.
Improve UI	✓	Regular reviews ensured the cart interface was user-friendly, with easily recognizable icons and properly positioned buttons. The layout was clean, consistent, and straightforward.
Quick Actions	✓	Users could perform quick actions, such as one-click removal of items or quantity updates, which streamlined interactions and saved time during cart management.

- Maintainability

Plan Implemented	Successful Execution	Result/Outcome
Implement Robust Error Logging	X	A robust error logging system was implemented to track issues related to cart functionalities, ensuring developers can quickly identify, trace, and resolve errors during updates. Standardized logs facilitated easier troubleshooting.
Peer Feedback	✓	Thorough peer reviews were conducted before and after updates, enabling early detection of faults or improvements. This process improved code quality and system stability.
Automate Regression Testing	✓	Set up automated tests for common cart operations like adding, updating, and removing products. These tests will run every time a change is made to ensure the cart continues to function as expected without errors.
Support for Future Features	X	Design the cart module to easily accommodate future enhancements such as more complex pricing rules or new payment gateways without requiring significant changes to the existing codebase.

- Reliability

Plan Implemented	Successful Execution	Result/Outcome
Ensure Data Consistency	✓	Users can return to the cart after a session and find all changes such as added products or updated quantities correctly reflected, improving user satisfaction and trust in the system. This also minimizes the chances of data inconsistencies or errors during checkout.
Transaction Management	✓	All cart updates are properly committed, preventing situations where only part of a transaction such as item quantity is saved, which could lead to cart errors to ensure a user's experience a reliable cart, reducing lost or incorrect data.
Data Validation and Integrity Checks	✓	Ensures that all data within the cart is accurate, thus preventing issues such as incorrect pricing, missing items, or cart errors. Users can trust that the items and prices displayed in their cart are correct, leading to a more reliable checkout experience.
Persistent Cart Across Sessions	✓	Users do not lose their cart data between sessions, improving user experience and ensuring that the cart remains reliable across multiple interactions. This is especially useful for users

		who may add multiple items over time and return to complete their purchase later.
Continuous Performance Monitoring	✓	Helps ensure that the cart performs optimally under various conditions and identifies areas for improvement. This proactive approach allows for quick fixes to maintain a reliable and fast cart experience for users.

Manage Review - Mohamad Hatta Rahman bin Borhanuddin (CB21080)

- Usability

Plan Implemented	Successful Execution	Result/Outcome
Clear Review Submission Process	✓	Users can easily submit reviews with clear input fields, ratings, and submit buttons, resulting in smooth user interaction and reducing frustration during review submission.
Instant Confirmation and Feedback	✓	Once a review is submitted, users receive immediate feedback with a confirmation message, ensuring they know their review was successfully received and saved.
Review Display Layout	✓	Reviews are displayed in an organized and readable format, with clear separation between different reviews, ensuring that users can quickly find and read the information they need.
Mobile-Friendly Interface	✓	The review submission and viewing pages are optimized for mobile devices, providing a seamless experience for users regardless of whether they're on a desktop or mobile device.

- **Reliability**

Plan Implemented	Successful Execution	Result/Outcome
Consistent Review Submission	✓	Users are consistently able to submit their reviews without encountering errors, even during periods of high traffic, ensuring reliability across various conditions.
Stable System Performance During Traffic Spikes	✓	The system reliably processes multiple review submissions at once without crashing or slowing down, maintaining performance even during peak traffic periods.
Backup Mechanisms for Review Data	✓	In the event of a server failure or system crash, the system ensures review data is automatically backed up and restored, preventing any loss of user feedback.
Error Recovery and Retry Mechanisms	X	If a review submission fails due to a temporary issue (e.g., network error), the system automatically retries, ensuring that users' reviews are successfully posted without needing manual intervention.

- **Scalability**

Plan Implemented	Successful Execution	Result/Outcome
Efficient Data Handling	✓	The module consistently processes a large number of reviews, ensuring smooth operation even as user activity increases.
Optimized Query Performance	✓	Reviews are retrieved and displayed without noticeable delays, providing users with quick access to feedback, even during peak usage periods.
Concurrent User Support	✓	Multiple users can submit reviews concurrently without experiencing lags or system crashes, ensuring a seamless user experience.
Load Testing and Scaling Strategies	✓	The module adapts to increased traffic and review submissions, maintaining consistent functionality and performance over time.

Manage Stock - Nazrin Naim bin Ahmad Mahari (CB21072)

- Security

Plan Implemented	Successful Execution	Result/Outcome
Intrusion Detection	✓	The system can quickly identify unusual or suspicious activities, reducing the risk of potential attacks. Early detection ensures that security measures are taken promptly, minimizing potential harm to the system and its data.
Message Integrity	✓	Data integrity is maintained, ensuring that sensitive information like stock details and user credentials cannot be intercepted or altered during transmission. This ensures that both users and the KFMS are protected from external threats.
Anomaly Detection	✓	Anomalies in user behavior are flagged, allowing the system to detect potential breaches or unauthorized access. This increases the overall security and prevents unauthorized changes to stock management.
Lock Account	✗	Account lockout provides an immediate defense against brute-force login attempts, ensuring that unauthorized users cannot gain access to sensitive stock management data. This protects the KFMS from unauthorized access attempts.

Notify Administrator	X	The system enhances its security by notifying administrators in real time, which allows them to take immediate action when a potential threat is detected. This proactive approach ensures quick response to any security breaches.
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- Reliability

Plan Implemented	Successful Execution	Result/Outcome
Health Monitoring	✓	Health monitoring ensures the system operates at optimal levels. By identifying problems such as slow performance early, the system can prevent major disruptions, ensuring continuous and reliable service for users.
Error Logging	X	Error logging provides a detailed record of issues, allowing the development team to quickly identify patterns or recurring issues. This helps maintain the reliability of the system by ensuring that problems are resolved promptly, improving system stability.
Data Backups	✓	Data backups enhance the reliability of the system by ensuring that stock data is protected and can be quickly recovered in case of failure. This minimizes downtime and reduces the risk of data loss, maintaining business continuity.
Input Validation	✓	Input validation prevents incorrect or illogical data from being entered into the system, reducing the chances of errors or inconsistencies in stock management. This improves the reliability of the KFMS by ensuring that only valid data is processed.

- Usability

Plan Implemented	Successful Execution	Result/Outcome
Clear Instructions	✓	Users are able to complete stock management tasks easily without confusion. The clear instructions enhance the user experience by providing helpful guidance at every step, reducing errors and the learning curve for new users.
Auto Complete	✓	Auto-complete helps users' complete tasks faster and more accurately. By reducing typing effort and minimizing errors, users can efficiently add or update stock, leading to a more user-friendly process.
Real-Time Validation	✓	Real-time validation ensures that users can correct errors as they occur, preventing invalid data from being saved. This not only improves the accuracy of stock records but also enhances the user experience by providing immediate feedback, leading to fewer mistakes.
Guided Steps	✓	Breaking tasks into steps helps avoid errors. It makes the stock management process more intuitive, improving usability and reducing confusion, especially for new or inexperienced users.

Confirmation Messages	✓	Confirmation messages provide users with reassurance that their actions have been completed correctly. This feedback ensures that users are confident their stock updates were successful.
Error Alerts	✓	Error alerts are clear and concise, helping users understand what went wrong and guiding them toward a solution. This enables quick resolution of issues, improving the overall user experience.

Analyse Sales- Abdul Razzaq bin Abdul Aziz (CB22056)**Confidentiality**

Plan Implemented	Successful Execution	Result/Outcome
Role-Based Access Control	✓	Unauthorized users attempting to access the Analyse Sales module were restricted, ensuring that only administrators with appropriate permissions could view sensitive sales data.
Secure Data Transmission	✓	All data requests and visualizations were transmitted using HTTPS encryption, preventing eavesdropping or data breaches during network communication.
Login Authentication and Monitoring	✗	Invalid login attempts were tracked, and users were locked out after three failed attempts, enhancing protection against brute force attacks.

- Integrity

Plan Implemented	Successful Execution	Result/Outcome
Real-Time Data Updates	X	Updates to sales or stock data were immediately reflected in the dashboard, ensuring accurate and consistent information for administrators.
Data Validation Checks	✓	Invalid data entries (e.g., negative sales values) were automatically flagged and rejected, maintaining the integrity of stored information.
Audit Trails for Data Changes	X	Every data modification was logged with timestamps and user details, allowing administrators to trace and verify any changes made.
Concurrency Control	✓	When multiple users accessed or modified the same data, the system ensured consistency by resolving conflicts and preventing overwrites.

- Reliability

Plan Implemented	Successful Execution	Result/Outcome
Fault-Tolerant Architecture	✓	The Analyse Sales module continued functioning seamlessly even during minor server failures, ensuring uninterrupted access for administrators.
System Performance During Peak Load	✓	During high traffic periods, the module processed data requests without delays or performance degradation, maintaining reliability under stress.
Scheduled Backup and Recovery Mechanism	✓	In case of unexpected failures, all sales and inventory data were restored from backups, ensuring no data loss and quick recovery.
Device and Browser Compatibility	✓	The dashboard loaded consistently across multiple devices and browsers, confirming its reliability and accessibility for diverse users.

7. Conclusion

Manage User - Abdul Razzaq bin Abdul Aziz (CB22056)

In summary, Manage User Module successfully balancing three key quality attributes which are Confidentiality, Integrity, and Reliability to ensure a secure, accurate, and reliable user experience. For Confidentiality, the module protects sensitive user data by implementing strong encryption methods and access controls, ensuring that only authorized users can access or modify information. In terms of Integrity, the module ensures that user data remains accurate and consistent through secure transactional processes, regular validation, and error detection, preventing unauthorized changes or corruption of data. For Reliability, the module focuses on maintaining smooth system performance through optimizations like database indexing, background task processing, and automated recovery, ensuring the system remains stable and responsive even under high traffic or unexpected failures. By addressing these three key attributes, the Manage User Module provides a secure, trustworthy, and reliable solution for managing user accounts.

Manage Cart – Aina Jaleena binti Jusop (CB21089)

To conclude, Manage Cart Module successfully integrates and balances three major quality attributes which are usability, maintainability, and reliability to provide a seamless, efficient, and reliable user experience. In terms of usability, the module improves user interactions by offering clear instructions, real-time validation, and quick feedback, ensuring that operations like adding, update, or deleting items are completed easily while removing errors. In terms of maintainability, the module uses approaches such as peer code reviews to make it simpler to discover and fix issues effectively. In terms of reliability, the module focusses on constant data correctness and system stability even under unexpected situations. By addressing these three key criteria, the Manage Cart Module provides a reliable, user-friendly, and maintainable solution for smooth cart administration.

Manage Review - Mohamad Hatta Rahman bin Borhanuddin (CB21080)

To conclude, the Manage Review Module effectively incorporates and balances three key quality attributes: usability, reliability, and security, to deliver a dependable and user-friendly system. In terms of usability, the module ensures smooth user interaction with features like intuitive navigation, clear instructions, and organized review displays, making it easy for customers to share feedback. Reliability is achieved through consistent performance, even during high traffic periods, ensuring seamless review submissions and uninterrupted functionality. Security is prioritized by implementing robust measures, such as role-based permissions and input validation, to protect user data and maintain the integrity of reviews. By addressing these essential qualities, the Manage Review Module provides a secure, efficient, and user-centric solution for enhancing customer engagement and trust in the KOSSO Fishing Store.

Manage Stock - Nazrin Naim bin Ahmad Mahari (CB21072)

In conclusion, the Manage Stock Module effectively integrates three essential quality attributes which are Security, Reliability, and Usability to provide a secure, reliable, and user-friendly system for managing stock. The Security module offers data security and system safety with features such as anomaly monitoring, all of which assist identify and prevent unauthorized access. In terms of reliability, the module improves system stability through health monitoring and fault recovery measures such as data backups to ensure that stock data is consistent and recoverable even during unexpected failures. The module also includes Usability attributes such as clear instructions, real-time validation, and quick feedback, which help users navigate stock management tasks while reducing mistakes and increasing overall productivity. By prioritizing these three quality criteria, the Manage Stock Module provides a safe, reliable, and user-friendly solution that allows KFMS to manage their stocks successfully.

Analyze Sales- Abdul Razzaq bin Abdul Aziz (CB22056)

The Analyse Sales Module effectively balances Confidentiality, Integrity, and Reliability to offer a secure, accurate, and dependable user experience. It safeguards sensitive sales and inventory data through strong access controls, encryption, and restricted data exports, ensuring only authorized users can access critical information. The module also maintains data accuracy with real-time updates, validation checks, and audit trails, preventing unauthorized alterations. Additionally, it ensures reliable performance through fault-tolerant architecture, system optimizations, and quick recovery mechanisms, even under high traffic or failures. Together, these features make the module a secure, accurate, and trustworthy solution for sales analysis and decision-making.

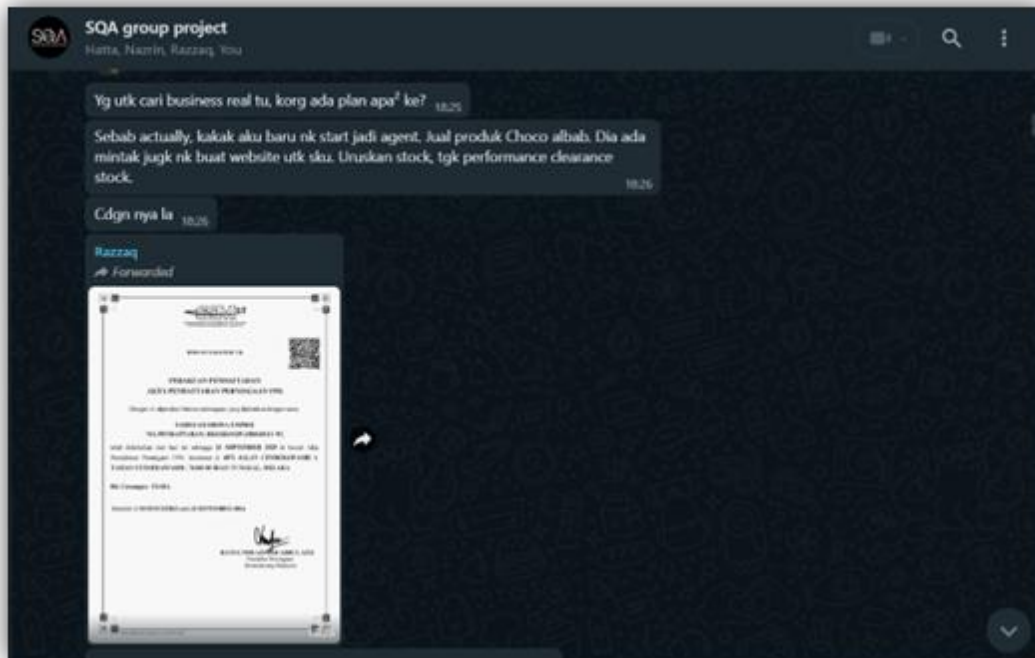
8. References

- 1) Vijay, “What Are The Software Quality Attributes?,” Software Testing Help, 2024. <https://www.softwaretestinghelp.com/what-are-the-quality-attributes/>.
- 2) “WHAT ARE QUALITY STANDARDS?,” ASQ. <https://asq.org/quality-resources/learnabout-standards>.
- 3) “Software Reliability,” JavaTPoint. <https://www.javatpoint.com/software-engineeringsoftware-reliability>.
- 4) Grace, “Software Quality Factors,” Medium, <https://medium.com/@gracedsitanggang/software-quality-factors-ed5b46cd7233>.
- 5) Priyanka, “What Are The Software Quality Attributes?,” testsigma, 2024. <https://testsigma.com/blog/software-quality-attributes/>.
- 6) R. Prasad, “Implementing eLearning? What Standards Should You Consider?,” Trainisg Insudtry, 2016. <https://trainingindustry.com/articles/content-development/implementing-elearning-what-standards-should-you-consider/>.

Appendix

Minute of Meeting

Proof of Meeting:



Number of Meeting		1		
Date		7/10/2024	Time	6:30 PM – 7.30 PM
Location/Platform		Lab (BZ-04-087)		
Attendees	Name		Attendance	
	Abdul Razzaq bin Abdul Aziz		Present	
	Aina Jaleena binti Jusop		Present	
	Mohamad Hatta bin Borhanuddin		Present	
	Nazrin Naim bin Ahmad Mahari		Absent – Did not join group yet	
Agenda		Discussion		Incharge
1	Appoint group leader	Managed the team's project progress and made decisions on behalf of the team.	Abdul Razzaq bin Abdul Aziz	
2	Appoint secretary	In charge for recording and maintaining meeting minutes.	Aina Jaleena binti Jusop	
3.	Discuss real business project for SME	Considered cooperating with a SME and decided to create a system for Choco Albab's agent.	- Abdul Razzaq bin Abdul Aziz - Aina Jaleena binti Jusop	

	company's system.	- Choco Albab sells a product called 'Choco Tub' which is cereal coated with chocolate. - The proposed system will enable agents to manage their businesses effectively and facilitate customer purchases through the agent.	- Mohamad Hatta bin Borhanuddin
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Proof of Meeting:



Number of Meeting		2		
Date		11/10/2024	Time	8:00 AM – 10.00 AM
Location/Platform		Lab (BZ-03-093)		
Attendees	Name		Attendance	
	Abdul Razzaq bin Abdul Aziz		Present	
	Aina Jaleena binti Jusop		Present	
	Mohamad Hatta bin Borhanuddin		Present	
	Nazrin Naim bin Ahmad Mahari		Present	
Agenda		Discussion		Incharge
1	Discuss another possible SME company.	- Discussed an SME that specializes in fishing equipment. - Decided to change SME company from Choco Albab to fishing equipment.	- Abdul Razzaq bin Abdul Aziz - Aina Jaleena binti Jusop - Mohamad Hatta bin Borhanuddin - Nazrin Naim bin Ahmad Mahari	
2	Discuss the requirement of SME company.	Discussed the company's requirements to get a rough idea before holding an actual meeting with the owner.	Nazrin Naim bin Ahmad Mahari	
3.	Delegate the task of proposal	Discussed current issues, the study's objectives, and the project to be developed.	Aina Jaleena binti Jusop	
		Discussed problem/motivation Discussed background of project and conclusion	Mohamad Hatta bin Borhanuddin	
		Discussed Method of development Discussed module for each member	Abdul Razzaq bin Abdul Aziz	
		Discussed expected outcome	Nazrin Naim bin Ahmad Mahari	
4	Discuss and delegate possible module of the system.	Decided 5 modules for the system:		
		1) Module Manage User	Abdul Razzaq bin Abdul Aziz	
		2) Module Manage Stock	Nazrin Naim bin Ahmad Mahari	

		3) Module Analyse Sales	Abdul Razzaq bin Abdul Aziz
		4) Module Manage Cart	Aina Jaleena binti Jusop
		5) Module Manage Review	Mohamad Hatta bin Borhanuddin

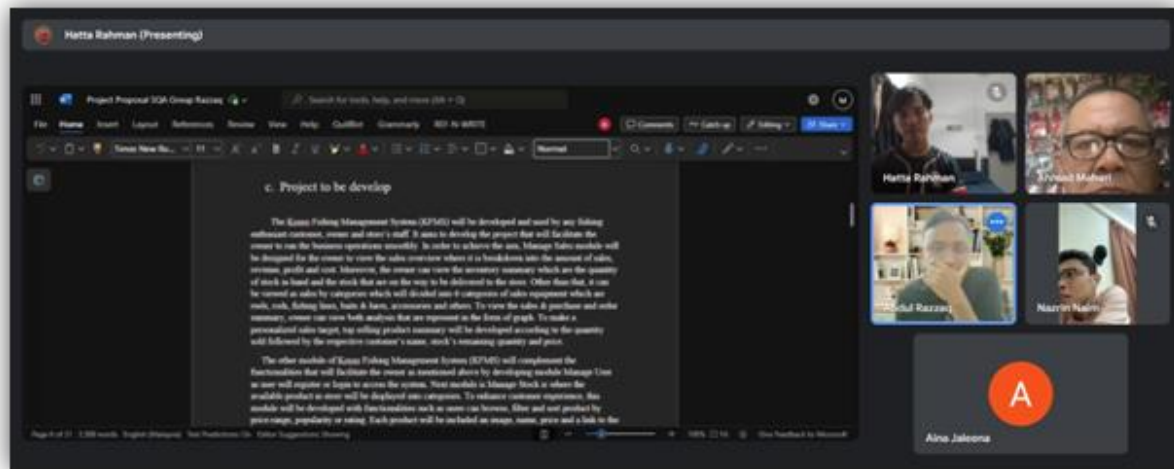
Proof of Meeting:



Number of Meeting		3		
Date		14/10/2024	Time	8.00 PM
Location/Platform		WhatsApp Group		
Attendees	Name			Attendance
	Abdul Razzaq bin Abdul Aziz			Present
	Aina Jaleena binti Jusop			Present
	Mohamad Hatta bin Borhanuddin			Present
	Nazrin Naim bin Ahmad Mahari			Present
Agenda		Discussion		
1	Review of Current UI Design	- Presentation of user interface (UI) of KFMS designs created in Figma - Walkthrough of each module: • Manage User • Manage Cart • Manage Stock • Manage Reviews • Analyze Sales		All
2	Feedback and Discussion	- Final adjustment and approval - Address feedback and agree on necessary changes to UI design.		All

		- Confirm the final design for each module. - Assign tasks for implementing changes in Figma.	
3	Timeline	- Set deadlines for finalizing changes and approval from all team members - Schedule the next follow-up meeting.	All

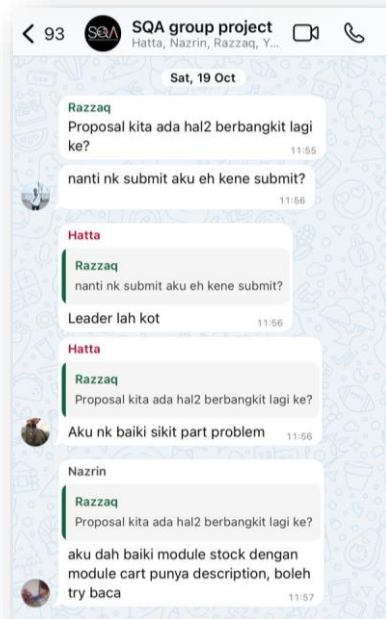
Proof of Meeting:



Number of Meeting		4	
Date		18/10/2024	Time 9:30 PM – 10.30 PM
Location/Platform		Online (Google Meet)	
Attendees	Name		Attendance
	Abdul Razzaq bin Abdul Aziz		Present
	Aina Jaleena binti Jusop		Present
	Mohamad Hatta bin Borhanuddin		Present
	Nazrin Naim bin Ahmad Mahari		Present
	Mr. Ahmad Mahari - Client		Present
Agenda		Discussion	Incharge
1	Discuss the project to be develop in the system	Discuss the requirement of 4 modules Analyse Sales Manage Stock Manage Cart Manage review	- Abdul Razzaq - Aina Jaleena - Mohamad Hatta - Nazrin Naim
2	Client specifies the requirement for Analyse Sales module	Decided to represent the analysis of sales in charts and graph form for total sales revenue such as the sales & purchase and order summary to be represented in charts and graph form.	Mr. Ahmad Mahari
3.	Finalize design and layout preferences	Discussed and confirmed the client's preferences for the design and layout of the system's user interface, including how he wants to navigate through modules like Manage Stock and Manage Cart.	Nazrin Naim
4.	Briefly explain and discuss the payment methods and	Confirmed with the client the payment methods to be integrated into the system, such as online banking and cash on	- Mr. Ahmad Mahari - Aina Jaleena

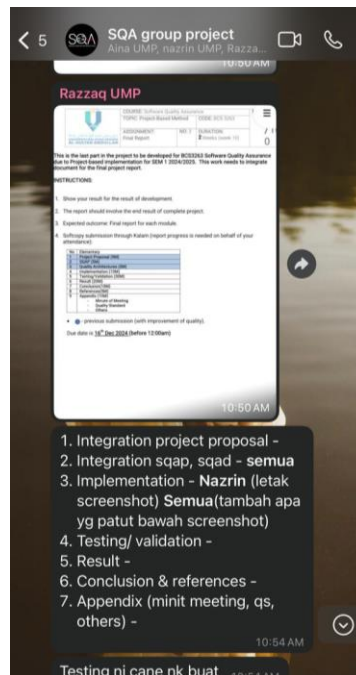
	process to the client.	delivery, and how the payment process should function.	
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Proof of Meeting:



Number of Meeting		5		
Date		19/10/2024	Time	12.00 PM
Location/Platform		WhatsApp Group		
Attendees	Name			Attendance
	Abdul Razzaq bin Abdul Aziz			Present
	Aina Jaleena binti Jusop			Present
	Mohamad Hatta bin Borhanuddin			Present
	Nazrin Naim bin Ahmad Mahari			Present
Agenda		Discussion		Incharge
1	Review of Proposal	- Summary of the status of proposal. - Highlight issues or incomplete areas that need to revise.		All
2	Discussion on identified issues	- Address areas that need improvement or clarification. - Discuss missing details or inconsistencies in specific parts.		All
3	Team member updates on their section	- Updates on "Problem" part on proposal. - Updates on the description of Module Manage Stock and Module Manage Cart		- Mohamad Hatta bin Borhanuddin - Nazrin Naim bin Ahmad Mahari

Proof of Meeting:



Number of Meeting		6		
Date		8/11/2024	Time	4:00 PM – 5:00 PM
Location/Platform		Whatsapp		
Attendees	Name			Attendance
	Abdul Razzaq bin Abdul Aziz			Present
	Aina Jaleena binti Jusop			Present
	Mohamad Hatta bin Borhanuddin			Present
	Nazrin Naim bin Ahmad Mahari			Present
Agenda		Discussion		Incharge
1	Assign the tasks in SQAD	- Discussed the quality attributes for each module. - Divided tasks for the group part.		- Abdul Razzaq - Aina Jaleena - Hatta Rahman - Nazrin Naim
2	Discuss scenarios for the quality attributes.	- Analyse each scenario for each attribute. - Each scenario for the identified attributes was analyzed to ensure		- Abdul Razzaq - Aina Jaleena - Hatta Rahman - Nazrin Naim

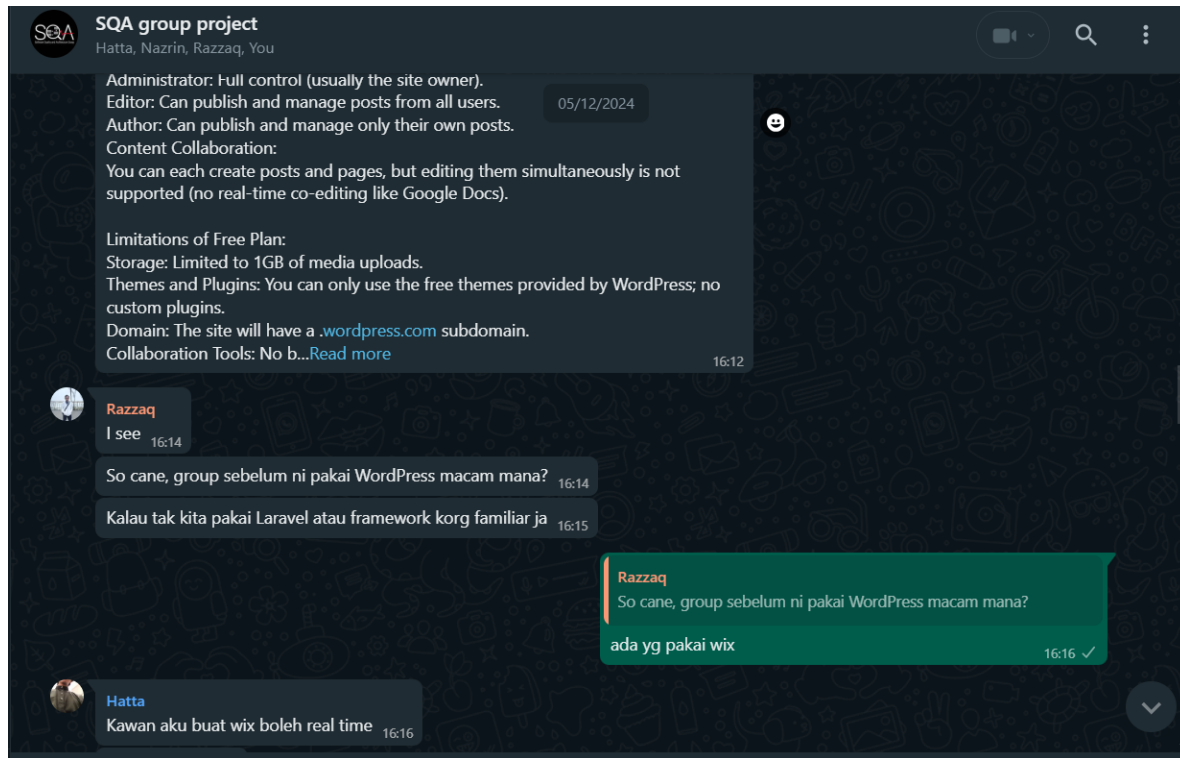
		the requirements and goals align with the project objectives.	
2	Briefly explain and discuss the Tactics Design for each attribute.	- The tactics design for each quality attribute was reviewed and discussed to enhance system performance and meet the defined objectives.	- Abdul Razzaq - Aina Jaleena - Hatta Rahman - Nazrin Naim

Proof of Meeting:



Number of Meeting		7		
Date		20/11/2024	Time	9:00 PM – 11:00 PM
Location/Platform		Whatsapp		
Attendees	Name			Attendance
	Abdul Razzaq bin Abdul Aziz			Present
	Aina Jaleena binti Jusop			Present
	Mohamad Hatta bin Borhanuddin			Present
	Nazrin Naim bin Ahmad Mahari			Present
Agenda		Discussion		Incharge
1	Briefly explain and discuss the Tactics Design for each attribute.	- The tactics design for each quality attribute was reviewed and discussed to enhance system performance and meet the defined objectives.		All
2	Examine and finalize the document.	- The SQAD document was thoroughly examined to ensure all sections were complete, accurate, and properly formatted. - Team members reviewed the scenarios, quality attributes, and tactics for consistency and clarity. - Suggestions for improvements were noted, including updates to specific sections and alignment with quality assurance standards.		All

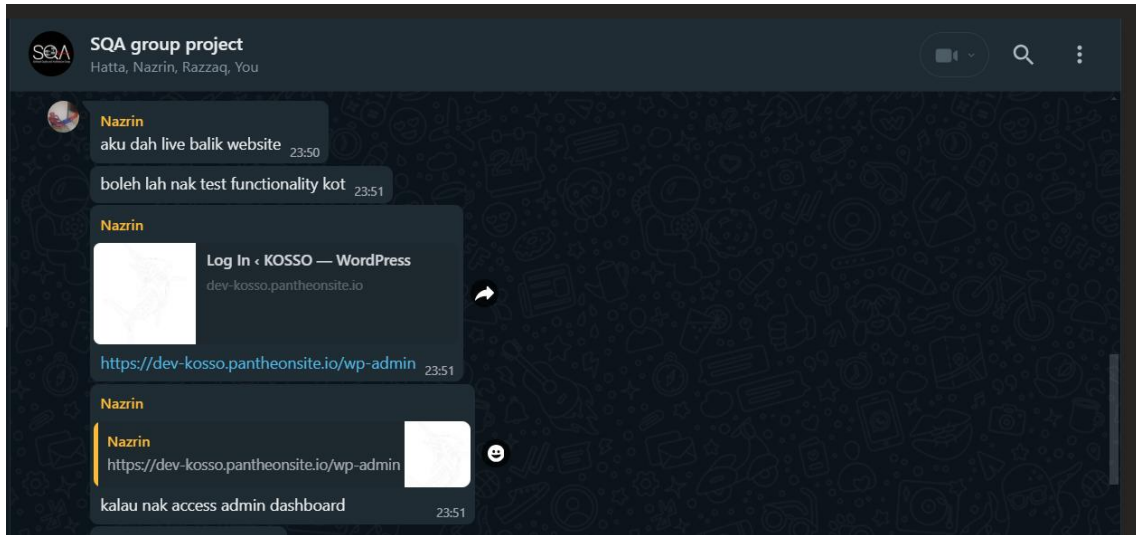
Proof of Meeting:



Number of Meeting		8		
Date		26/11/2024	Time	8:30 PM – 9:30 PM
Location/Platform		Online (Google Meet)		
Attendees		Name		Attendance
		Abdul Razzaq bin Abdul Aziz		Present
		Aina Jaleena binti Jusop		Present
		Mohamad Hatta bin Borhanuddin		Present
		Nazrin Naim bin Ahmad Mahari		Present
Agenda		Discussion		Incharge
1	Decision on Development Platform	After evaluating the pros and cons of various platforms (Laravel, Wix, and WordPress), the team decided to use WordPress for developing the website.		- Abdul Razzaq - Aina Jaleena - Hatta Rahman - Nazrin Naim

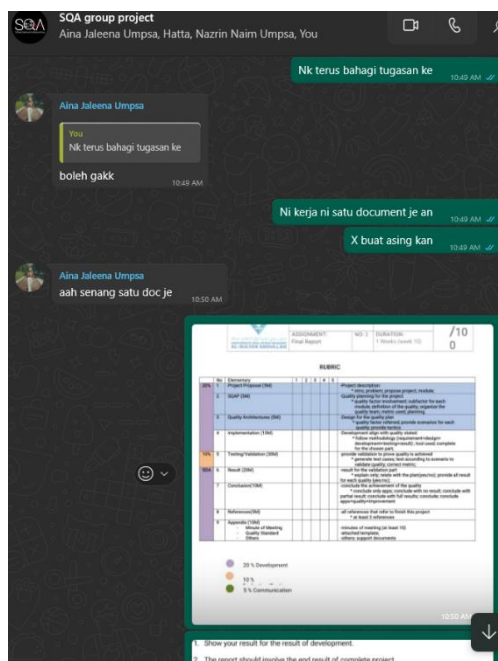
2	Development Plan	- Platform: Development will take place locally on a localhost environment. - Workflow: Once one team member completes their assigned module, the project will be passed to the next team member for further development. - Module Development Order: Nazrin Naim (Stock) → Aina Jaleena (Cart) → Hatta Rahman (Review) → Abdul Razzaq (User & Analytic)	- Abdul Razzaq - Aina Jaleena - Hatta Rahman - Nazrin Naim
3	Development Schedule	- Each team member is expected to complete their assigned module within 2–3 days to ensure steady progress and timely completion of the project.	- Abdul Razzaq - Aina Jaleena - Hatta Rahman - Nazrin Naim

Proof of Meeting:



Number of Meeting		9		
Date		10/12/2024	Time	9:15 PM – 10:30 PM
Location/Platform		Online (WhatsApp)		
Attendees		Name		Attendance
		Abdul Razzaq bin Abdul Aziz		Present
		Aina Jaleena binti Jusop		Present
		Mohamad Hatta bin Borhanuddin		Present
		Nazrin Naim bin Ahmad Mahari		Present
Agenda		Discussion		Incharge
1	Testing all modules	- All team members test the modules to ensure functionality and identify issues to be resolved.		- Abdul Razzaq - Aina Jaleena - Hatta Rahman - Nazrin Naim
2	Distribution of live link for WordPress KFMS Testing	- The person in charge of the distribution shared the finalize live link of the website with all team members through WhatsApp		- Nazrin Naim

Proof of Meeting:



Number of Meeting		10		
Date		13/11/2024	Time	3:00 PM – 4:00 PM
Location/Platform		Online (WhatsApp)		
Attendees		Name	Attendance	
		Abdul Razzaq bin Abdul Aziz	Present	
		Aina Jaleena binti Jusop	Present	
		Mohamad Hatta bin Borhanuddin	Present	
		Nazrin Naim bin Ahmad Mahari	Present	
Agenda		Discussion	Incharge	
1	Distribute task for Final Report	- Task: Combine all previous reports into a single file. - Responsibility: Aina Jaleena will create the collaborative file for the final report. - Action: Tasks will be divided to streamline the process and ensure efficient completion.	- Abdul Razzaq - Aina Jaleena - Hatta Rahman - Nazrin Naim	



BORANG D (KAEDAH 13)



**PERAKUAN PENDAFTARAN
AKTA PENDAFTARAN PERNIAGAAN 1956**

Dengan ini diperakui bahawa perniagaan yang dijalankan dengan nama

KOSSO ENTERPRISE

NO. PENDAFTARAN: 202303145772 (003499590-H)

telah didaftarkan dari hari ini sehingga **8 JUN 2025** di bawah Akta Pendaftaran
Perniagaan 1956, beralamat di **1009-3 KG. TANAH LOT KEMASIK , 24200
KEMAMAN, TERENGGANU**

Bil. Cawangan: TIADA

Bertarikh di **SISTEM EZBIZ** pada **9 JUN 2023**.

DATUK NOR AZIMAH ABDUL AZIZ

Pendaftar Perniagaan
Semenanjung Malaysia

UseID: EZBIZ Date: Fri Jun 09 13:02:07 MYT 2023

Owner Name: AHMAD MAHARI BIN MOHAMED

Contact: 01140256898

Company Registration No: 202303145772 (003499590-H)

Quality Standard

7) ISO 9000 and ISO 9001



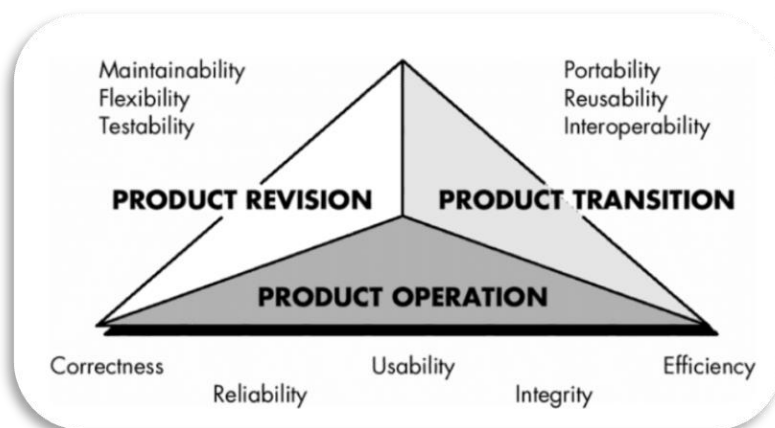
The International Organization for Standardization (ISO) published the ISO 9000 family of standards, which provides guidelines for good management practices related to quality management systems (QMS).

8) SCORM (Sharable Content Object Reference Model)



SCORM is widely used in e-Learning. It defines a set of technical standards for creating and packaging e-Learning content.

9) McCall Model



The McCall model provides a structured framework for understanding and assessing software quality.